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Historical Performance in European Football



A data-driven analysis of elite performance, recurrence and evolution in the best European league

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1. Introduction

Football analytics can help distinguish exceptional results from broader patterns of performance. Historical comparisons are particularly challenging because domestic competitions, scoring systems and playing conditions have changed substantially over time. Rather than treating modern football as the only reference point, this project examines elite team-season performances across eight European leagues and several historical periods.

The project focuses on four dimensions: overall points per match, points per match among non-champions, goals scored per match and goals conceded per match. These indicators capture different aspects of achievement: sustained results, outstanding seasons without a league title, attacking output and defensive efficiency. Keeping them separate makes it possible to examine where they overlap and where they tell different stories about football history.

The dataset covers All-Time and 1955+ periods, as well as a Since 3 Points period for each league; the Premier League also has a distinct historical window beginning in 1992 although the 3-points format started in 1981 in England. Because competitions did not all begin at the same time, these periods do not represent identical samples. The methodology therefore uses per-match measures and a normalization of historical points, while retaining the historical context of each competition.

The analysis proceeds from individual records to the depth of elite performance, comparisons between leagues, all-time rankings, decade distributions, club recurrence and relationships between performance indicators. The central question is not which league or club is definitively the strongest, but how exceptional performances differ by metric, competition and historical period, and what these differences can reasonably tell us about the evolution of elite football.

2. Data and Methodology

1. Dataset

The dataset used in this project brings together historical league performance data from eight European domestic championships: the Premier League, La Liga, Serie A, Bundesliga, Ligue 1, Primeira Liga, Eredivisie and the Hungarian top division. For each league, the dataset contains team-level season statistics covering three historical periods: from the adoption of the three-point system to the present day, a period since 1955 and an all-time period with a bonus period between 1955 and 1992 for the Premier League. The main variables used throughout the analysis are points, matches played, goals scored and goals conceded. These variables are used to construct the four performance indicators studied in this project: points per match, points per match among non-champions, goals scored per match and goals conceded per match. The analysis is conducted at team-season level, meaning that each observation represents the performance of a club during a specific league season. This structure makes it possible to identify individual historical records, construct Top 10 rankings for two periods and Top 15 all-

time period and study the recurrence of clubs among the strongest performers across different periods. The dataset therefore provides both a club-level and historical perspective, allowing individual performances to be studied while also examining broader changes across leagues and eras in relation to their context.

Domestic League	Since 3 pts	1955+	All-Time	1955-1992
Premier League	✓	✓	✓	✓
Ligue 1	✓	✓	✓	—
Serie A	✓	✓	✓	—
Liga	✓	✓	✓	—
Bundesliga	✓	✓*	✓	—
Eredivisie	✓	✓*	✓	—
Primeira Liga	✓	✓	✓	—
Hungary	✓	✓	✓	—

Table 1: Repartition of the historical periods for each domestic league. * Since Eredivisie et Bundesliga were created after 1955, the 1955+ and All-Time periods are the same for both leagues.

2. Historical Point Normalization

One of the main methodological challenges of this project is comparing teams from different eras, as the points system has not always been the same across European football. Before the adoption of the modern three-point system, a win was generally worth two points.

Two approaches can be considered when dealing with this historical difference. The first approach is to reconstruct historical points using the modern three-point system, by assigning three points to each historical win and one point to each draw:

Modern-equivalent Points per Match = (3 × Wins + Draws) / Matches

This approach answers a specific question: how many points per match would this team have obtained if the modern three-point system had been used?

The second approach is to normalize the original historical points relative to the maximum possible points under the scoring system of the time:

Normalized Points per Match = (Historical Points / Maximum Possible Historical Points) × 3

This is the approach used in this project. Rather than rewriting historical results according to a modern scoring system, it measures how close a team came to the maximum possible performance within its own historical scoring system, while expressing that performance on a common 0–3 scale. This distinction is important because the two methods do not measure exactly the same thing. The first method creates a hypothetical modern equivalent, whereas the second preserves the historical performance itself and only changes its scale. For a project focused on comparing relative levels of dominance across different eras, the latter approach was therefore selected.

For example, consider a team that recorded 20 wins, 10 draws and 8 defeats over 38 matches under a two-point system. Its original points total would be 50, out of a possible 76. Using the normalization method, its performance becomes:

(50 / 76) × 3 = 1.974 points per match

Using the modern-equivalent method instead would give:

(20 × 3 + 10) / 38 = 1.842 points per match

The difference illustrates why the choice of methodology matters. The first value represents the team's performance relative to the maximum available under its historical scoring system, while the second represents a hypothetical reconstruction under modern rules. The dataset used in this project already provides normalized points-per-match ratios for historical seasons. These values are therefore preserved directly in the analysis. The normalization described above is used as the methodological framework and as a fallback when a normalized ratio is not available, rather than recalculating historical performances unnecessarily.

3. Historical Competition Context and Format Changes

Historical league performances must also be interpreted in the context of how domestic competitions were structured at the time. European championships have not always operated with the same number of teams, number of matches or competition format. These differences do not invalidate historical comparisons, but they are important when interpreting records across very long periods.

One of the most obvious changes concerns the number of teams participating in the top division. The Premier League, for example, began in 1992 with 22 clubs, meaning that each team played 42 league matches. The competition was reduced to 20 clubs from the 1995–96 season onwards, resulting in the 38-match format that is now familiar. Similar changes occurred in other European leagues. La Liga, for example, has operated with different numbers of clubs throughout its history, including periods with 10, 12, 14, 16, 18 and 20 teams, as well as two seasons with 22 clubs in 1995–96 and 1996–97.

Other championships have also experienced changes in league size and season structure over time. In Hungary the league started with 6 teams and the number of games changed a lot

throughout the seasons. The number of matches is particularly relevant when comparing historical records. A team playing 42 matches in a season has more opportunities to accumulate points or score and concede goals than a team playing 34 or 38 matches. For this reason, the analysis relies primarily on per-match indicators rather than raw seasonal totals. Points per match, goals scored per match and goals conceded per match provide a more consistent basis for comparison when the number of fixtures varies between periods.

Competition formats have also changed in other ways. Some historical seasons included different phases, regional or group structures, or final stages used to determine the champion or other positions. These formats mean that not every historical championship season represents exactly the same competitive structure as the modern double round-robin league. Such seasons are therefore considered within their historical context rather than being treated as perfectly equivalent to contemporary seasons. The French division provides another particularly interesting example of historical variation. During the 1970s, France experimented a bonus-point systems designed to encourage attacking football. In 1973–74, teams could receive an additional point when a match produced at least three goals, while a different bonus system based on winning by at least three goals was subsequently introduced. These experiments were eventually abandoned, and France later adopted the modern three-point system permanently in 1994.

There is also an important distinction between the creation of a competition and the introduction of a scoring system. In England, for example, the three-point system was introduced in the top division before the creation of the Premier League. The Premier League itself began in 1992, while three points for a win had already been used in English top-flight football since 1981. Consequently, the “Since 3 points” period used for the Premier League in this dataset corresponds to the Premier League era beginning in 1992, rather than to the complete period during which three points were used in the English top division. This distinction is important when interpreting the historical periods in the dataset.

Changes to the Laws of the Game, including the 1925 offside-law change and later interpretations, also form part of the context in which scoring rates evolved. These developments are not modelled directly in the dataset, so they should inform historical interpretation rather than be treated as proven explanations for individual records. The analysis consequently compares per-match outcomes while acknowledging that league formats and the game itself were not constant over time.

Overall, these differences mean that historical football should not be viewed as a completely static environment. The analysis therefore focuses on indicators that are as comparable as possible while acknowledging that league formats and the rules of the game have changed over time.

3. Historical Records

1. Historical records by league

The first step of the analysis is to identify the highest levels of performance statistics reached in each domestic championship. It also might not be the best team that reached the highest performance because of the different context and we could also argue that it's even more difficult to achieve these kinds of performances when the domestic league is at its best overall. The objective here is also not to determine which league or club is "better", but to establish the historical performance ceiling observed within each competition.

Four distinct performance dimensions are analysed separately:

- overall points per match;
- points per match among non-champions;
- goals scored per match;
- goals conceded per match.

The distinction between these indicators is important. A team may have achieved an exceptional points-per-match record without holding the highest attacking record, while another team may stand out primarily through its defensive performance. Keeping the four dimensions separate therefore makes it possible to preserve the different meanings of each record rather than reducing performance to a single composite measure.

The heatmaps below present the highest value recorded for each league and period as mentioned in the methodology. For points and goals scored, a higher value represents a stronger performance. For goals conceded, the interpretation is reversed: a lower value represents a stronger defensive performance.

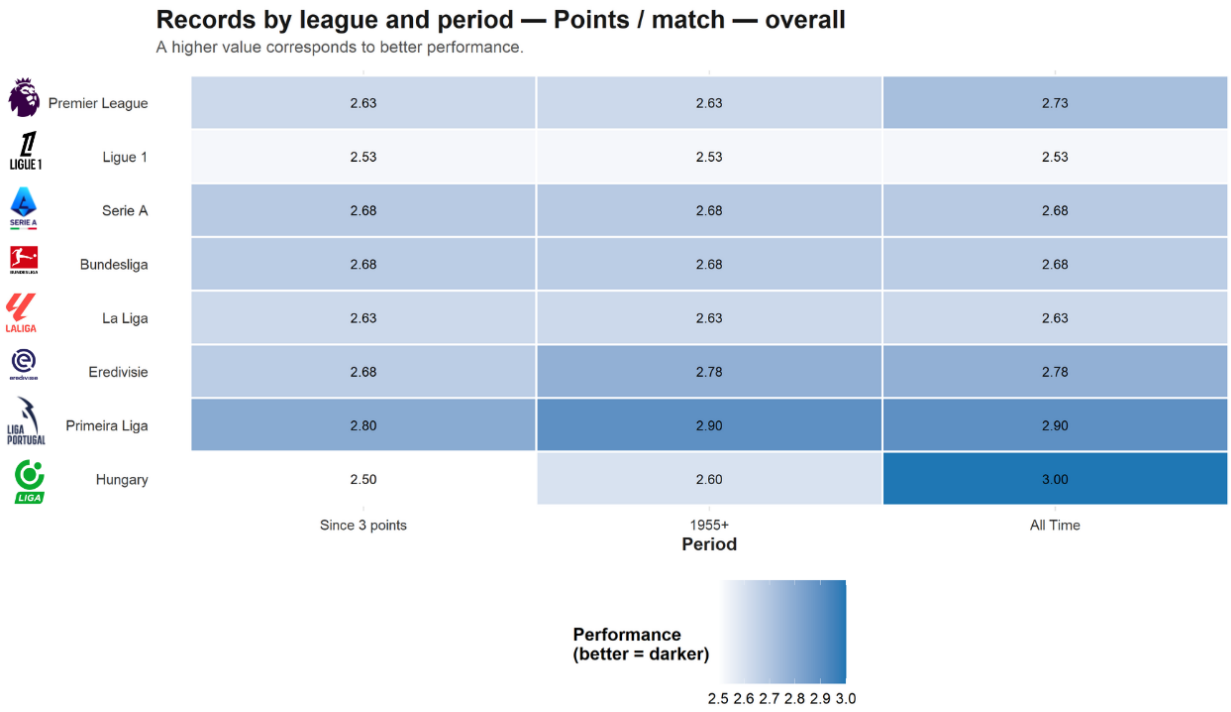


Fig. 1: Heatmap for the record of points per match for each league and period.

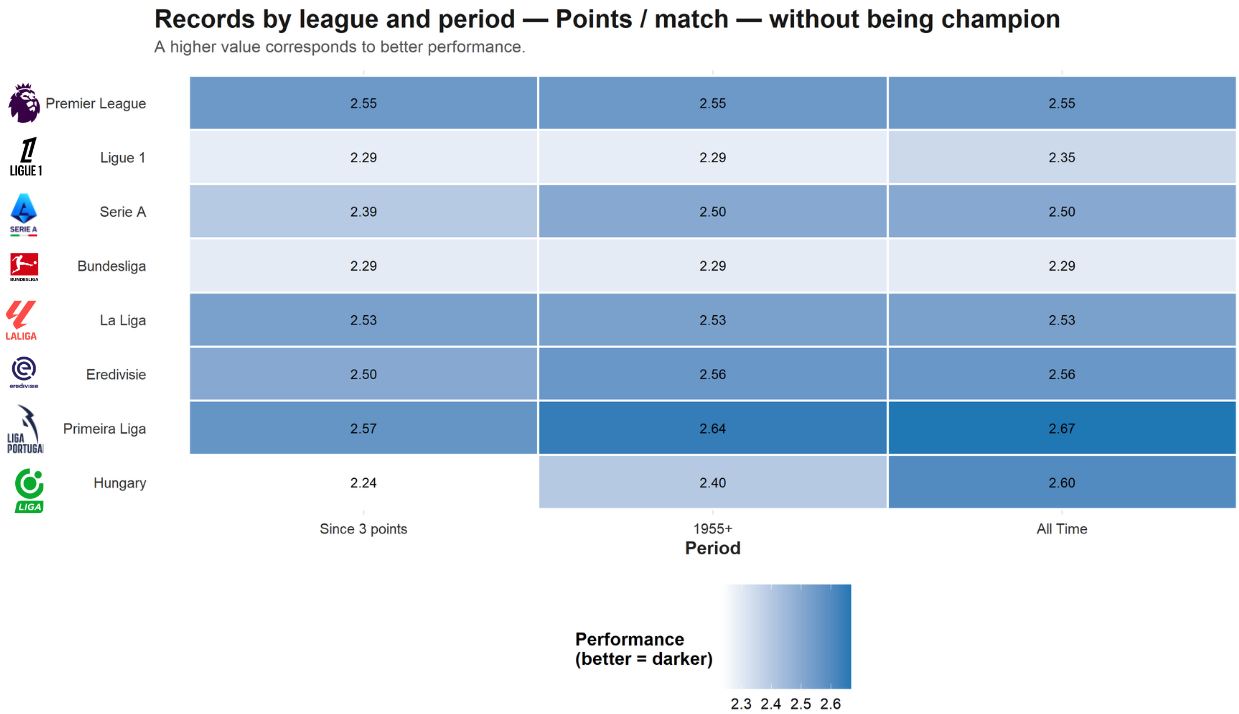


Fig. 2: Heatmap for the record of points per match without being champion for each league and period.

The records for points per match show that the highest levels of league performance are relatively consistent across several European championships (Fig. 1). The strongest performance is found in Hungary, reaching the maximum ratio of 3.00 points per match during the All-Time period. Primeira Liga, Serie A, Bundesliga and Eredivisie also stand out, with records reaching or approaching 2.8 points per match. Primeira Liga is particularly notable, as its record remains above 2.8 points per match regardless of the period considered. The Premier League, La Liga and Ligue 1 also show very high records. Among non-champions it's consistently lower than the overall records, which is expected since this category excludes the league winner (Fig. 2). However, the values remain remarkably high across all eight championships, showing that historically dominant teams did not always win their domestic league. Primeira Liga records the highest values in this category, while La Liga and the Premier League also show strong performances. This metric therefore highlights that elite-level performances can occur outside championship-winning seasons.

The records for goals scored per match show much larger differences between championships than those observed for points per match (Fig. 3). The highest records are found in Hungary, where the All-Time record reaches 6.68 goals per match which is astonishingly high, especially in comparison to the records of the classic top 5 domestic leagues. Primeira Liga also stands out, with its All-Time record reaching 4.83 goals per match. La Liga, Bundesliga and Eredivisie also show particularly high records, generally above 3.5 goals per match. By comparison, the Premier League, Ligue 1 and Serie A display lower maximum values, especially in the most recent period.

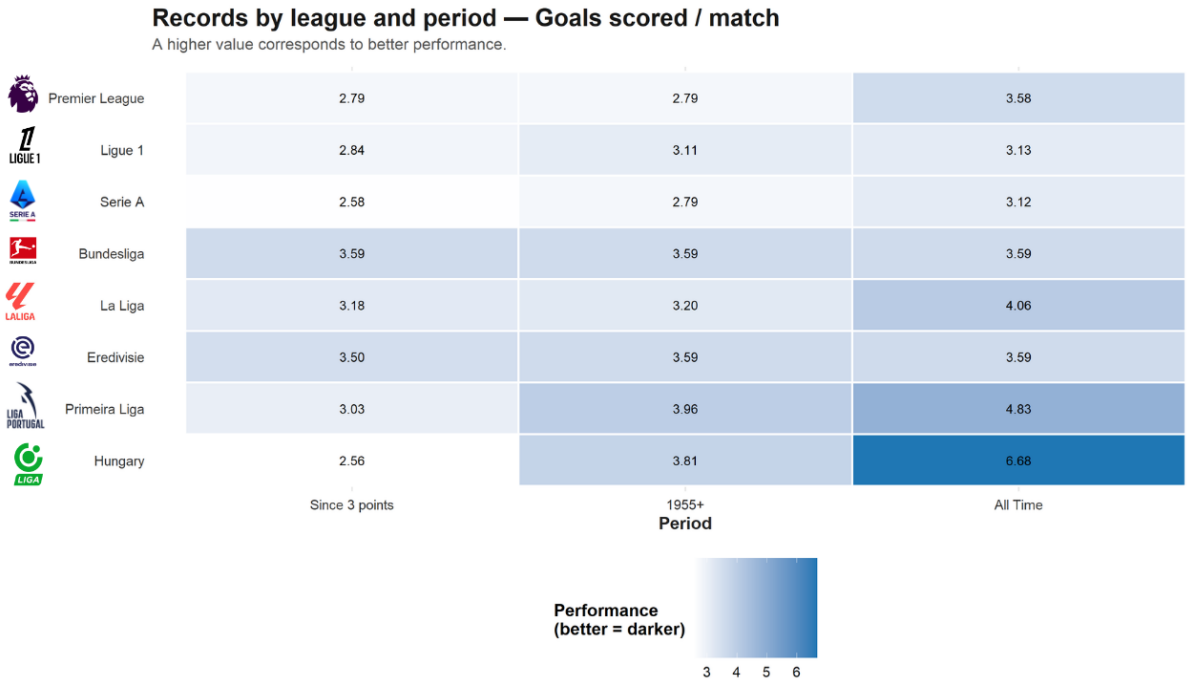


Fig. 3: Heatmap for the record of goals scored/match for each league and period.

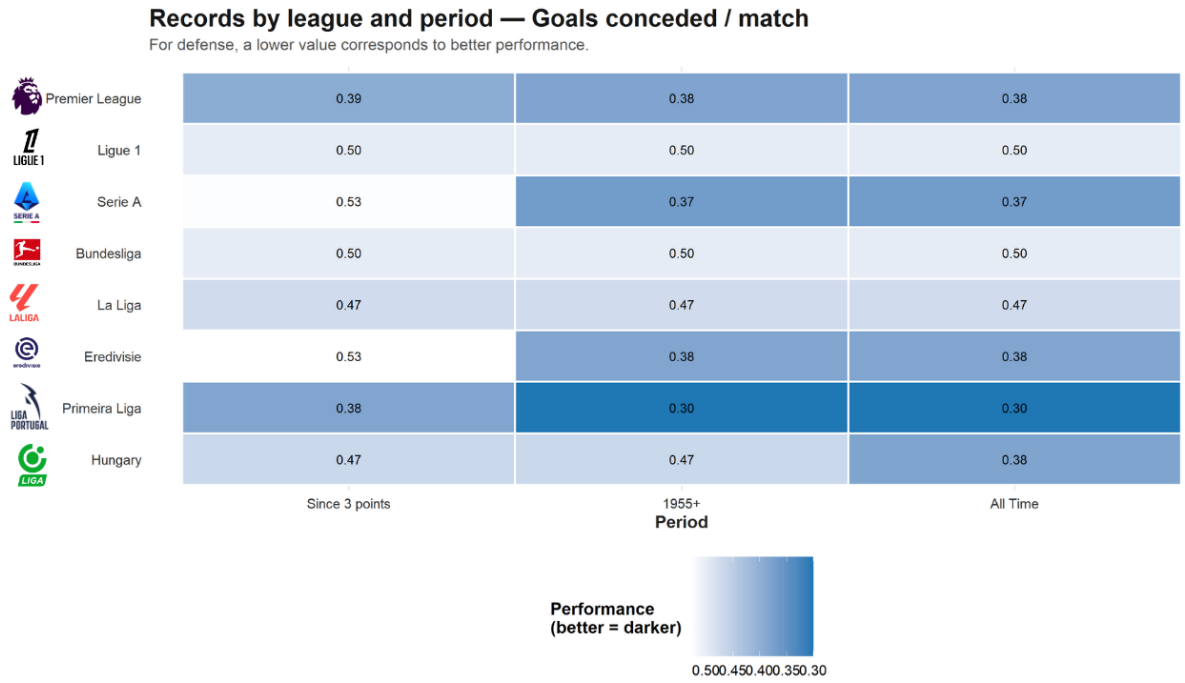


Fig. 4: Heatmap for the record of least goals conceded per match for each league and period.

The records for goals conceded per match show a different pattern, with lower values indicating stronger defensive performance (Fig. 4). Primeira Liga records the lowest value overall, at just 0.30 goals conceded per match in the 1955+ and All Time periods. The Premier League also stands out, with a record of 0.38 goals conceded per match across the historical periods considered. Serie A and Eredivisie show particularly strong defensive records in their longer historical periods, while the other championships generally remain around 0.47–0.50 goals conceded per match.

2. *Who Holds the Records?*

Now let's present and talk about some of the teams behind these incredible records:

Metric	League	Club	Period	Season	Record
points per match	BUNDESLIGA	FC BAYERN MÜNCHEN	All-Time	2012-2013	2.68
	LIGA PORTUGAL		All-Time	1972-1973	2.90
points per match among non-champions		L.F.C.	All-Time	2018-2019	2.55
	eredivisie	FEYENOORD ROTTERDAM	All-Time	1972-1973	2.56
goals scored per match	LIGA	MTK BUDAPEST	All-Time	1917-1918	6.68
	LALIGA	ATHLETIC CLUB BILBAO	All-Time	1930-1931	4.06
goals conceded per match	SERIE A	CAGLIARI	All-Time	1969-1970	0.37
	LIGUE 1	PARIS SAINT-GERMAIN	All-Time	2015-2016	0.50

Table 2: Selected historical records across the eight European championships.

For the all-time records in points per match, the first team selected is Bayern Munich in the 2012–2013 season. This was one of, if not the, greatest seasons in the club's history. Bayern won the domestic league, the DFB-Pokal and the Champions League, completing a historic treble. They defeated their domestic rivals Borussia Dortmund in the first Champions League final between two German clubs. During the season, Bayern also produced one of the highest goal totals and one of the lowest numbers of goals conceded in Bundesliga history, finishing with a goal difference of +80. This exceptional balance was built on the quality of their goalkeeper Manuel Neuer, a defence led by captain Philipp Lahm, and an outstanding attacking unit featuring Arjen Robben, Franck Ribéry, Thomas Müller and their two main strikers, Mario Mandžukić and Mario Gómez. The second record goes back much further, to the 1972–1973 Benfica side led by Eusébio. Benfica remained unbeaten throughout the domestic league season and were top of the table from the first to the last matchday, making this undoubtedly one of the greatest seasons in Portuguese football history. Although the club associated with the so-called Béla Guttmann curse was eliminated early in the season from the European Cup by Derby County and from the Taça de Portugal by Leixões, Benfica dominated the domestic championship in extraordinary fashion. They scored 101 goals in just 30 matches while

conceding only 13, placing this team among the historical record holders for both attacking and defensive performance. The season was also the final outstanding campaign of Benfica's greatest legend, Eusébio, who enjoyed one of his best scoring seasons, with 42 goals in 33 official matches.

For the all-time records in points per match without being champion, the first team selected is Liverpool during their legendary season 2018-2019 under Jurgen Klopp. This season saw the club winning their sixth champion's league and finishing second in Premier League only point away from Manchester City with the impressive amount of 97 points in 38 matches. This is an outstanding season which also remains one of the very best in terms of points per match during the Premier League era (Manchester City did better in 2017-2018 and in 2018-2019). The second team selected is Feyenoord Rotterdam also in 1972-1973 during the last season of the legendary coach Ernst Happel. Few seasons after winning the European cup and under the domination of the Ajax of Rinus Michels, led by their best player Willem Van Hanegem, Feyenoord achieved one of their best seasons and finished second, two points away from this outstanding Ajax Amsterdam of Johan Cruyff.

For the all-time records in goals scored per match, let's begin with the most outstanding record of this series and of the whole portfolio and the MTK Budapest in 1917-1918. It's of course a very far era since it's more than 100 years ago, during the first world war. The football was far less developed in Europe and professional football nearly didn't exist. Although, the Hungarian domestic league was one of the first in Europe like in Italy and started in 1901, the football changed a lot since these days. This season was the second one since the stop of the Hungarian league in 1914. The MTK Budapest was already dominating the league especially since the best Hungarian player of the decade joined them: Alfred Schaffer. In 1916-1917, the club ended up champion while losing only one game but in 1917-1918, they ended up unbeaten with only a tie game and twenty-one wins. But the most impressive statistic here is obviously these 147 goals scored in 22 matches which result in a frightening ratio of 6.68 goals scored per match, totally out of this world. Less impressive in comparison to this statistic, we have the Athletic Club in 1930-1931 which is the record for the most goal scored/match in the Big 5 leagues (Premier League, Liga, Serie A, Bundesliga and Ligue 1). They ended up champion for the second time after last season and scored 73 goals in only 18 matches which is amazing.

For the all-time record in goals conceded per match, the first team selected is Paris Saint-Germain during the 2015–2016 season under Laurent Blanc. This was genuinely one of, if not the, greatest seasons in the history of Ligue 1. What makes this performance particularly interesting is that PSG combined an extraordinary attacking output with an equally impressive defensive record, conceding only 19 goals throughout the league season. PSG won all four domestic trophies and broke several records, including the record for most goals scored in a Ligue 1 season and the highest points total, with 96 points from 38 matches. Led by Zlatan Ibrahimović, who scored 38 league goals and 50 across all competitions, PSG produced an exceptional season on both sides of the ball. Despite their domestic dominance, however, they were unable to progress beyond the quarter-finals of the Champions League. The second team selected is Cagliari in the 1969–1970 season, which produced one of the greatest defensive performances in European football history and the best record among the Big Five leagues in

this dataset. Although the most remembered player from this team is undoubtedly their legendary striker Luigi Riva, Cagliari were also remarkably well organised defensively. Their style was influenced by the defensive principles associated with *Catenaccio*, while being adapted to the team's own approach. The term does not simply mean playing with an ultra-defensive or "locked-down" system; rather, it refers to a broader defensive organisation based on structure, discipline and tactical coverage. Cagliari ultimately won the Serie A title, scoring 42 goals in 30 matches while conceding only 11.

3. How Records Changed Across Eras

Records are often broken over time, although some have remained unbeaten for decades or even remain standing today. This is particularly true for records related to champions and teams with the highest points-per-match ratios, both overall and among non-champions. Indeed, some of the very best performances date back to the very beginning of domestic league football. Preston North End, for example, achieved an exceptional points-per-match record in the first season of the English Football League in 1888–1889, which still stands among the highest records in the Big Five leagues. A similar example can be found in Hungary, where Budapesti TC became one of the country's earliest champions after winning every match of the season. Most of the other records, however, date either from the 1930s or from the modern era. Since the 2010s, the Big Five leagues have seen the emergence of several dominant teams that could be described as "superteams", driven by the rapid growth of broadcasting rights, the transfer market and the arrival of major investors. As a result, many of the record performances in these leagues are relatively recent, with examples such as PSG in 2015–2016 and Bayern Munich in 2012–2013, which have already been discussed, but also Juventus in 2013–2014 in Serie A, FC Barcelona in 2012–2013 and Real Madrid in 2011–2012 in La Liga.

A similar pattern can be observed for goals scored per match, with peaks in the 2010s, particularly in Spain and Germany. However, many of the highest attacking records still date from the period before the Second World War, especially in Hungary, where some of the strongest performances were achieved before 1920, as well as in England. The pattern is largely reversed for goals conceded per match. Most of the strongest defensive records date from the 1960s to the 1980s, although some exceptional modern performances can also be found, such as PSG in 2015–2016 and Bayern Munich.

These differences between periods are particularly interesting when looking at points per match. The evolution of this indicator does not simply reflect an increase in the number of points obtained by the best teams. It also reflects changes in the competitive environment, the points system and the overall structure of domestic competitions.

Season length and competition format are important context for historical records. Per-match indicators reduce the direct effect of differing numbers of fixtures, but they cannot make seasons fully equivalent when leagues used regional phases, final groups or unusual bonus-point rules. The records should therefore be interpreted within the competition structure in which they were achieved.

Points-per-match records also need to be read alongside the normalization method described in Section 2.2. The normalized measure compares performance relative to the maximum available under the original scoring system; it does not reconstruct modern points totals. This distinction matters when interpreting apparent stability in points performance across eras.

Attacking and defensive records show larger historical variation than points-per-match records. Changes in rules, tactics, professionalization and the competitive environment are plausible parts of that story, but the rankings cannot isolate their individual effects. The decade analysis later in the report provides a more direct view of when these exceptional performances occurred.

4. Depth of Elite Performance

Historical records identify the highest level reached by a team, but they do not indicate whether such performances were isolated or whether similarly strong performances were achieved repeatedly. This section therefore looks beyond the single best performance by examining the depth of the Top 10 rankings for each of the four performance dimensions. The average of the ten best performances is compared with the record for each league and period, providing an indication of how concentrated elite performance was behind the historical peak.

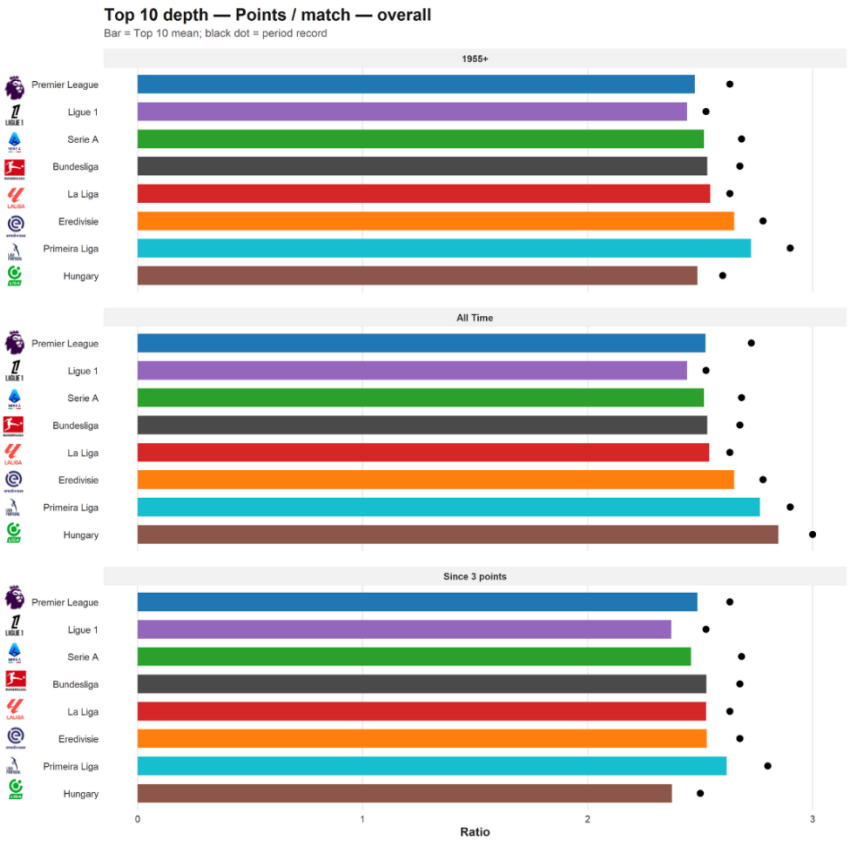


Fig. 5: Average of the Top rankings for points per match for each domestic league and period.

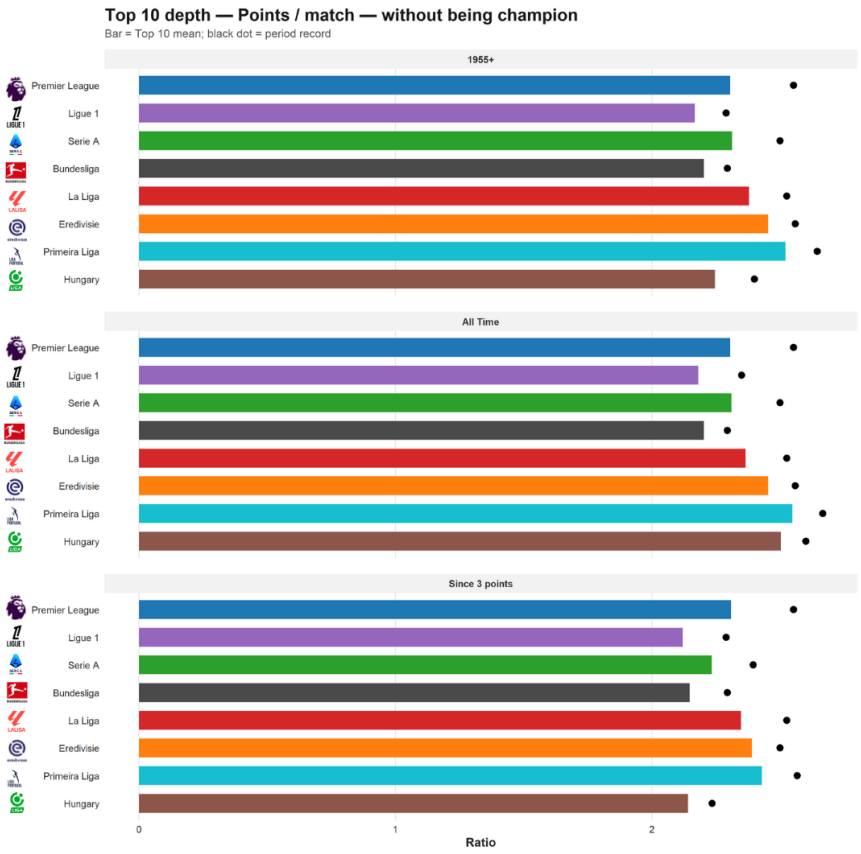


Fig. 6: Average of the Top rankings for points per match without being champion for each domestic league and period.

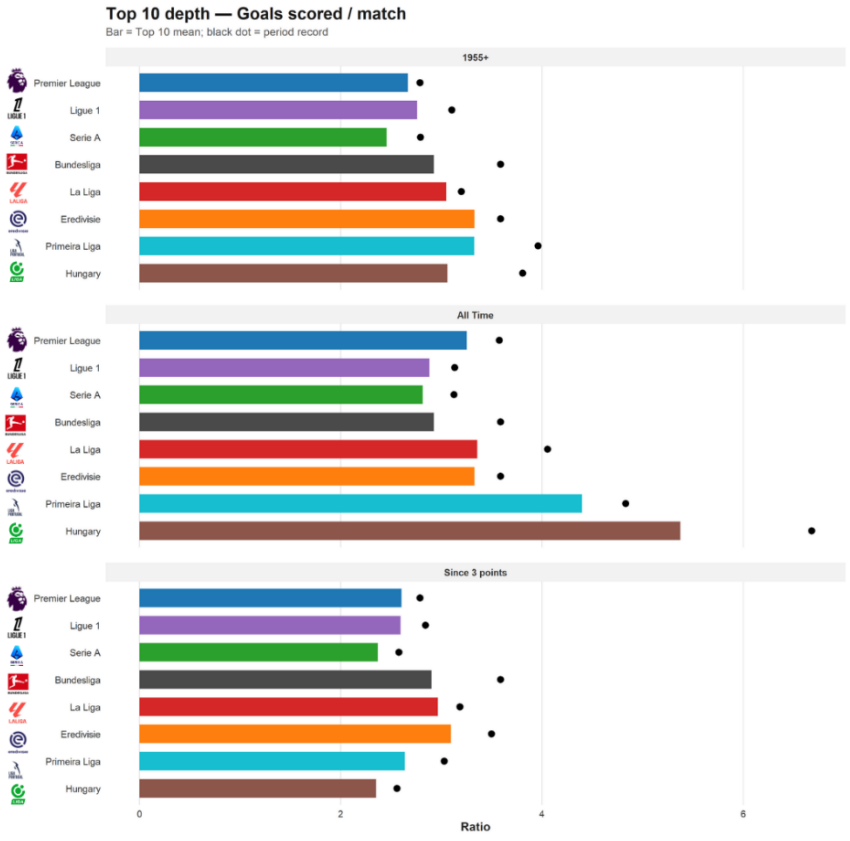


Fig. 7: Average of the Top rankings for goals scored/match for each domestic league and period.

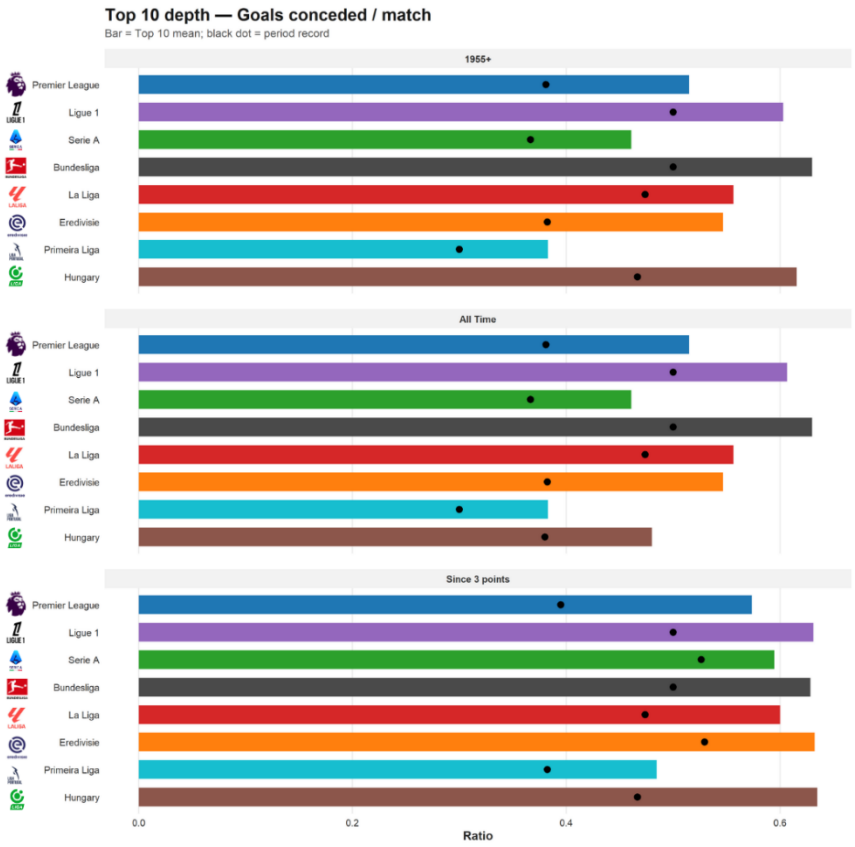


Fig. 8: Average of the Top rankings for goals conceded per match for each domestic league and period.

The comparison between the record and the Top 10 average provides an important distinction between an isolated historical peak and a deeper concentration of elite performances. This is also what we can observe with the boxplots (See Fig. S1, Fig. S2, Fig. S3, Fig. S4). A relatively small gap suggests that several teams reached similarly high levels, whereas a large gap indicates that the record stands further apart from the rest of the Top 10. This distinction is particularly relevant for the historical records identified in the previous section, as some of the most extreme performances may represent exceptional individual seasons rather than a broader pattern of elite performance.

The results show that the depth of elite performance in the top 10 and top 15 varies considerably between leagues, periods and performance dimensions. First of all, it's very interesting to look at the variation between the three periods when we look at the points per match. Indeed, it doesn't change very much overall when we look at the average of the elite performances and in the big 5 it remains close to the record team except maybe in England, especially among the non-champions. We can explain that by the exceptional performances that we've seen recently especially with Arteta's Arsenal in 23-24 or Klopp's Liverpool in 2018-2019 and 2021-22 combining with more ancient performances like Leeds United under Don Revie in 1970-1971. This can lead to a bigger gap overall but we still can observe that it's mostly similar across the periods. On the opposite, we can see that in the other three leagues, especially in Hungary, there are more gap. We can observe that the all-time average is far above the two other periods whether we talk about champions or non-champions (Fig. S5, Fig. S6). This indicates what was already discussed: the best performances in Hungary are mostly dated before 1955 and before

the World War II. Overall, as we can see on the Fig. S1 and Fig. S2, there's not much variation between the three periods for each league which isn't the case for the two other metrics.

This pattern is very different for the metrics related to goals scored and goals conceded. We can observe much greater variations, which is what we could expect since points per match is a metric bounded by a maximum of three points. First, the values do not change substantially between the Since 3 Points and 1955+ periods for most of the Big Five leagues. In the Bundesliga, however, we observe a larger gap between the average and the record, represented by Bayern Munich in 2025–2026 and their astonishing season, during which they scored 122 goals in 34 matches. More generally, there is no particularly large gap between the records and the Top 10 averages, suggesting that multiple outstanding attacking performances were achieved throughout the history of these championships.

However, two leagues stand out because of the magnitude of the changes observed: Primeira Liga and the Hungarian league. Although the values generally increase when moving to the All-Time period, the Hungarian league shows the most striking evolution. It goes from having the lowest average for goals scored per match in the Since 3 Points period to by far the highest average in the All-Time period. In addition to the extraordinary record achieved by MTK Budapest in 1917–1918, already discussed in the previous section, the average itself is remarkably high. This indicates that performances above 4.5 goals per match were achieved repeatedly in Hungarian football, rather than being the result of a single exceptional season. This pattern is also visible in the boxplots (Fig. S3). However, this exceptional attacking environment did not remain constant throughout Hungarian football history. The 1956 Hungarian Revolution represented an important turning point. Following the uprising and its suppression by Soviet forces, many Hungarian footballers emigrated including several members of the generation that had dominated European football in the previous decade (Ferenc Puskas, Sandor Kocsis, László Kubala, Zoltán Czibor), while political support for football also declined. The decline was not immediate, however, as Hungary still produced outstanding players during the 1960s, including Flórián Albert, Ferenc Bene and Lajos Tichy but it came later in the 1970s whether we talk about clubs or national team.

Primeira Liga also shows a major increase, with its Top 10 average almost doubling between the Since 3 Points and All Time periods. This change may partly be linked to the exceptional Sporting teams of the 1930s and the extraordinary scoring performances of Fernando Peyroteo. In Hungary, the pre-World War II period is particularly important, but another notable period can be identified between the late 1940s and the mid-1950s. This was the era in which Gusztáv Sebes led the Hungarian national team, which became one of the strongest sides in international football, while several Hungarian clubs, including Honvéd and MTK Budapest, also featured many of the country's leading players. These historical contexts provide possible explanations for the concentration of exceptional attacking performances observed in the data, although the statistics alone cannot establish a direct causal relationship. The pattern for goals conceded is different again. Most of the strongest defensive performancess are relatively recent, with Serie A standing out as an exception, as many of its best records date from the second half of the twentieth century while some remain relatively recent (Fig. S4).

These results provide an initial overview of elite performance across different leagues and periods. The next section therefore focuses on broader statistic differences between these domestic leagues.

5. League Comparison

1. Big Five vs Other European Leagues

The comparison between the Big Five and the other European leagues provides further evidence of the patterns observed in the previous sections. The Big Five generally display fewer extreme records, but their Top 10 performances tend to be more closely clustered. One possible explanation is the overall competitiveness of these championships, which have remained among the strongest domestic leagues across most periods, with Ligue 1 being a notable exception. This may have limited the extent to which individual teams could dominate their domestic league over long periods, while also allowing several different clubs to produce outstanding performances relative to the competitive level of their respective championships. However, this interpretation should be treated with caution, as some Big Five leagues have also experienced periods of strong domestic dominance, such as Bayern Munich's long-term dominance in the Bundesliga.

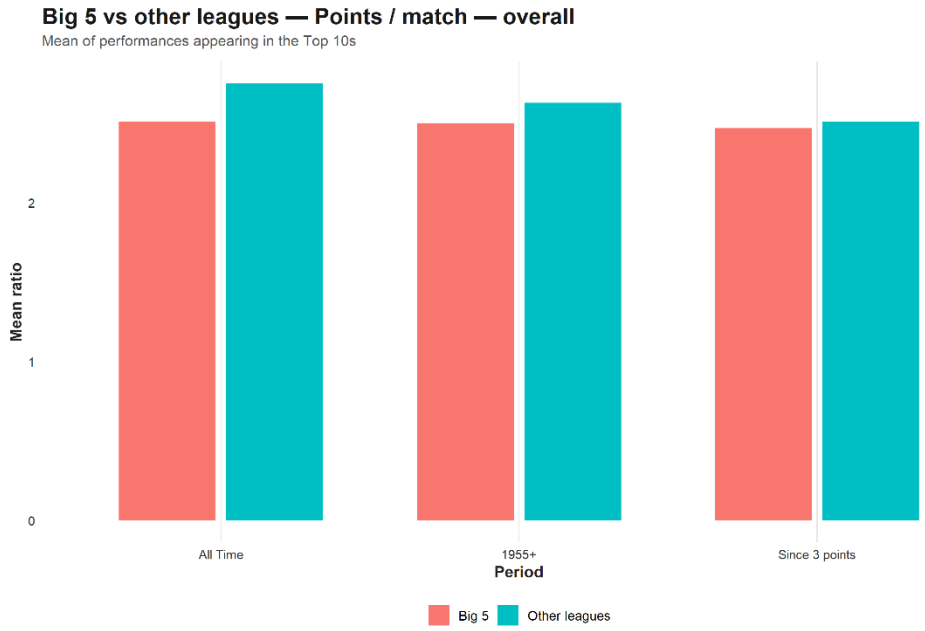


Fig. 9: Comparison between the Big 5 leagues and other leagues for points per match

Across the four performance dimensions, the Big Five do not systematically outperform the other leagues when considering the average of their ten best performances like we could expect since extreme performances are less likely to happen in more competitive league.

For points per match, the other leagues group reaches slightly higher values than the Big Five across all three periods, although the difference becomes smaller in the Since 3 Points period.

During the All-Time periods, we observe the patterns of the outstanding performances before World War II in Portugal and especially Hungary (Fig. 9). The same pattern is even more visible when looking at points per match among non-champions, where the other leagues consistently record higher averages especially in All-Time (2.29 vs 2.50) (Fig. 10).

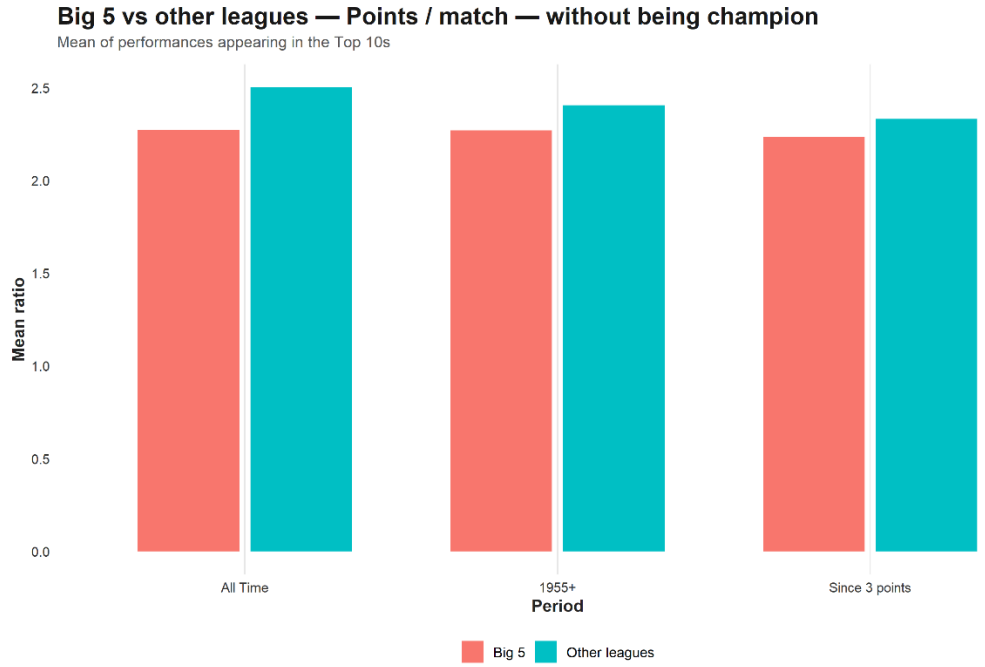


Fig. 10: Comparison between the Big 5 leagues and other leagues for points per match among non-champions

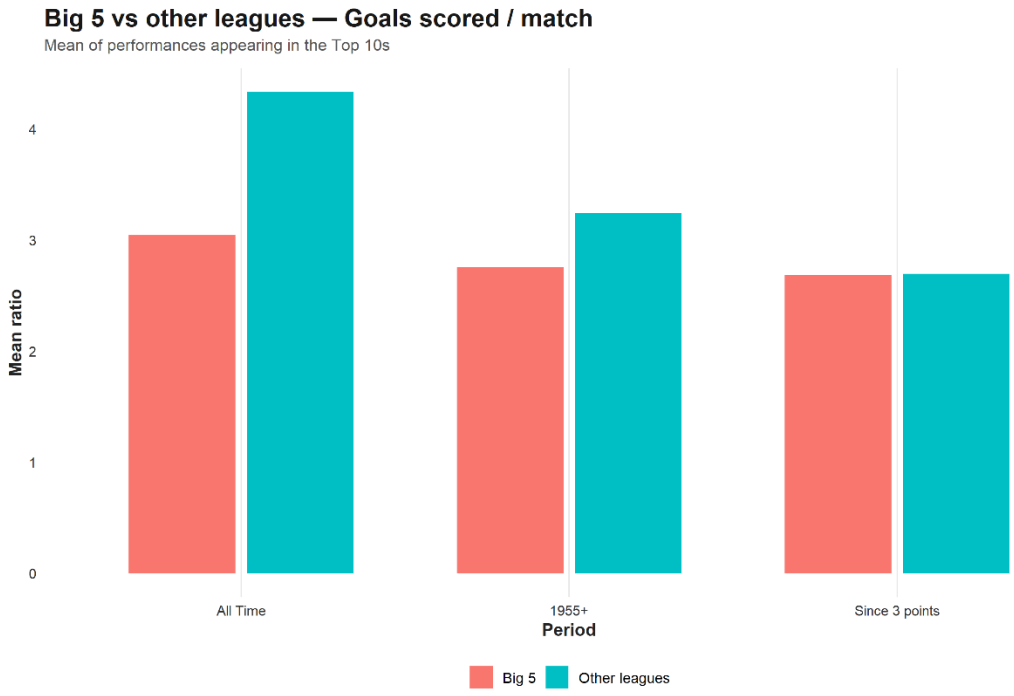


Fig. 11: Comparison between the Big 5 leagues and other leagues for goals scored per match

The largest difference appears for goals scored per match (Fig. 11). While the two groups are relatively close in the Since 3 Points period, the gap increases substantially when earlier seasons are included. In the All-Time period, the other leagues reach a much higher average than the Big Five (3 vs 4.63). This result is consistent with the exceptional attacking performances identified earlier, particularly in Hungary and Primeira Liga, and illustrates how strongly historical context can influence comparisons across very long periods. This is also consistent with what happens in Netherlands and especially the domination of few clubs on these domestic leagues with a certain gap at times with the other teams of the championships (Ajax Amsterdam, PSV Eindhoven and Feyenoord in Netherlands, SL Benfica and FC Porto in Portugal or even Sporting CP, MTK Budapest, Újpest and Honved Budapest until 1956 in Hungary).

Goals conceded show a similar but opposite pattern (Fig. 12). Since lower values represent better defensive performance, the other leagues again obtain the stronger average across all three periods. However, the difference becomes relatively small in the Since 3 Points period, suggesting that the gap between the two groups is much less pronounced in the modern era.

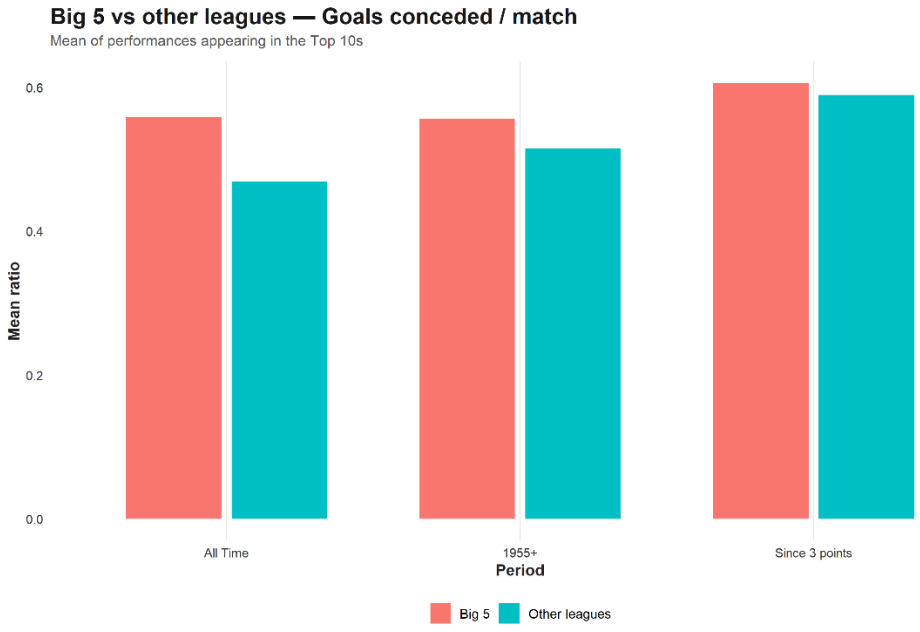


Fig. 12: Comparison between the Big 5 leagues and other leagues for goals conceded per match

Beyond the comparison between the Big Five and the other leagues, it is also interesting to examine how diverse elite performance has been within each championship.

2. Diversity of Elite Performance

The number of distinct clubs represented in a Top 10 provides a measure of how widely elite performances are distributed within a league. It complements the later recurrence analysis: diversity asks how many clubs share the selected places, while recurrence identifies the clubs that appear most often. Neither measure alone establishes league-wide competitive balance, but together they distinguish a concentrated elite from one drawn from a wider range of clubs.

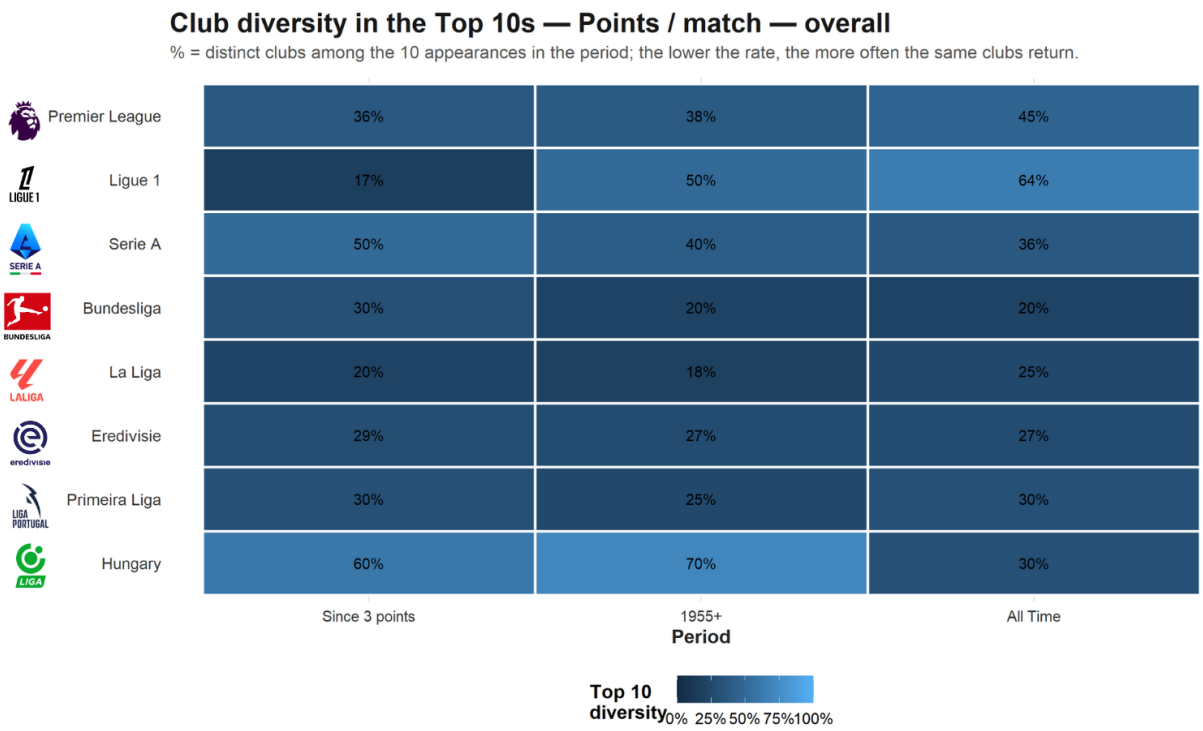


Fig. 13: Comparison of the diversity of clubs in the Top 10 most points per match for each league and period

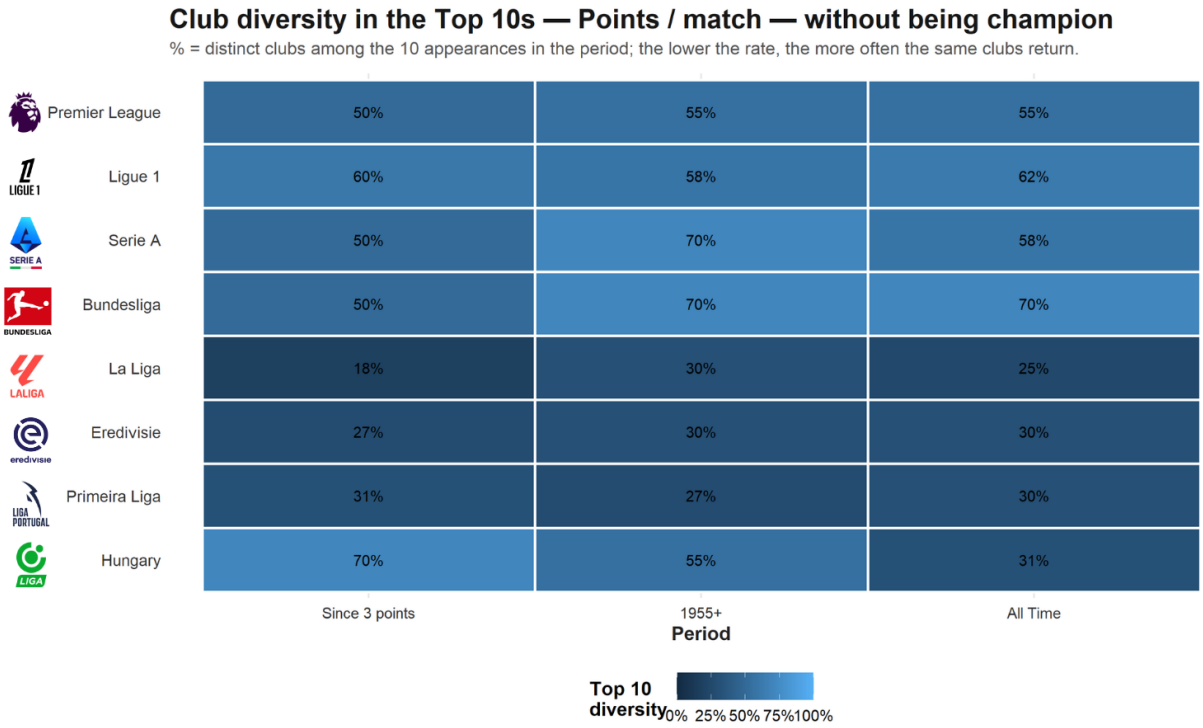


Fig. 14: Comparison of the diversity of clubs in the Top 10 most points per match among non-champions for each league and period

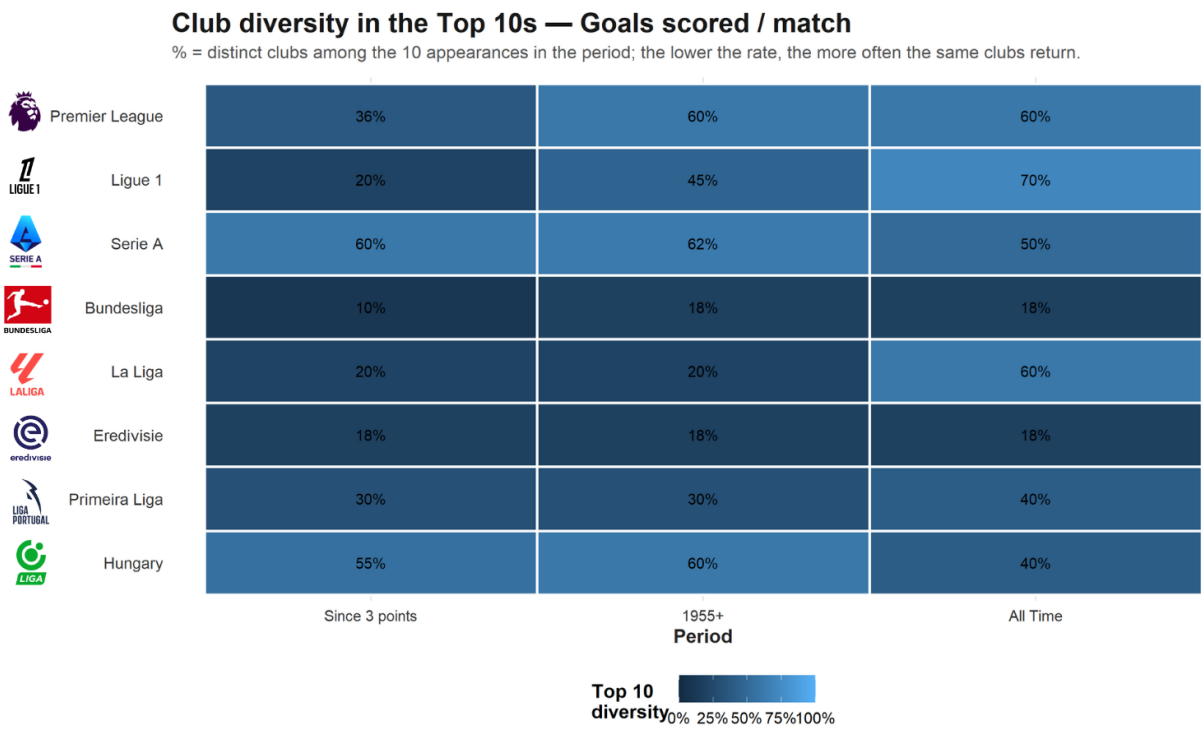


Fig. 15: Comparison of the diversity of clubs in the Top 10 most goals scored per match for each league and period

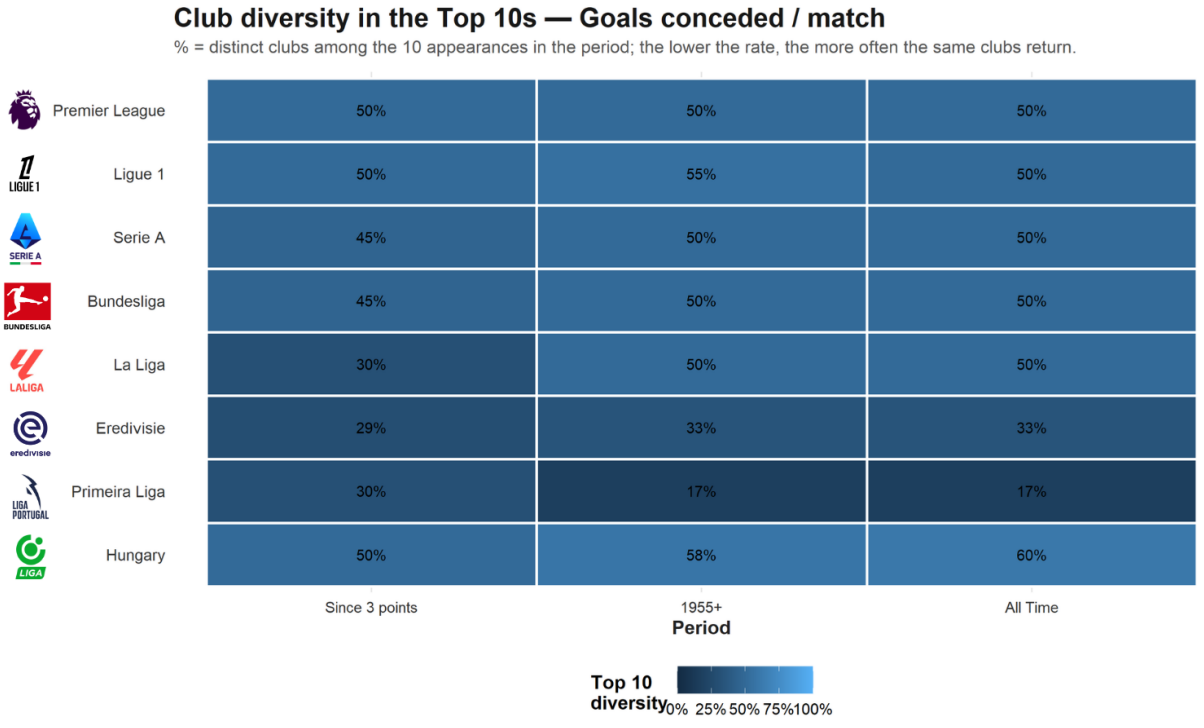


Fig. 16: Comparison of the diversity of clubs in the Top 10 least goals conceded per match for each league and period

First of all, we can observe that, as expected, the Since 3 Points period generally shows less diversity across most leagues and performance metrics than the 1955+ and All-Time periods.

For goals scored, points without being champion and overall points, the lower diversity can partly be explained by the dominance of specific clubs. Bayern Munich and Paris Saint-Germain are particularly prominent in Germany and France respectively, while FC Barcelona and Real Madrid account for a large share of the strongest performances in Spain (Fig. 13, Fig. 14, Fig. 15). This is somewhat different for points per match among non-champions (Fig. 14), particularly in the Bundesliga and Ligue 1. Since the champion is excluded from this metric, the dominance of a single club has less influence on the results. In Ligue 1, for example, the teams finishing behind the champion have changed considerably over time, with six different clubs finishing second over the last ten seasons alone. This rotation becomes even more pronounced when considering the entire history of the championship, reflecting the significant changes in the competitive landscape of French football over time. As a result, elite performances among non-champions are more widely distributed across different clubs, even during periods of strong dominance by a particular champion.

In the Eredivisie and Primeira Liga, diversity remains relatively stable and generally below 30% in several comparisons, indicating that the selected elite performances are concentrated among a recurring group of clubs (Fig. 13–16). This is evidence of concentration within the rankings, not by itself proof that the leagues as a whole were less competitive.

The Premier League/English top-flight series and Serie A show comparatively high diversity in several periods and metrics. This suggests that the clubs represented among the strongest performances changed more across eras, although the exact pattern varies by indicator. Diversity should therefore be read alongside the recurrence charts in Section 8 rather than as a standalone measure of competitive balance.

For English football, the long historical window captures successive cycles of dominance that would be obscured by looking only at the modern Premier League era. More generally, differences in historical coverage and the thresholds used for each ranking affect the diversity percentages, so comparisons are most informative when made within the same metric and period framework.

6. European Elite

1. European Elite - Big Five

For overall points per match, Preston North End's 1888–1889 season stands at the top of the ranking, with 2.73 points per match (Fig. 17). This illustrates the lasting importance of early English football in the historical record, despite the substantial changes the game has undergone since then. The rest of the ranking features a mixture of modern superteams like most of them are from the 2010s and historically dominant sides, including Juventus, Bayern Munich, invincible Bayer Leverkusen 2023-2024, Real Madrid, Manchester City and Barcelona. Several clubs appear multiple times, particularly Bayern Munich, Barcelona and Real Madrid, demonstrating their ability to produce exceptional league campaigns across different periods.

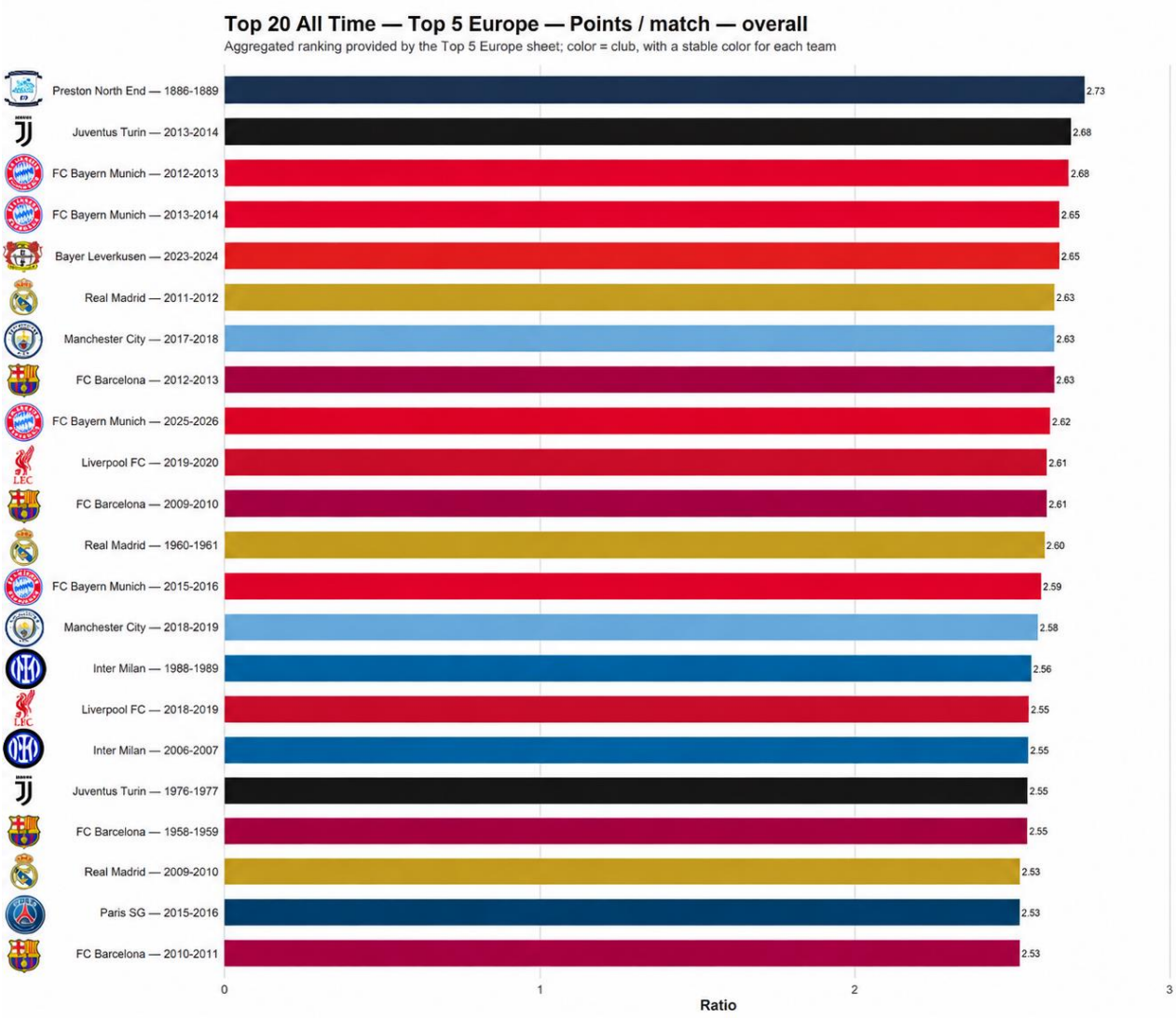


Fig. 17: Top 20 all-time performances in the Big Five leagues by points per match.

The ranking for points per match among non-champions offers a different perspective. Liverpool's 2018–2019 season leads the list, followed by Real Madrid in 2009–2010 and Torino in 1976–1977 (Fig. 18). These examples illustrate that exceptional points-per-match ratios do not necessarily result in a league title. The presence of several Real Madrid and Barcelona seasons also highlights how outstanding performances can occur even when another club finishes above them in the standings. This shows how their rivalry during the 2010s enhances the competition and it shows how extraordinary season could be done by one of the two clubs even though the other club is champion. For example, Real Madrid only won 2 La Liga during the 2009-2018 period but still had an average point of 90 which is enormous. On the two graphs of points per match, we can observe that the least represented league is Ligue 1 with only Paris Saint Germain and Strasbourg (Fig. 17 and Fig. 18).

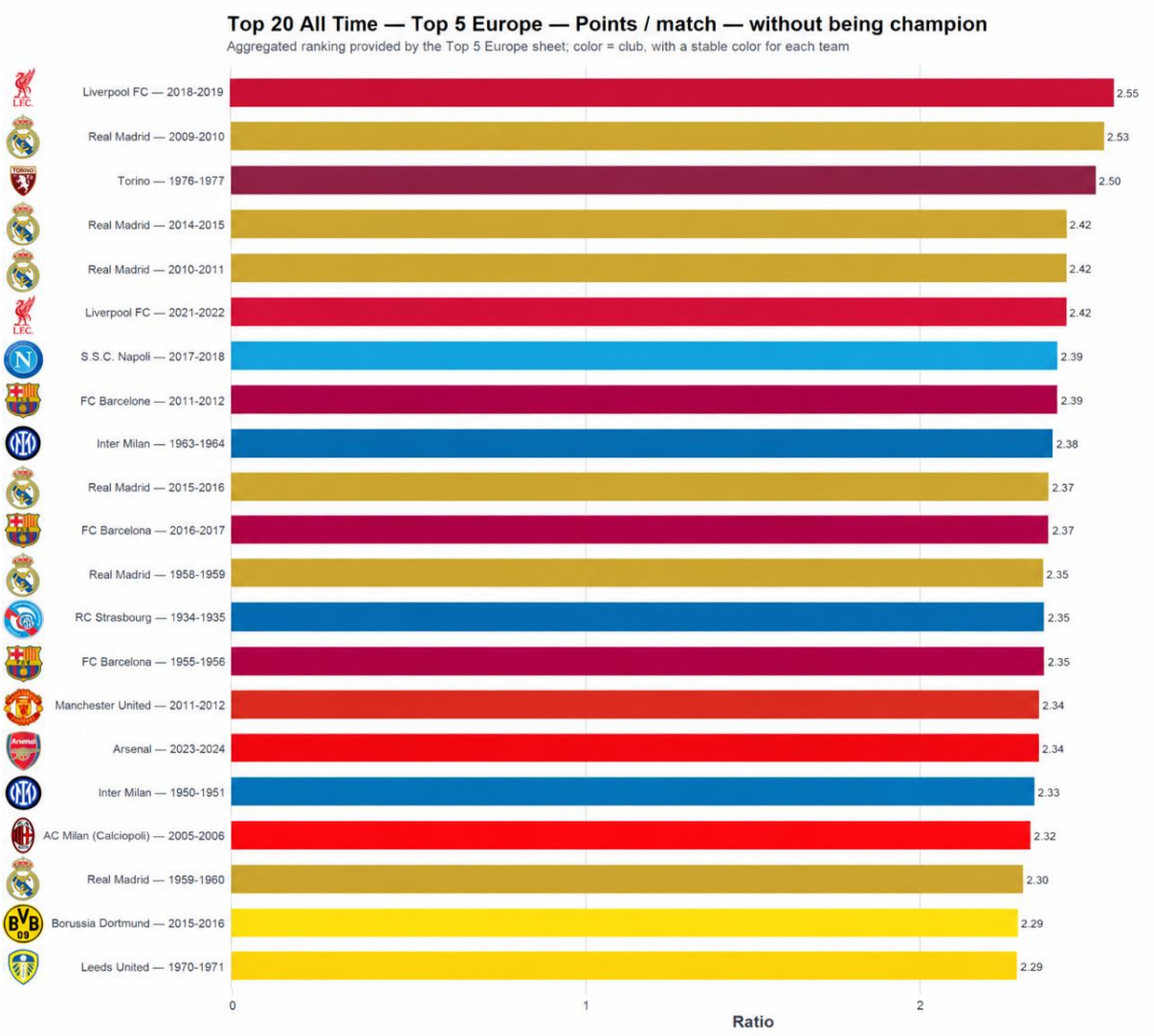


Fig. 18: Top 20 all-time performances in the Big Five leagues by points per match among non-champions

The goals-scored ranking reveals a stronger historical dimension. Athletic Bilbao's 1930–1931 season leads the list with 4.06 goals per match, followed by Bayern Munich in 2025–2026 and several English and Spanish sides from the late nineteenth and early twentieth centuries (Fig. 19). The repeated appearance of Athletic Bilbao in this ranking underlines the club's remarkable attacking output during the early 1930s, especially since they have 4 appearances in the Top 8. The club is an amazing model of local club which sticks to the values of its region and still find a way to dominate even though they never came back at this level since then.

At the same time, the presence of modern teams such as Bayern Munich shows that exceptional attacking performance is not exclusive to to old eras even though we can observe patterns. Indeed, the best performances in England are during the 19th century, in Spain it's before World War II. Finally in Serie A there are the Milan of Gren, Liedholm, Nordahl which has been dominating the early 1950's in Italy just after the Superga's tragedy which took the life of Valentino Mazzola and most of Torino's team and staff. Torino is also present here with their outstanding season 1947-1948 where they scored 125 goals in 40 matches which is still the record of goal scored in one season of the big 5 leagues.

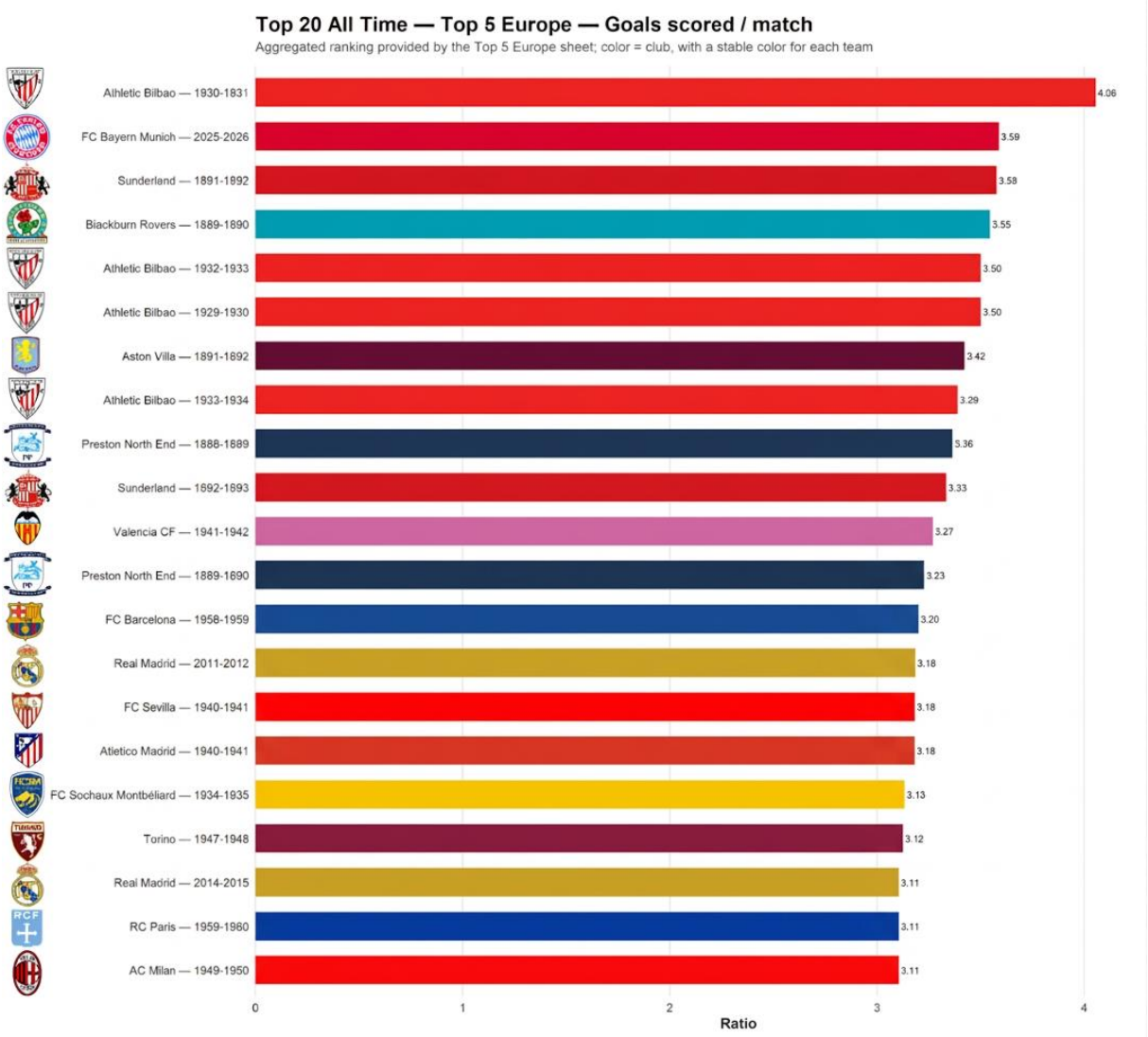


Fig. 19: Top 20 all-time performances in the Big Five leagues by goals scored per match

Finally, the goals-conceded ranking highlights a different group of teams (Fig. 20). Cagliari's 1969–1970 season leads with just 0.37 goals conceded per match, followed closely by Liverpool in 1978–1979 and Chelsea in 2004–2005. Several other defensive records come from Italian clubs, particularly AC Milan and Juventus, reflecting the presence of historically outstanding defensive seasons in Serie A. Unlike the attacking ranking, which features many very old records, the strongest defensive performancess in this list are concentrated more heavily in the second half of the twentieth century and the modern era. It also comes from less dominant teams like Deportivo 1993-1994 although the team led by his Brazilian star Bebeto was close to be champion ahead of Cruyff's Barcelona.

Overall, the Big Five rankings show that the historical elite is made up of both clubs that repeatedly produced outstanding seasons and teams whose place in the rankings comes from a particularly exceptional campaign. The differences between the four metrics also demonstrate that historical greatness can take several forms: consistently accumulating points, achieving extraordinary attacking output or maintaining exceptional defensive solidity.

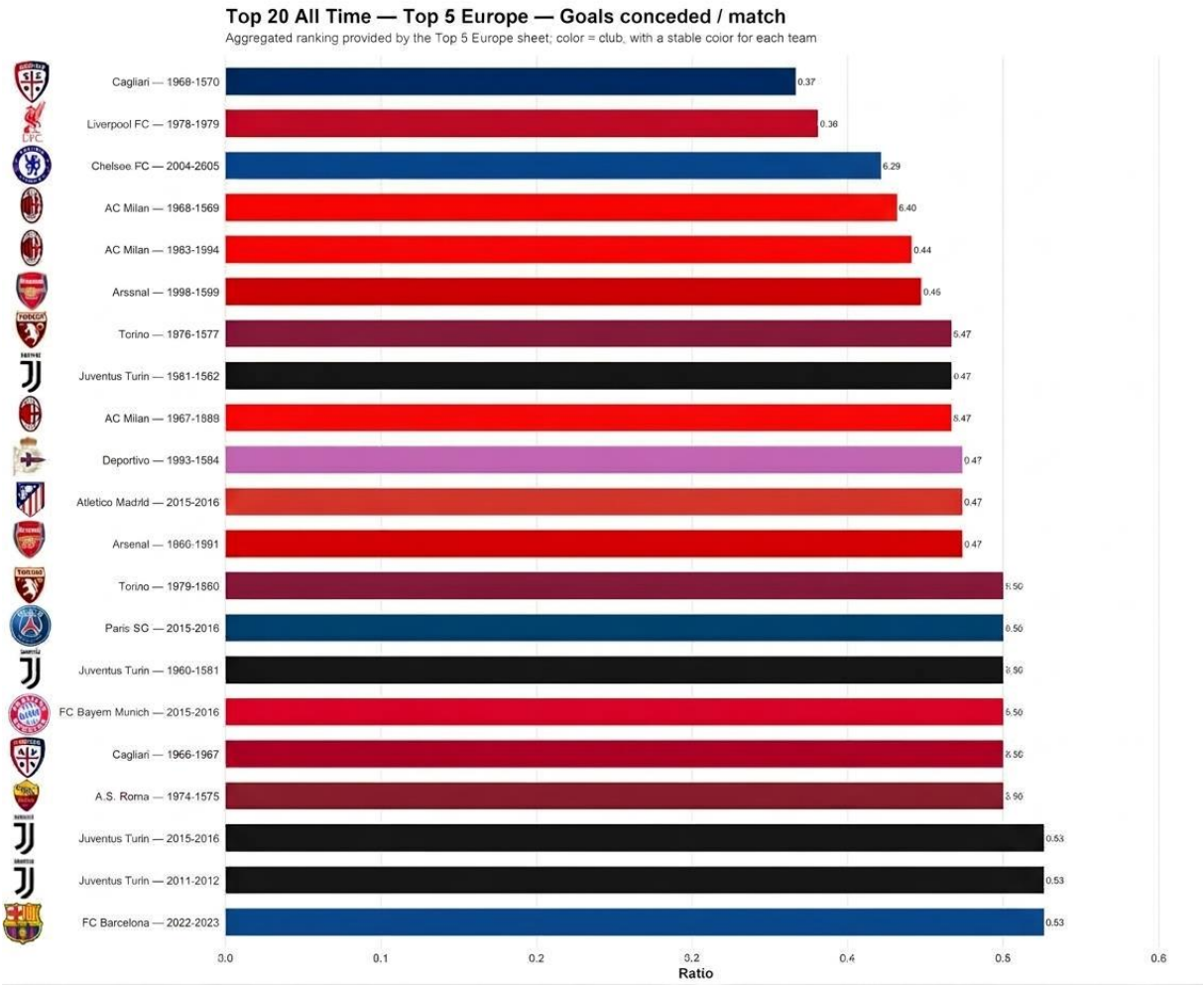


Fig. 20: Top 20 all-time performances in the Big Five leagues by goals conceded per match

2. European Elite – 8 domestic leagues

Expanding the analysis from the Big Five to all eight domestic leagues significantly changes the picture of historical elite performance. While the previous section highlighted exceptional seasons from clubs in Europe's five major championships, the broader rankings reveal the importance of Portuguese, Dutch and particularly Hungarian football in the historical record. This comparison is especially relevant because the current status of a league does not necessarily reflect the level of performance achieved by its clubs throughout history and that as we talked about, we expect more exceptional performances in these less competitive leagues but it could also be the opposite because of the superteams present in the Big 5 leagues.

For overall points per match, the difference between the top 20 big 5 and this one including the 8 leagues is striking (Fig. 21). Ferencváros in 1931–1932 and Budapesti TC in 1901 both reach the maximum ratio of 3.00 points per match, while MTK Budapest appears repeatedly among the highest-ranked performances and is the most present with 7 appearances. Benfica, Porto and other Portuguese clubs also feature prominently. By comparison, the Big Five ranking was led by Preston North End's 1888–1889 season at 2.73 and here it ends up at the 19th place. The

expanded ranking therefore reveals that the very best historical performances exceed the strongest records in the Big Five, particularly from Hungary and Portugal.

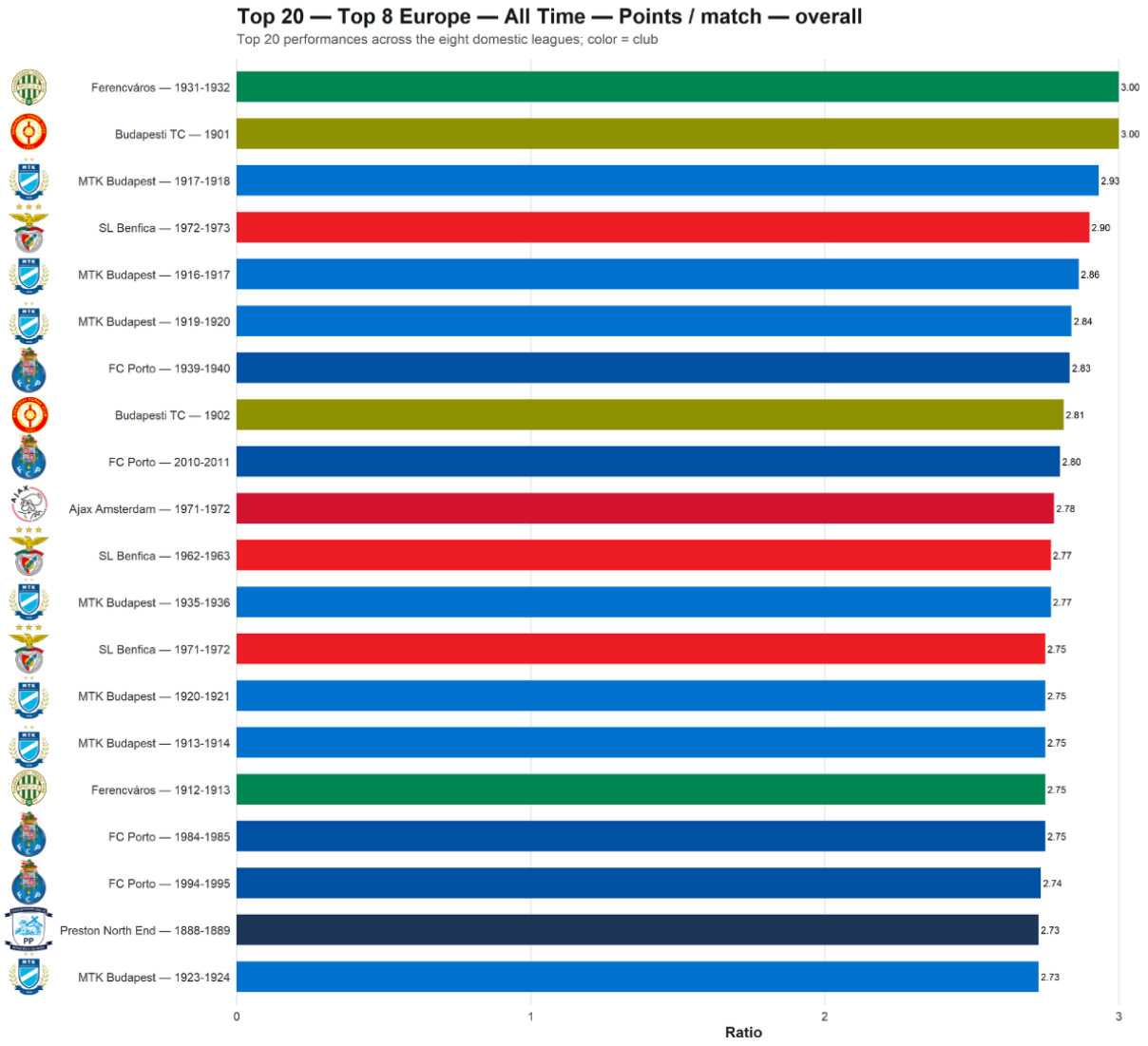


Fig. 21: Top 20 all-time performances in the 8 domestic leagues by points per match.

A similar pattern appears for points per match among non-champions (Fig. 22). Sporting CP's 1939–1940 season leads the ranking with 2.67, followed by Porto in 1990–1991. Hungarian clubs such as Ferencváros, MTK Budapest and Újpest also feature prominently, alongside Benfica and several Big Five teams. This demonstrates that exceptional points-per-match performances outside the Big Five were not limited to league champions and could also be achieved by clubs that finished behind another team in the domestic standings. We can finally observe that apart from Liverpool 2018-2019 and Real Madrid 2009-2010, there are not much big 5 team seasons left.

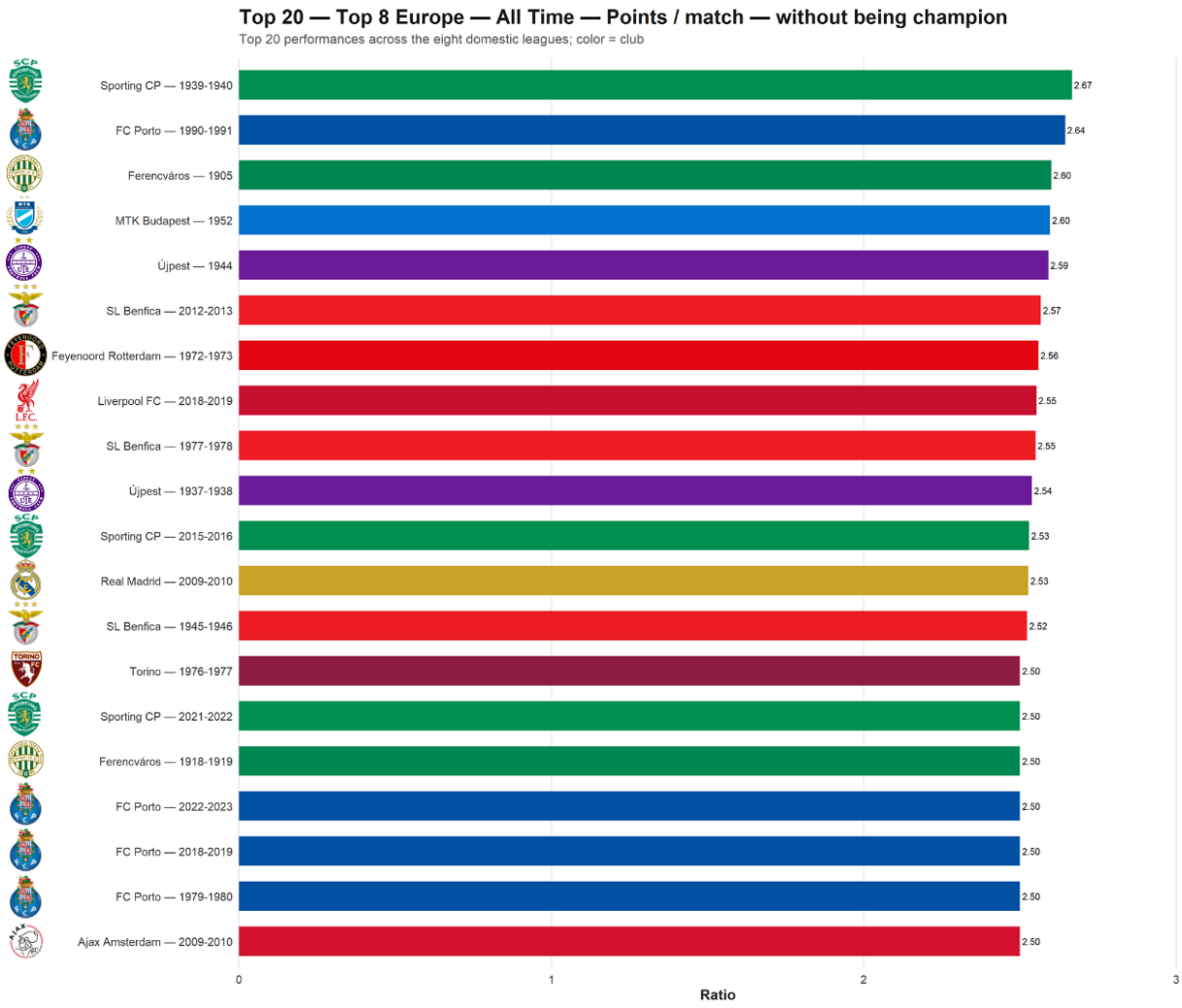


Fig. 22: Top 20 all-time performances in the 8 domestic leagues by points per match among non-champions

The difference becomes even more pronounced for goals scored per match (Fig. 23). MTK Budapest's 1917–1918 season leads with an extraordinary 6.68 goals per match, followed by Újpest in 1945 and 1945–1946. Hungarian clubs dominate much of the ranking, while Sporting CP and Ferencváros also appear repeatedly. Compared with the Big Five ranking, where Athletic Bilbao's 1930–1931 season leads at 4.06 goals per match, the expanded ranking highlights just how exceptional the attacking records in Hungarian football were during the first half of the twentieth century. It’s impressive how there are 15 teams from Hungary in the top 20 rankings with all performances coming from 1950 and before. Sporting's repeated appearances also reinforce the importance of Portugal's historical attacking performances and goes with the Peyroteo presence.

Finally, the goals-conceded ranking reveals another important difference. Porto occupies the first three positions, with records of 0.30, 0.30 and 0.32 goals conceded per match (Fig. 24). Benfica also appears repeatedly, alongside MTK Budapest, Ferencváros and several Big Five clubs. The strongest defensive performances in the expanded ranking therefore come from a broader range of leagues, showing that outstanding defensive records were not exclusive to Europe's five major championships.

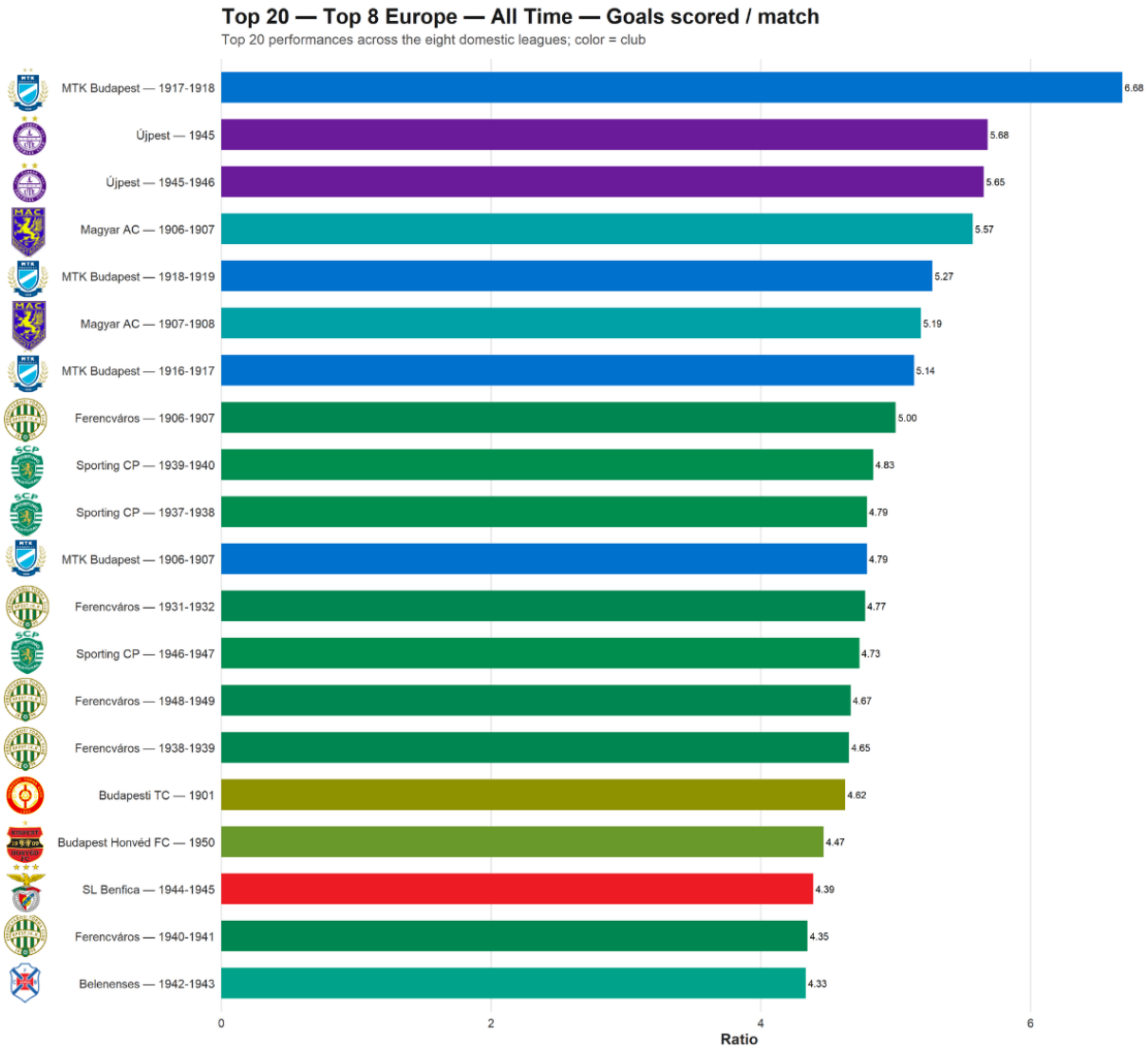


Fig. 23: Top 20 all-time performances in the 8 domestic leagues by goals scored per match

Overall, expanding the scope to eight leagues substantially changes the historical picture of European elite performance. Hungary stands out particularly for points per match and goals scored, while Portuguese clubs feature strongly across several metrics. These results highlight how the geography of elite performance has changed over time, with certain leagues becoming major centres of European football during specific historical periods. Hungarian football, in particular, occupied a prominent position in the European game during the first half of the twentieth century, producing exceptionally dominant teams and remarkable attacking performances. However, these rankings must be interpreted with an important limitation in mind: exceptional individual performances do not necessarily reflect the overall strength of a league or even the level of the team in every aspect of the game. A club may achieve extraordinary statistics not only because of its own quality, but also because of the gap separating it from its domestic opponents. This is particularly relevant when comparing leagues with different competitive structures and levels of development. The rankings capture the highest levels of performance recorded in each competition, but they do not measure the strength of the entire league or the quality of its average team.

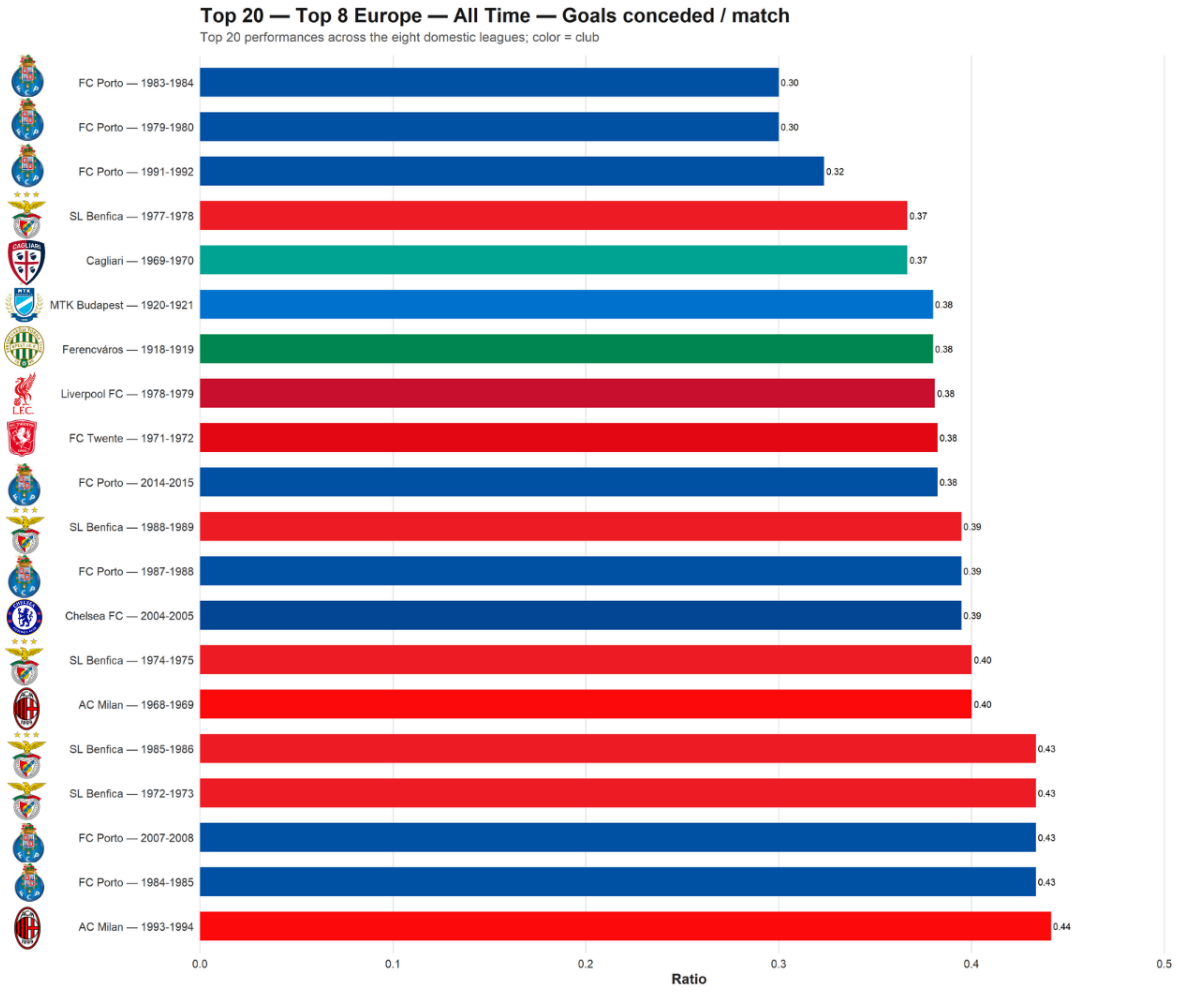


Fig. 24: Top 20 all-time performances in the 8 domestic leagues by goals conceded per match

Furthermore, the European football hierarchy has evolved considerably over time. The growing financial and sporting concentration of talent within the leading leagues, reinforced by developments such as the Bosman ruling in 1995, contributed to a changing competitive landscape. The historical prominence of Hungarian and Portuguese clubs should therefore be understood within the context of their respective eras, rather than as evidence that these leagues were consistently stronger than the Big Five.

Ultimately, the purpose of expanding the analysis beyond the Big Five is not to establish a definitive hierarchy between championships, but to identify exceptional performances that would otherwise be overlooked and to understand how these performances were distributed across different leagues and historical periods.

7. Historical Concentration Across Decades

This section examines the distribution of Top 10 performances across decades. By comparing different historical periods and examining the results for each championship, it aims to identify when elite performances were most concentrated, whether these patterns differed across leagues, and how they varied between points per match, goals scored and goals conceded.

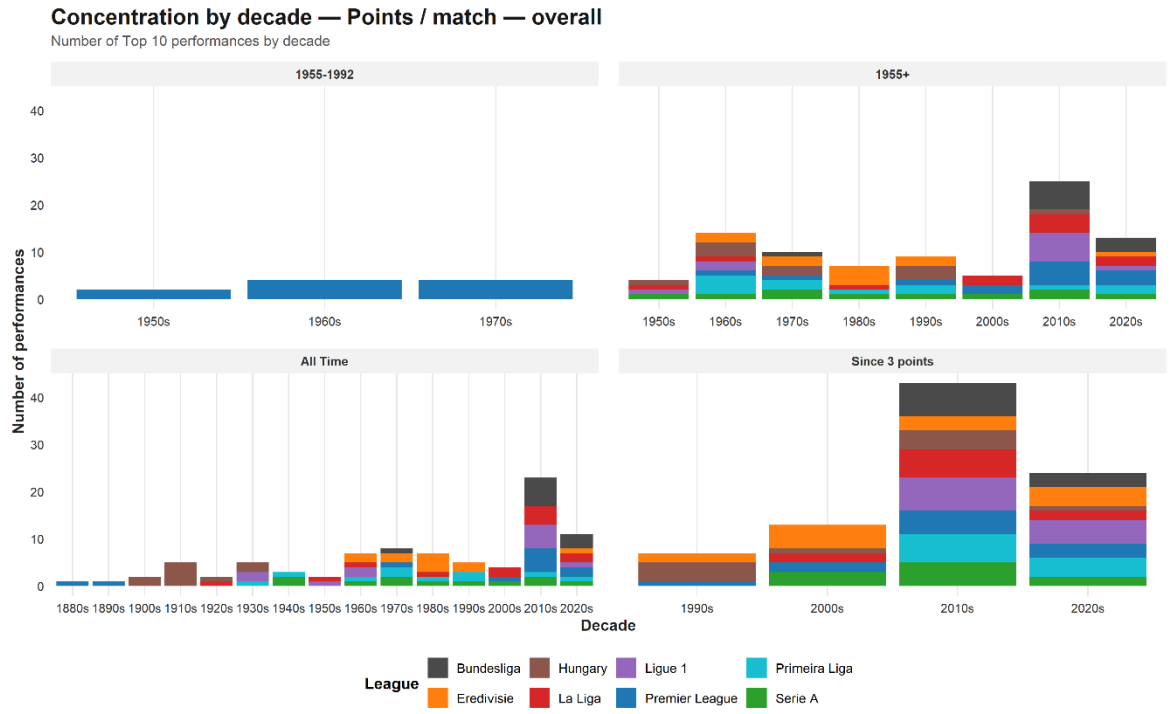


Fig. 25: Distribution of points-per-match performances across the eight domestic leagues by historical period.

Like we already observed, we can see a similar pattern for points per match and goal scored with a great dominance of the 2010s (Fig. 25, Fig. 26, Fig. 27). Indeed, we can see that for each period, this is where we can find most of the performances for points per match with a very high domination of the Big 5 leagues and especially Bundesliga, Ligue 1 and Premier League (Fig. 25). In Premier League and the other domestic leagues, we can find again the importance of the pre–World War II period almost for all the performances of their top 15 rankings. The pattern is kind of similar among non-champions even though it's not exactly the same dominance and is consistent with our past findings and the context of these eras. Finally, we observe the 1970's high period in Premier League when we look at the 1955-1992 period.

The distribution of goals scored per match follows a similar pattern (Fig. 27). A substantial share of the highest attacking performances dates from earlier decades that we already discussed, particularly in the All-Time rankings, while the more recent periods also feature prominently in several leagues and is by far the most represented decade in the big 5.

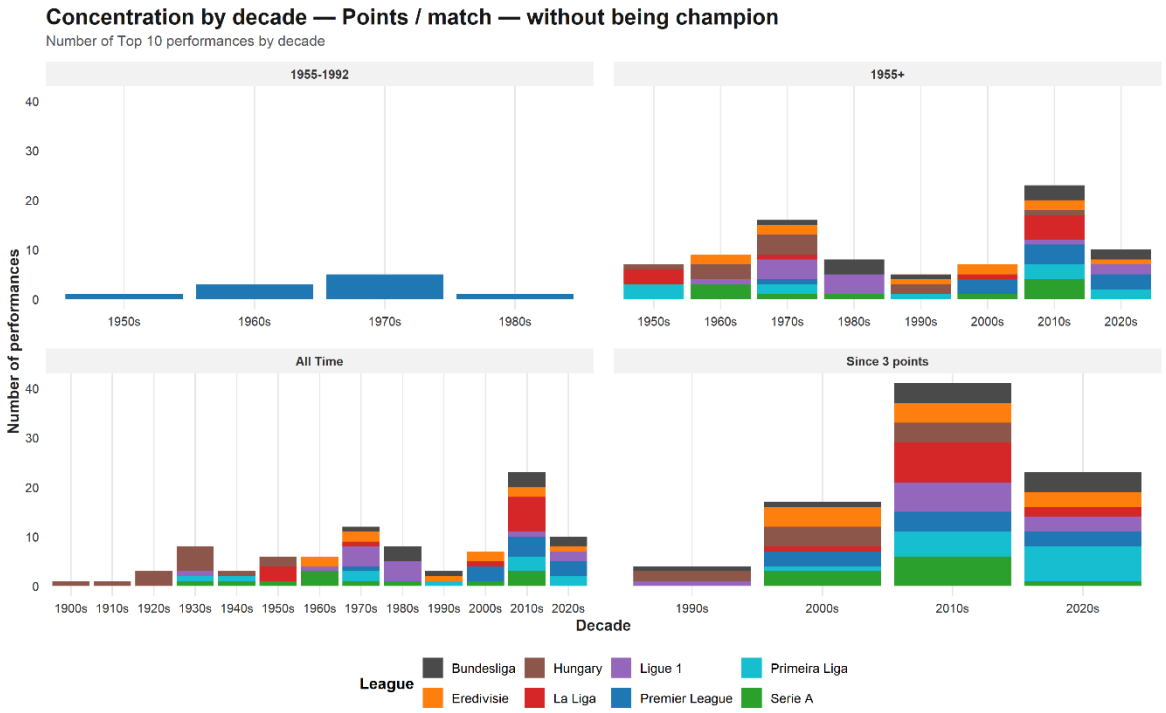


Fig. 26: Distribution of points-per-match performances among non-champions across the eight domestic leagues by historical period.

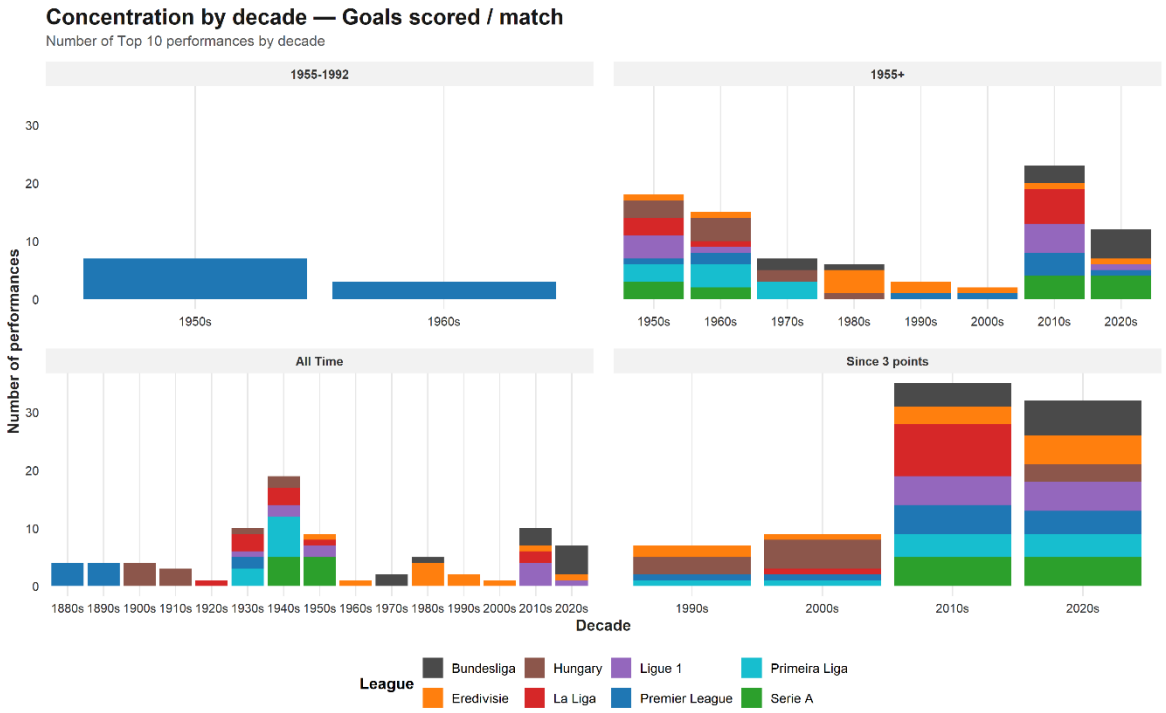


Fig. 27: Distribution of goals scored per match performances across the eight domestic leagues by historical period.

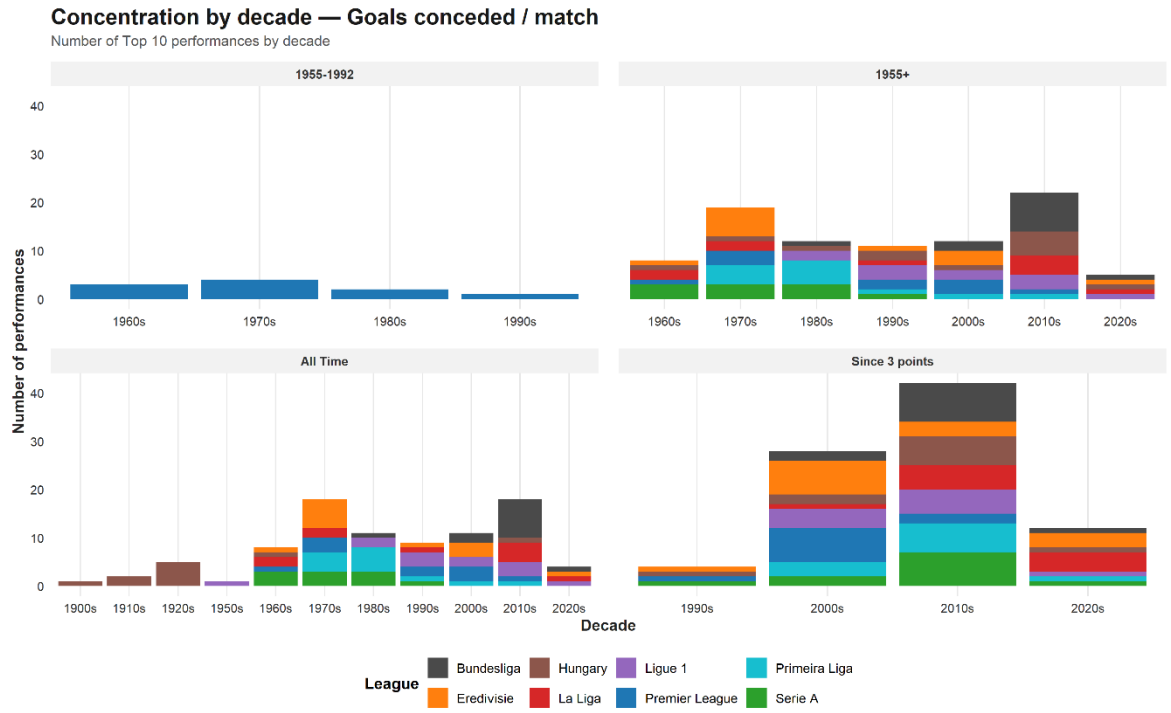


Fig. 28: Distribution of goals conceded per match performances across the eight domestic leagues by historical period.

Goals conceded show another distribution which is very different from the three others, with strong defensive performances spread across different periods (Fig. 28). Indeed, the 1940's and 1950's completely disappeared even for the 1955-1992 period in Premier League and we observe a distribution across the period we discussed especially in Serie A, Primeira Liga, Eredivisie and Premier League. Finally in Ligue 1, Bundesliga and Liga, the historical pattern is less obvious but we observe the dominance of the 2010s with the 4 superteams we already discussed (Real Madrid, FC Barcelona, Bayern Munich and Paris Saint Germain).

These differences suggest that the historical concentration of elite performance cannot be described through a single overall trend.

The championship-level distributions help explain these broader patterns, revealing that the historical concentration of elite performances differs substantially between leagues.

Within the Big Five, the distributions reveal distinct trajectories rather than one common historical peak. The strong presence of points-per-match performances in the 2010s is particularly apparent in the combined ranking (Fig. S13), but the championship-level figures help distinguish this broader pattern from the history of individual leagues.

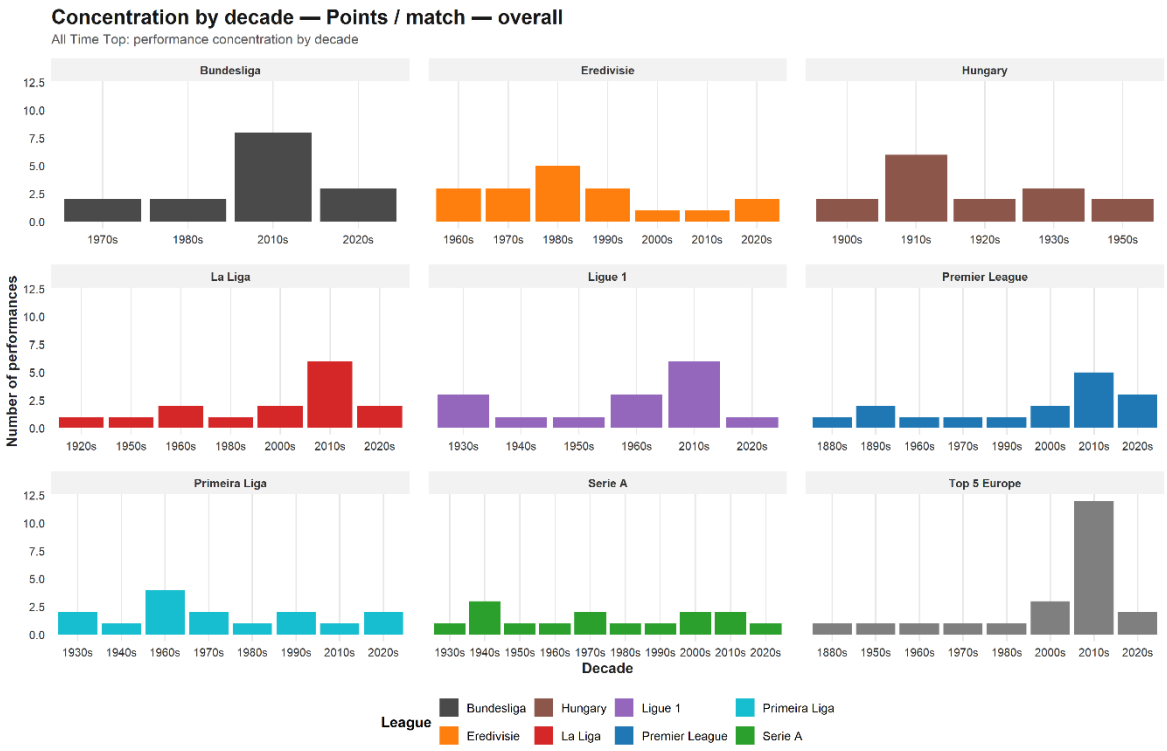


Fig. 29: Distribution of points per match performances by decade for each domestic league and the Big 5

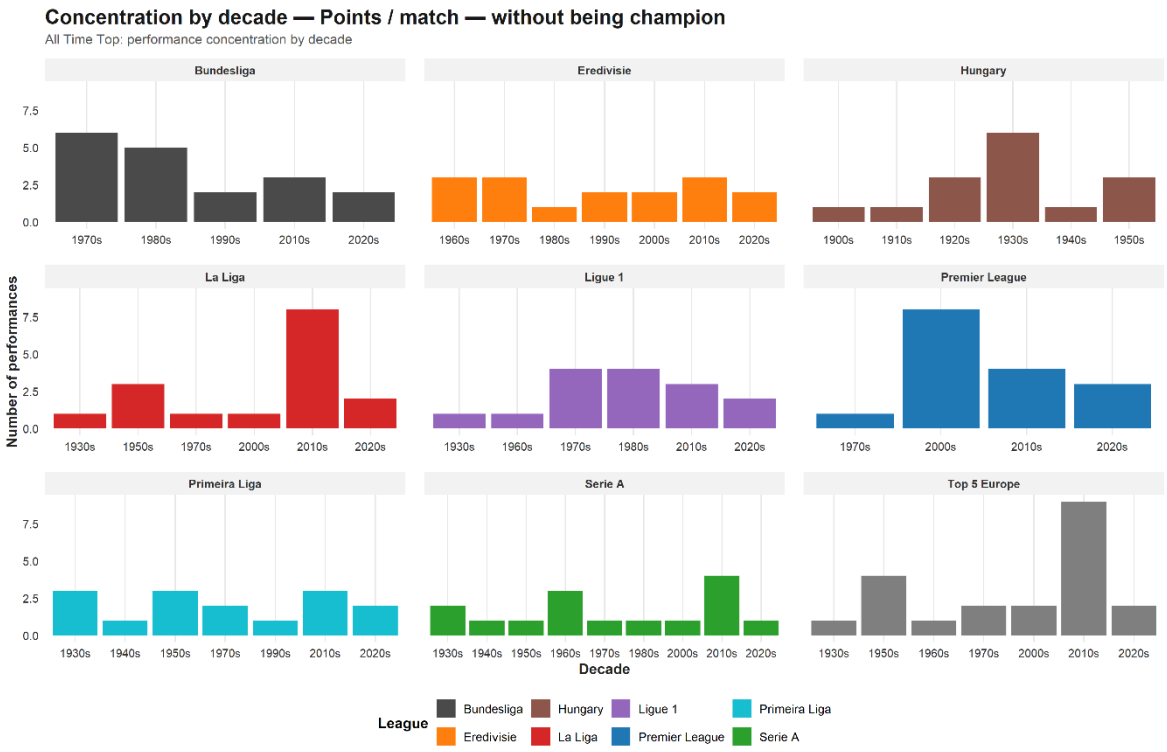


Fig. 30: Distribution of points per match performances among non-champions by decade for each domestic league and the Big 5

The Bundesliga displays a particularly concentrated distribution of elite points-per-match performances, among both champions and non-champions. Although its history begins in 1963, giving it a shorter historical window than several other championships, the strong concentration of performances in the 2010s and 2020s remains striking. The Premier League and the wider English top-flight record also show a notable presence of elite performances in the 2000s and 2010s. The 2000s were particularly significant, with the dominance of the traditional "Big Four" composed of Manchester United, Chelsea, Liverpool and Arsenal, and becoming a defining feature of the league's history (Fig. 29, Fig. 30). Among the other Big Five leagues, Serie A stands out for the prominence of the 1960s, a pivotal period both tactically, as discussed earlier, and in terms of domestic performance. Juventus enjoyed considerable success at the beginning of the decade, notably with the attacking partnership of Giampiero Boniperti and John Charles, supported by Omar Sívori in a more creative role. AC Milan subsequently rose to prominence, with the emergence of Gianni Rivera, before Helenio Herrera's Inter became one of Europe's leading teams, achieving major success in the European Cup. This period illustrates how several successive dominant teams contributed to Serie A's historical profile.

The Primeira Liga and Eredivisie also display distinct historical distributions, showing that exceptional points totals emerged across different competitive environments. Hungary presents the most clearly defined historical pattern, with its strongest performances concentrated in the 1950s and earlier. The 1956 Hungarian Revolution and the subsequent emigration of numerous players provide important historical context for this decline, although the decade distributions alone cannot establish a direct causal relationship.

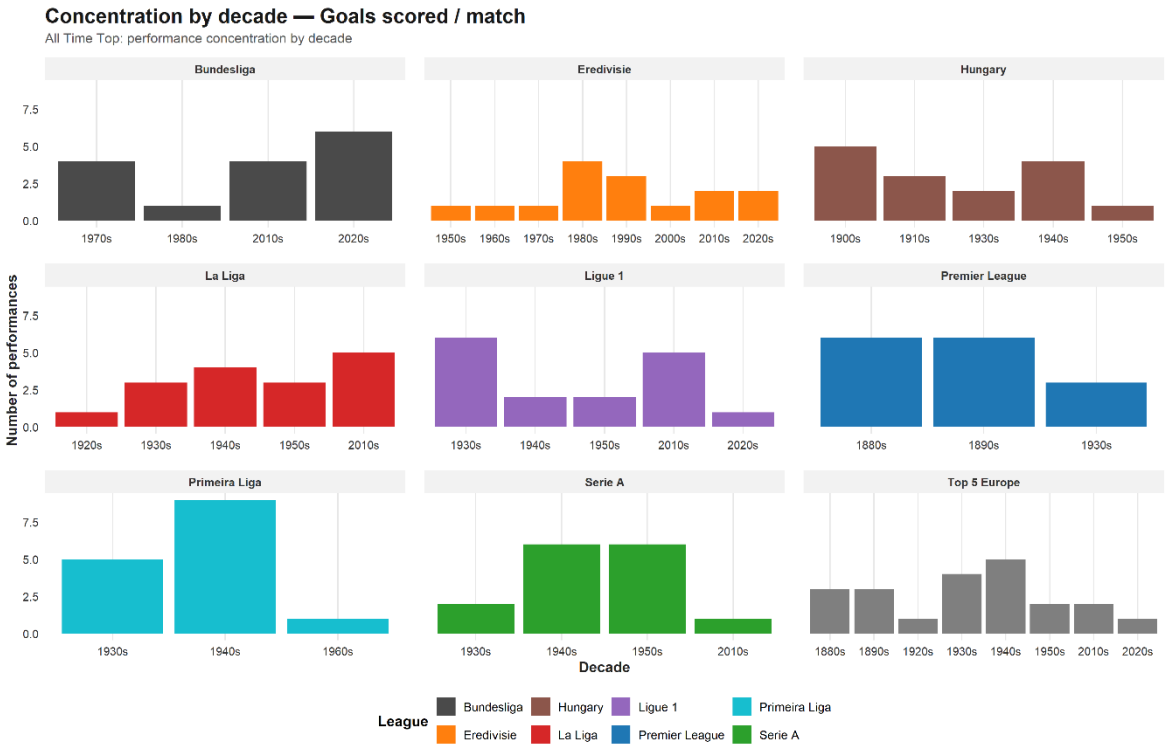


Fig. 31: Distribution of goals scored per match performances by decade for each domestic league and the Big 5

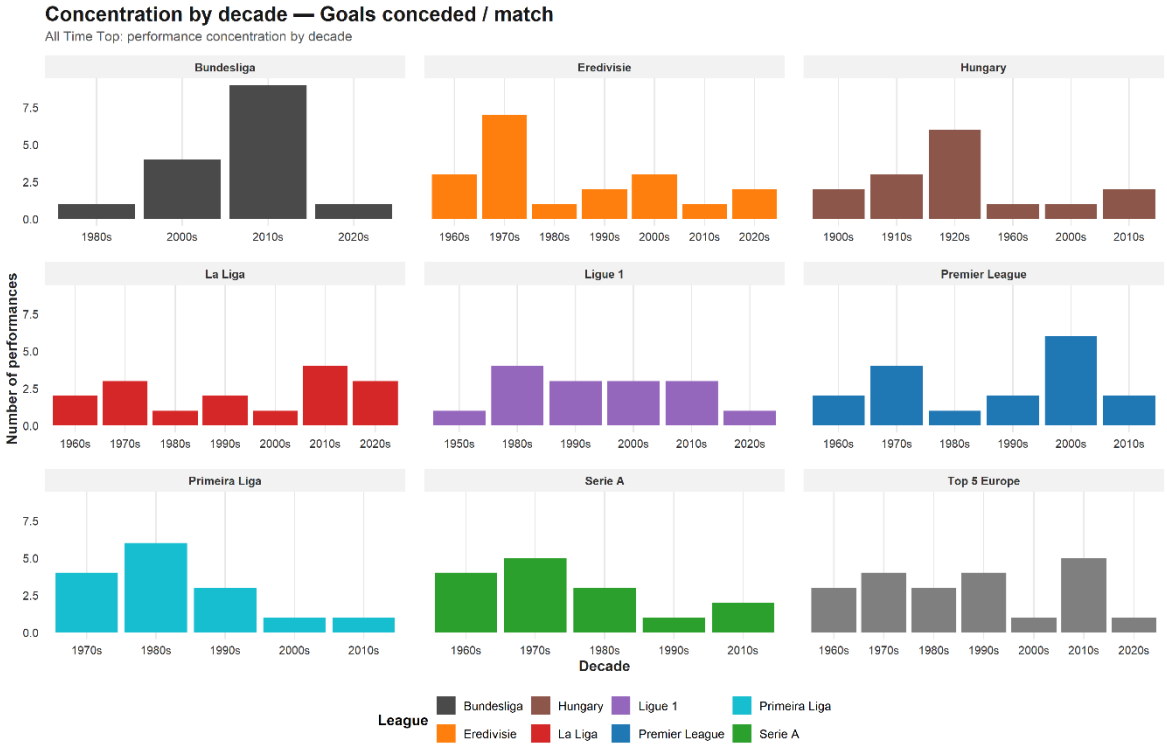


Fig. 32: Distribution of goals conceded per match performances by decade for each domestic league and the Big 5

Hungary stands out for its concentration of exceptional attacking performances in earlier decades, following a pattern similar to that observed for points per match (Fig. 31). The Primeira Liga displays a similarly distinctive concentration between the 1930s and 1950s. English football also experienced significant changes during this period, particularly following the 1925 offside law amendment, which influenced attacking play and contributed to the evolution of scoring patterns where we only got pre 1940s performances. Meanwhile, the other Big Five leagues and the Eredivisie feature outstanding attacking performances across both earlier and more recent periods, showing that exceptional scoring records are not confined to the past and that recent decades have also produced remarkable offensive output.

These patterns highlight the importance of considering each championship's historical context rather than assuming a common evolution in attacking performance. They also suggest that goals scored and goals conceded may be more sensitive to tactical approaches, coaching philosophies and changes in the game than points per match. Indeed, the strongest attacking and defensive performances tend to be concentrated in a smaller number of decades which is very expected. There are many different ways to win in football in terms of approach and this will always lead to a high ratio of points per match even though times showed that the best teams in domestic league are often the more offensive ones. But by the end, the tactical pattern, the philosophy of coaches or even of the league overall will often drive the goal scored and goal conceded statistics. You can observe that very easily when comparing for example the number of goals scored in Serie A during the 1990's and during the 2010s.

The Eredivisie is a notable exception, with elite attacking performances appearing across a broader historical range. This may reflect the repeated presence of highly dominant attacking teams, although the rankings alone cannot establish whether the league as a whole was consistently more offensive than its competitors.

Defensive records follow a different pattern (Fig. 32), with exceptional performances spread across most of the 1970-2010 period. Serie A features notable defensive achievements from earlier decades in the second half of the 20th century, while the other leagues display a very balanced across time like in Ligue 1 or even La Liga. Hungary's prominence in the attacking rankings, for example, does not automatically translate into an equivalent concentration of defensive records. Finally, we find again in Bundesliga, the superteam of Bayern Munich during the 2010s which drive nearly all the pattern. Overall, this confirms what we saw before and show a wide concentration even though we can observe few patterns. It also displays that even regarding only the very best performances of a league can be a starting point for analysis, even more with the knowledge and football context behind it because the best teams are often a driver of changes in a league or in Europe.

8. Recurrence and Persistence of Elite Performance

The previous section examined when elite performances occurred. This section focuses on the clubs behind these results, identifying which teams repeatedly appear in the Top 10 rankings. Recurrence is measured by the number of distinct seasons in which a club features in a given ranking. It therefore highlights the repeated presence of elite performances, rather than the overall success or title count of a club but the same season can't count multiple times for these figures. The appendix also includes figures showing which clubs appear in the rankings across the three historical periods, as well as the total number of appearances accumulated across all three periods. These additional visualisations provide a broader view of recurrence and persistence, highlighting clubs whose elite performances extend across multiple eras.

The recurrence of overall points-per-match performances reveals a strong concentration around a few dominant clubs in several championships (Fig. 33). Bayern Munich leads the Bundesliga ranking with nine appearances, illustrating its repeated presence among Germany's highest-performing teams. In Ligue 1, Paris Saint-Germain records eleven appearances, while PSV Eindhoven and Ajax dominate the Eredivisie ranking with eleven and nine appearances respectively. Hungary presents a different historical profile, with MTK Budapest and Ferencváros leading the ranking, reflecting the importance of earlier generations of Hungarian football.

Elsewhere, Barcelona and Real Madrid lead the La Liga ranking, while Juventus stands out in Serie A with eight appearances. In the Primeira Liga, Benfica and Porto record ten and eight appearances respectively. The Premier League displays a more balanced distribution among Manchester City, Liverpool and Chelsea, each featuring repeatedly in the ranking. Overall, these results show that recurrence is often concentrated around historically dominant clubs, although the extent of this concentration differs substantially between championships.

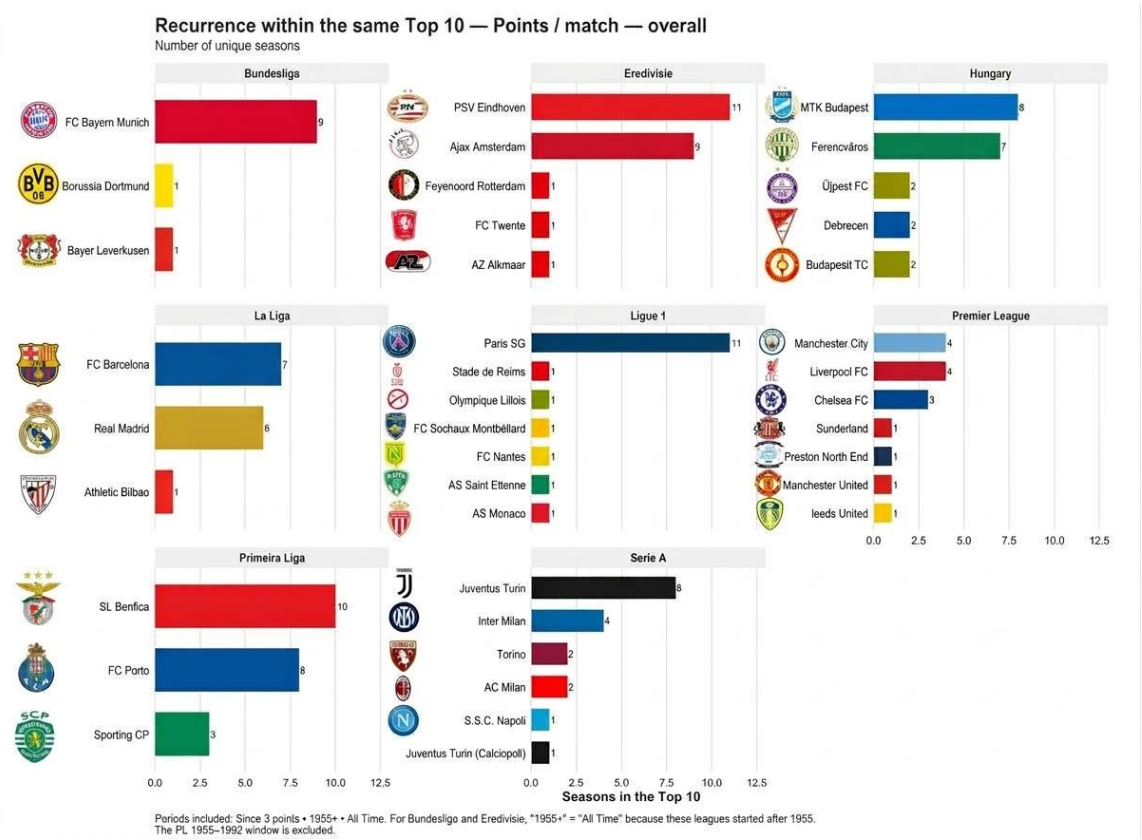


Fig. 33: Recurrence of clubs in the rankings of points per match for each period in each league

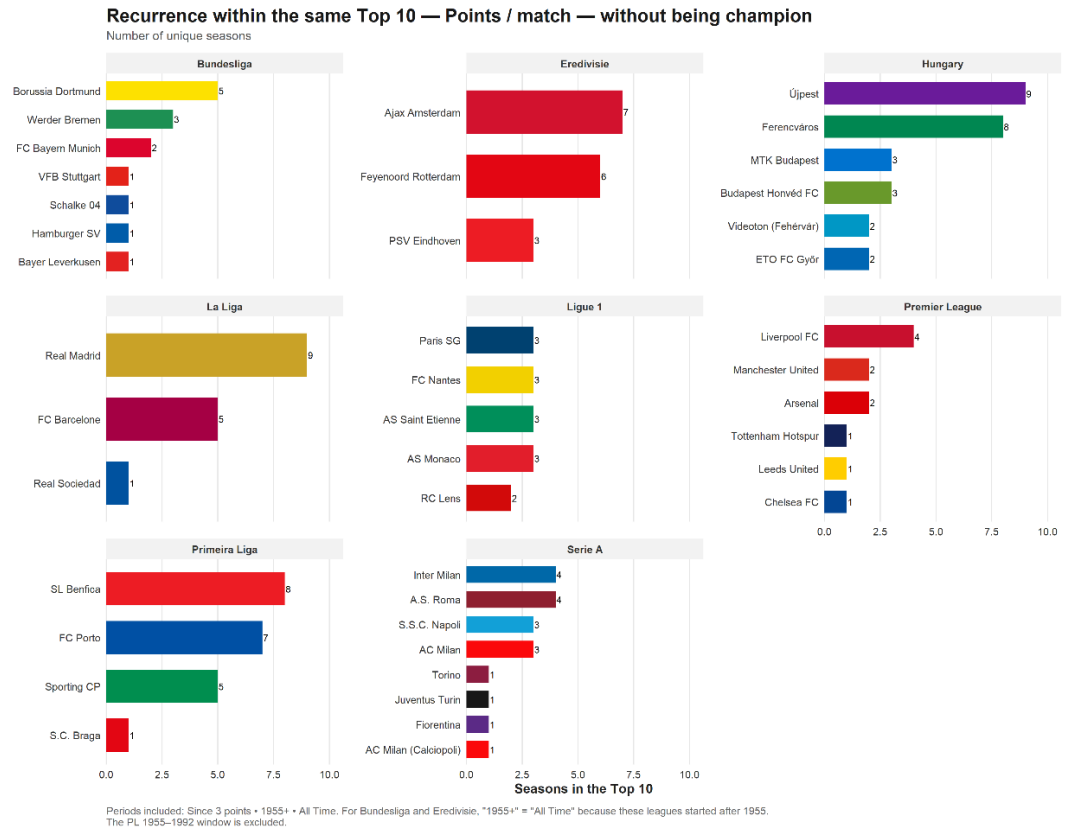


Fig. 34: Recurrence of clubs in the rankings of points per match among non-champions for each period and league

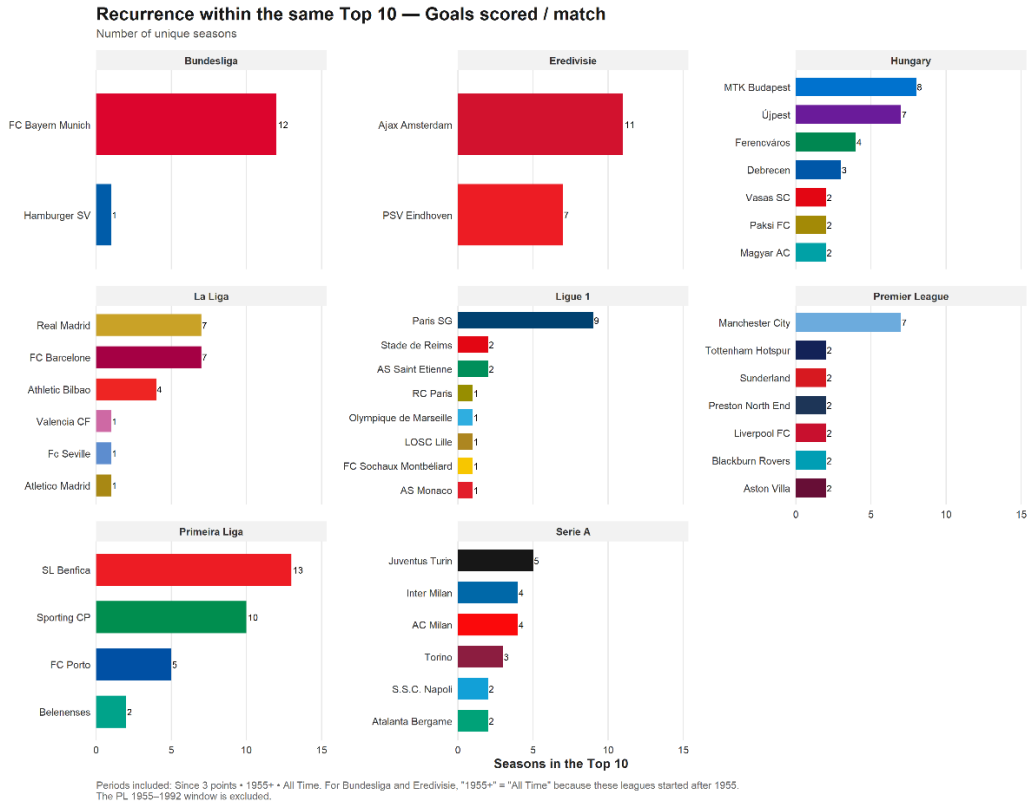


Fig. 35: Recurrence of clubs in the rankings of goals scored per match for each period in each league

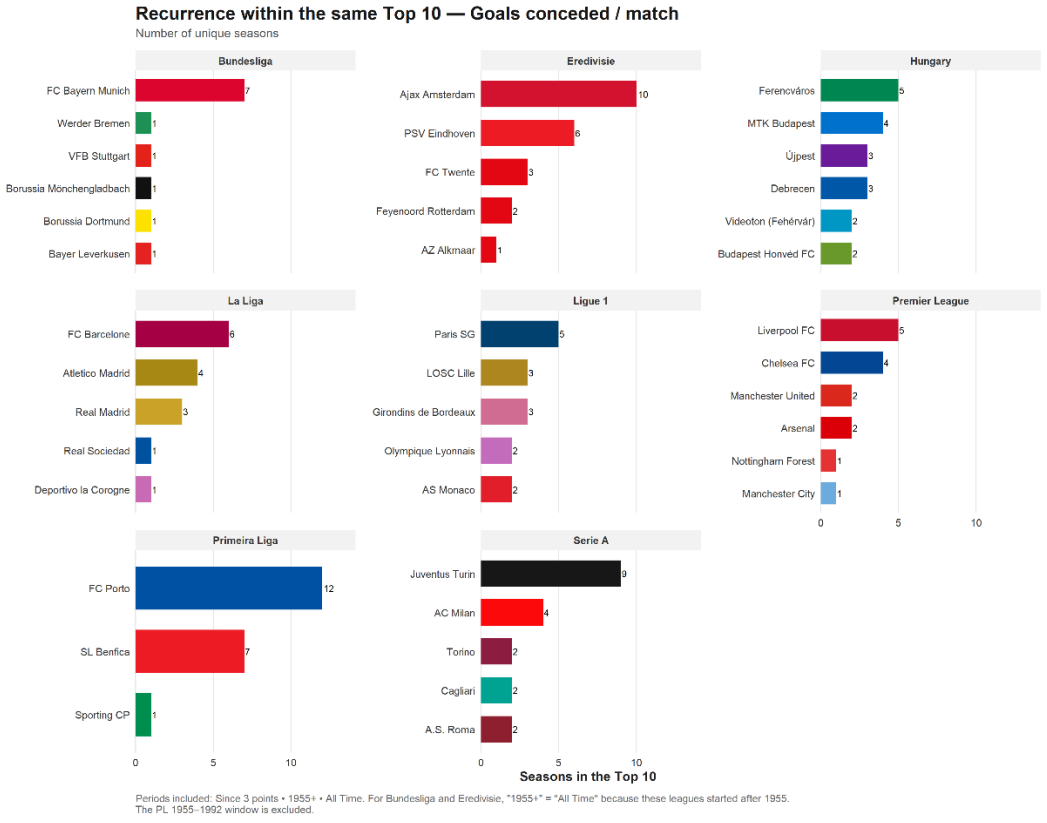


Fig. 36: Recurrence of clubs in the rankings of goals conceded per match for each period in each league

The ranking of non-champions provides a different perspective on recurrence, identifying clubs that repeatedly achieved exceptional points totals without winning the league (Fig. 34). Real Madrid leads La Liga with nine appearances and is as expected since there was an amazing competition between this club and FC Barcelona throughout times. It goes the same for Benfica and Porto which record eight and seven respectively in Portugal. In Hungary, Újpest and Ferencváros stand out with nine and eight appearances, while Ajax and Feyenoord lead the Eredivisie ranking.

For the Bundesliga which was often dominated by Bayern Munich, the league presents a more distributed pattern, with Borussia Dortmund appearing five times and Werder Bremen three times (Fig. 34). We can observe the same thing in Ligue 1 where the presence of PSG since 2012 enhanced the competition and led to great teams ending up second. In the Premier League, Liverpool leads with four appearances, while Serie A shows contributions from several clubs, including Inter, Roma, Napoli and AC Milan. These distributions illustrate that the clubs are mostly the same whether we talk about champions or not but it will differ especially when a team has a very high domination on a league (Fig. S13, Fig. S14).

Attacking recurrence reveals some particularly strong concentrations (Fig. 35). Benfica leads the Primeira Liga with thirteen appearances, followed by Sporting with ten, while MTK Budapest and Újpest dominate the Hungarian ranking with eight and seven appearances. Bayern Munich appears twelve times in the Bundesliga Top 10, and Ajax eleven times in the Eredivisie, highlighting the repeated attacking excellence of these clubs.

In Ligue 1, Paris Saint-Germain leads with nine appearances, while Real Madrid and Barcelona each record seven in La Liga (Fig. 35). Serie A displays a more distributed pattern, with Juventus leading on five appearances, followed by Inter and AC Milan. The Premier League ranking is led by Manchester City with seven appearances. These differences show that repeated attacking excellence can be concentrated in a small number of clubs reflecting the teams and the financial gap extending since 2010s, while other championships feature a wider range of teams across their historical rankings and reflect what mostly happen before 1960's especially in la Liga with Bilbao, Sevilla and Valencia.

Defensive recurrence produces another distinct picture (Fig. 36). Porto leads the Primeira Liga with twelve appearances, followed by Benfica with seven, while Ajax tops the Eredivisie ranking with ten. Juventus leads Serie A with nine appearances, and Bayern Munich dominates the Bundesliga ranking with seven. These results illustrate how much in the same league, a team can dominate nearly every aspect of the records and especially when it had an incredible run of domination like Bayern Munich during the 2010s. Bayern Munich was the best attacking team and was so dominant with the ball that combined with one of the best goalkeepers ever, it helped them conceding very few goals during a Bundesliga season.

The distribution is more balanced in several other championships. Barcelona leads La Liga with six appearances, while Paris Saint-Germain and Liverpool top Ligue 1 and the Premier League rankings with five each. Hungary also shows contributions from several historically successful clubs, led by Ferencváros with five appearances and MTK Budapest with four. These results demonstrate that recurring defensive excellence does not necessarily follow the same pattern as

attacking performance like in Liga but, in the end, we almost see the same clubs, especially in Bundesliga and the leagues outside of the big 5.

Across the four metrics, a recurring pattern emerges: a relatively small group of clubs account for a substantial share of elite performances in several championships (Fig. S9, Fig. S10, Fig. S11, Fig. S12). However, the degree of concentration varies by league and by performance dimension. Bayern Munich, Paris Saint-Germain, Benfica, Porto and Ajax all feature prominently in particular rankings, but their recurrence differs between points accumulation, attacking output and defensive solidity.

The main insight from these rankings is not simply that certain clubs appear more frequently than others, but that recurrence can reveal different forms of domestic dominance. When a club repeatedly features among the strongest performers in points per match, goals scored and goals conceded, its historical prominence extends beyond a single area of performance. Conversely, a club that appears predominantly in one ranking may be associated with a more specific form of excellence. Comparing the four metrics therefore provides a more nuanced understanding of how clubs established their place in domestic football history.

The distinction between overall points and points among non-champions also helps clarify the competitive context behind these performances. When the same clubs recur in both rankings, exceptional results are not limited to title-winning seasons. This may indicate that several teams were capable of reaching similarly high standards, even when only one could finish first. Where recurrence is more concentrated among overall points leaders, the strongest performances may instead be more closely associated with dominant title-winning teams. The relationship between these rankings is therefore as informative as the individual results.

Finally, recurrence should be understood as a pattern of repeated excellence rather than uninterrupted dominance (Fig. S13, Fig. S14, Fig. S15, Fig. S16). Many historically prominent clubs experienced several successful periods separated by less productive eras, reflecting the changing nature of teams, generations and competitive environments. The clubs appearing most frequently in these rankings have not necessarily remained dominant throughout their history; rather, they have repeatedly returned to the highest levels of performance (Fig. S13, Fig. S14, Fig. S15, Fig. S16). This distinction is central to understanding historical persistence: it is not only a question of how long a club dominates, but also of how often it manages to re-establish itself among the elite.

9. Relationships Between Results, Attacking Output and Defensive Solidity

This section examines the overlap between these exceptional performances across three dimensions: points per match, goals scored per match and goals conceded per match. By comparing team-seasons appearing in the Top 10 and Top 15 rankings for the corresponding metrics, the analysis explores whether outstanding results coincide with exceptional attacking output or defensive solidity, and how these combinations vary across championships and historical periods.

Its purpose is therefore not to establish a general relationship between goals and points, but to investigate how different performance characteristics combine among historically exceptional teams. The resulting patterns provide an additional perspective on the profiles of these performances, without implying that one metric necessarily causes another.

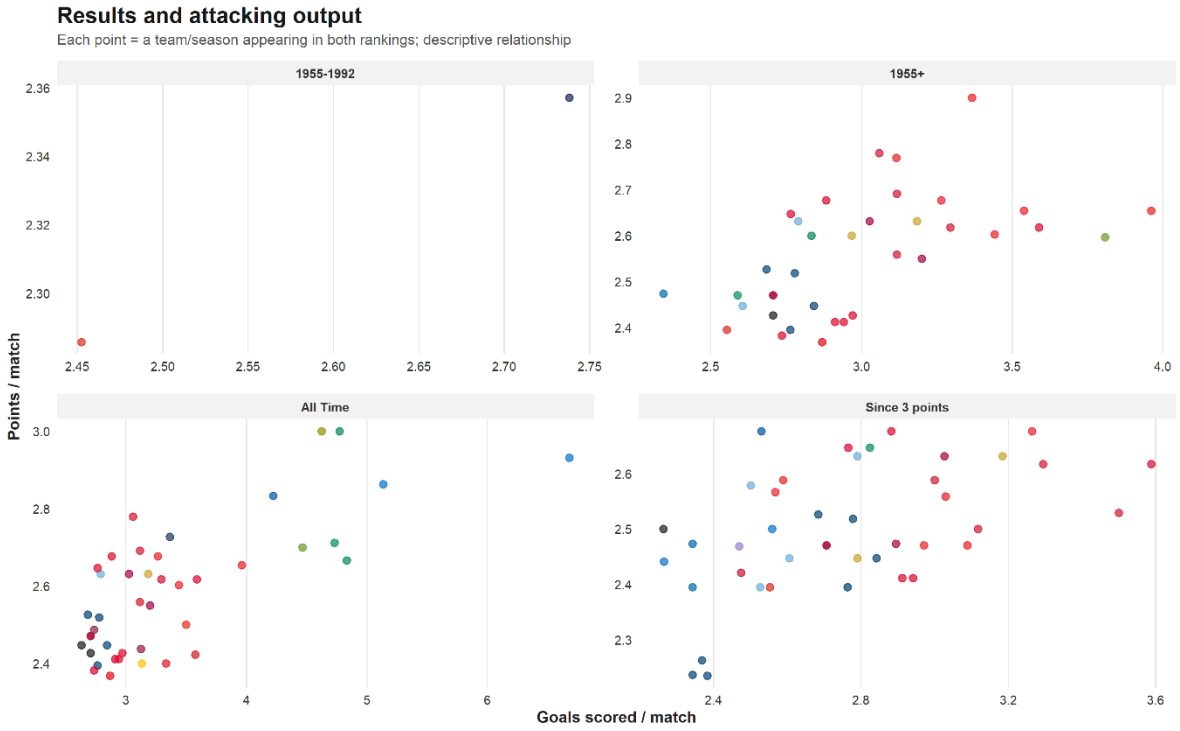


Fig. 37: Relationship between Points per Match and Goals Scored per Match Across Historical Periods

The first scatter plot compares points per match and goals scored per match among team-seasons appearing in both corresponding Top 10 rankings (Top 15 for the All-Time period). The objective is to determine whether the most successful performances in terms of points also feature among the strongest attacking records, and whether this overlap varies across historical periods.

The All-Time panel displays the widest range of attacking output (Fig. 37). While many observations remain between approximately 2.5 and 3.5 goals per match which is already amazing, several exceptional seasons exceed four goals per match, with the highest scoring rate approaching seven as we already know with MTK Budapest 1917-1918 and all the outstanding hungarian offensive performances (Fig. 23). However, these extreme attacking performances do not all correspond to the highest points-per-match values but are most of the times part of the top 15 or just ended up champion for the least.

This illustrates an important distinction within the historical rankings: an exceptional scoring record does not necessarily coincide with an equally exceptional points total.

The 1955+ and Since 3 Points panels contain a narrower range of attacking performances, although their distributions still show differences in points accumulation. In both periods, some team-seasons combine high scoring rates with strong points totals, while others achieve

comparable attacking output without reaching the same points-per-match level. Overall, a slight positive trend can still be observed in the scatter plots, with some of the highest-scoring performances also corresponding to the highest points-per-match totals. This is consistent with the expectation that strong attacking output will contribute to achieving better results and with the contexts and findings around the best offensive performances (rules, teams before World War II, the 1950's and the superteams during the 2010s and since then).

To compare these relationships more systematically across championships and historical periods, Pearson correlation coefficients are used to measure the strength and direction of the linear association between goals scored per match and points per match (Fig. 38). Values closer to +1 indicate a stronger positive association, values closer to −1 indicate a stronger negative association, and values around zero indicate a weaker linear relationship. This complements the scatter plots by making differences between championship-period combinations easier to identify.

Several combinations display strong positive associations, notably the Premier League in 1955+, Hungary since three points, and La Liga and Ligue 1 since three points (Fig. 38). Within these selected samples, team-seasons with higher scoring rates tend to have higher points-per-match values.

However, the relationship is not consistent across the dataset. La Liga in the All-Time period displays a strong negative correlation, while Serie A since three points and the Primeira Liga in 1955+ also show notable negative associations. The Eredivisie and Ligue 1 All Time distributions are negative as well. Crucially, these negative or weak associations reflect distinct tactical and regulatory environments across eras. In Serie A, this aligns with Italy's historical tactical culture—defined by Catenaccio, defensive solidity, and result-oriented minimalism (the classic 1-0 win)—where raw attacking volume was rarely the primary determinant of league dominance. In La Liga's All-Time sample, pre-1950 seasons (such as Athletic Bilbao in the 1930s) were characterized by chaotic, high-scoring matches before modern defensive structures matured. Furthermore, the 1925 offside law modification (reducing the required defender count from 3 to 2) triggered an explosion in scoring across Europe that predated modern defensive balancing. When high attacking output was widespread, extreme goalscoring became less distinctive as a predictor of a maximum points-per-match ratio. Consequently, some of the highest-scoring team-seasons did not necessarily correspond to the highest points totals, explaining why the relationship appears weaker or negative in certain All-Time samples.

These results do not imply that scoring more goals reduces a team's points total. Rather, they indicate that, among the selected team-seasons appearing in both rankings, the highest attacking output does not consistently coincide with the highest points totals. Also, it doesn't say that higher attacking output doesn't coincide with higher points totals because it obviously does most of the times.

The contrast between periods is therefore a key finding. The relationship observed in a modern ranking can differ substantially from that found when performances from several historical eras are combined. Pearson coefficients help quantify these differences, but they remain descriptive measures: they do not establish causality and may be sensitive to the number of observations

and the presence of extreme performances in each sample. Overall, the results suggest that the overlap between exceptional attacking output and exceptional points totals varies according to the championship and historical period, rather than following a universal pattern. We still observe a substantial number of teams which were present in both rankings during the periods. This is also impressive because it shows how extraordinary some teams have been throughout history and especially in multiple aspects of the game.

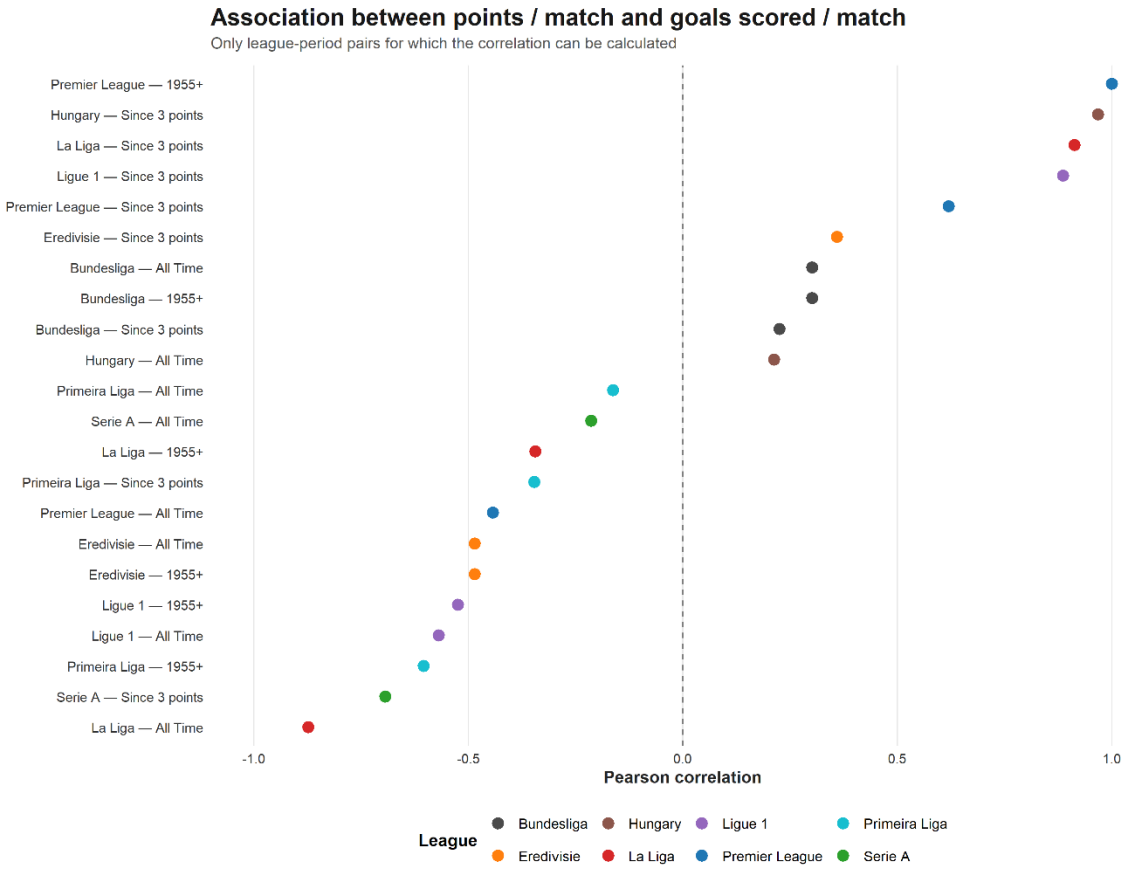


Fig. 38: Pearson Correlations Between Points per Match and Goals Scored per Match Across League-Period Combinations

The second scatter plot compares points per match with goals conceded per match among team-seasons appearing in both the points and defensive Top 10 rankings (Fig. 39). Unlike the attacking analysis, lower values on the horizontal axis represent stronger performances. The figure therefore examines whether the seasons achieving the strongest points totals also feature among the most defensively solid performances.

Across the All Time, 1955+ and Since 3 Points panels, the observations show that elite points totals can coexist with different goals-conceded rates. Several team-seasons combine relatively low goals-conceded rates with high points-per-match values, suggesting that exceptional defensive performances can accompany outstanding results. However, other selected seasons achieve high points totals despite conceding more frequently, while some particularly strong defensive records correspond to less exceptional points totals. We also observe that especially during the three points, that the teams which had the most points aren't those who conceded the

least in the rankings. Although, as discussed previously, the dominance of a team like Ajax Amsterdam, Bayern Munich or FC Porto can result in an amazing defensive season and outstanding performances. Nevertheless, neither the 1955+ nor the Since 3 Points panel establishes a perfectly consistent relationship between the two metrics.

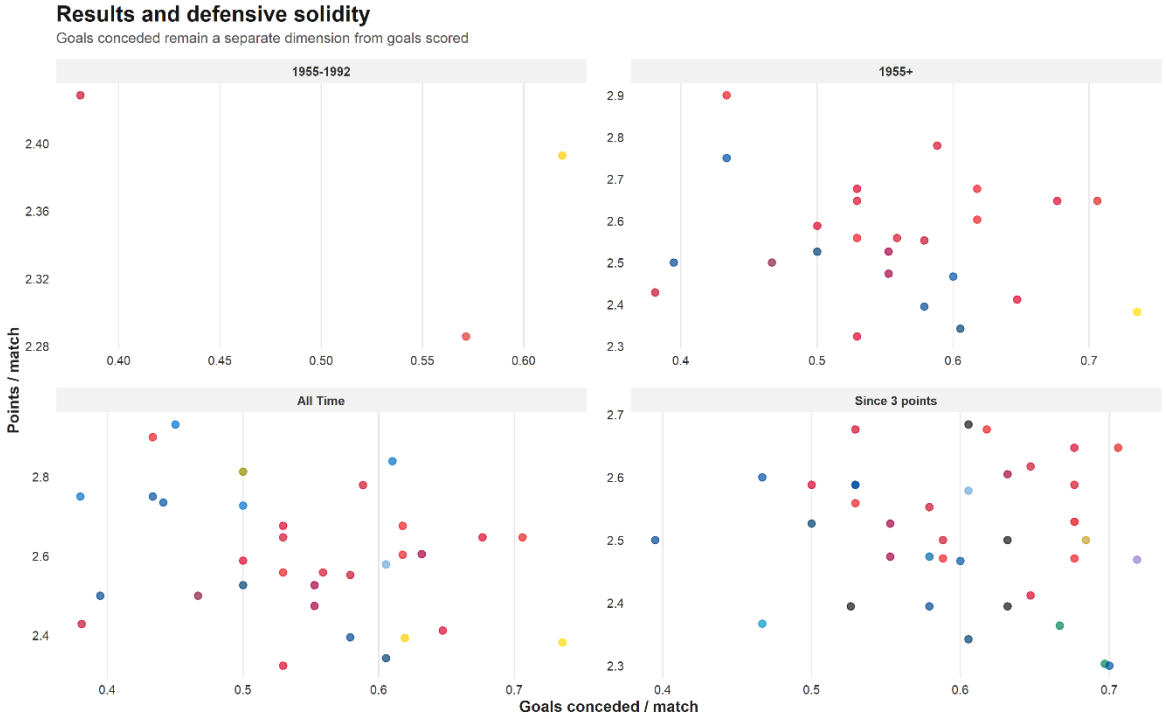


Fig. 39: Relationship between Points per Match and Goals Conceded per Match Across Historical Periods

Taken together, the three figures reveal that elite performance is not characterised by a single, consistent relationship between points accumulation and the other two dimensions. The attacking analysis shows substantial variation in the strength and direction of the correlations across championship-period combinations, while the defensive scatter plots illustrate that strong defensive records can accompany different levels of points-per-match performance.

The strongest positive attacking correlations suggest that, in certain selected rankings, exceptional scoring output and exceptional points totals tend to coincide, more than defensive ones as expected. By contrast, the negative correlations in several All Time and historical-period rankings indicate that the team-seasons appearing in both Top 10 lists do not necessarily follow the same pattern. However, these coefficients must be interpreted cautiously. The analysis includes only team-seasons present in both corresponding Top 10 rankings, rather than all seasons played in each championship. The number of overlapping observations therefore varies between championship-period combinations, and some strong correlations may be influenced by a small number of points or individual outliers. The correlation figure also excludes combinations for which a coefficient cannot be calculated. Its results should consequently be interpreted as descriptive evidence of how selected elite performances overlap, rather than as a general measure of the relationship between goals and points in a league around the top performances.

10. Discussion

1. What the records reveal about historical elite performance

The clearest result is that the historical identity of an exceptional season depends on the indicator being examined. The points rankings are comparatively stable across eras, whereas goals scored show much more extreme historical variation and defensive records follow a different temporal pattern (Figs. 1–4 and 25–32). This distinction is more informative than a single combined ranking would be: points per match captures the conversion of performances into results, while goals scored and conceded preserve different tactical and competitive dimensions. The project therefore does not identify one universal form of excellence; it shows that elite achievement has taken different forms across the eight leagues.

The all-time attacking rankings are particularly shaped by early Hungarian and Portuguese performances. MTK Budapest's 6.68 goals per match in 1917–18 is an extreme example, while the prominence of Hungarian clubs in the expanded all-time Top 20 demonstrates how much the historical window changes the picture (Figs. 3, 23 and 31). These historic attacking anomalies are deeply tied to the evolution of playing conditions and rules, most notably the 1925 offside rule revision which dismantled the old 2-3-5 Pyramid defenses and sparked a golden era of offensive productivity before Herbert Chapman's WM formation fully spread. In Hungary, this environment produced the legendary Danubian School and Gusztáv Sebes' Golden Team era, where attacking fluidities reached unprecedented peaks. These results should not be read as proof that early leagues were stronger overall than modern competitions; rather, they show that exceptional scoring rates occurred under specific historical tactical frameworks. The modern Bayern Munich example, with 122 goals in 2025–26, provides a useful contemporary contrast: an outstanding rate within modern tactical pressing systems, even though it remains below early 20th-century extremes. Today, most of the amazing performances will be more related to the points per match as we can observe that in the Big 5 leagues and in the rankings. For goalscoring, we can also mention the Real Madrid and FC Barcelona seasons during the first half of the 2010s until 2016, which led to amazing average goals scored per match close to 3.

Defensive records tell a different story. Porto's three leading all-time defensive seasons, each conceding around 0.30–0.32 goals per match, demonstrate a sustained presence at the extreme defensive end of the dataset (Fig. 24). The decade distributions place many exceptional defensive performances in the 1960s–1980s, rather than reproducing the strong 2010s concentration visible in several points rankings (Figs. 28 and 32). This concentration coincides with the tactical golden age of defensive organization in European football, from Helenio Herrera's Inter and Nereo Rocco's AC Milan perfecting Catenaccio to Giovanni Trapattoni's Juve and Bob Paisley's Liverpool. In these eras, defensive solidity was the explicit tactical foundation for domestic supremacy. This divergence suggests that offensive, defensive and points records do not share one common historical trajectory, highlighting how tactical paradigms directly governed performance output.

2. Elite depth, league comparisons and what a Top 10/15 can tell us

The comparison between the record season and the average of the Top 10 is important because it distinguishes an isolated peak from a deeper concentration of elite performances (Figs. 1–8). A league can produce one extraordinary outlier without producing many similarly strong seasons. Conversely, a relatively high Top 10 average suggests that exceptional performance is not limited to a single record-holder. This is a more useful distinction than treating the highest value as a direct ranking of league strength. The Big Five versus other-leagues comparison reinforces that point. The non-Big Five leagues are particularly prominent in the all-time attacking and defensive comparisons, while the gap narrows in the more recent period (Figs. 9–12). The finding is not that smaller leagues are categorically stronger; rather, historical extreme performances are distributed differently across the selected competitions. All-time rankings are sensitive to early seasons and unusual records, whereas more recent periods provide a more consistent competitive setting but cover fewer decades. The interpretation must therefore remain specific to the period and metric. The diversity analysis adds a second distinction. A Top 10 containing performances from many clubs indicates that the selected elite is spread across a wider set of teams; a low-diversity ranking indicates that fewer clubs occupy more of the available places (Figs. 13–16). Yet this is not equivalent to measuring competitive balance across an entire league. Because the analysis focuses on the extreme end of the distribution, it describes the composition of elite performances rather than the distance between all clubs. This is why diversity and recurrence are best interpreted together. Seeing that 15 Hungarian teams are present in the top 20 best performances accounting for the ratio of goals scored per match highlights how common extreme scoring was during the first 50 years of the Hungarian league.

However, interpreting these comparative statistics requires several critical analytical caveats regarding competitive imbalance, squad depth, substitution rules, and scoring environments. An extreme record reflects both the internal quality of the dominant team and the relative weakness or financial disparity of its domestic opponents. Without measuring league-wide variance, such as Gini coefficients of points or goals across a full season, it is impossible to separate genuine tactical superiority from structural disparity within a two-tiered league. Furthermore, statistical variance plays a crucial role when comparing seasons of different lengths. Maintaining an extreme ratio of 2.80 points or 0.35 goals conceded per match across a short 22-match schedule (common in early Hungarian or Spanish seasons) is sample-wise less demanding than maintaining that same pace over a modern 38-match marathon, where regression toward the mean exerts stronger pressure due to cumulative fatigue and injuries. Crucially, the evolution of modern squad construction and substitution regulations plays a pivotal role in mitigating these physical demands. Historical teams operated with minimal or no substitutions and smaller core squads, meaning an injury or physical decline directly suppressed team performance, especially when we see that offensive players were less protected by the referees and were more often subject to serious injuries like fracture than today. In contrast, the progressive integration of tactical substitutions culminating in the permanent adoption of five substitutes per match alongside the development of broader squad construction strategies, has allowed modern elite clubs to sustain high tactical intensity (such as aggressive pressing and high defensive lines) across 38 domestic fixtures and expanded European

calendars. Financial polarization enables modern superteams to build double-starter squads, rotating international-caliber players without experiencing performance drop-offs, thereby insulating their points-per-match ratios from physical regression. Finally, raw goal averages mask era-specific scoring baselines and tactical metas. Scoring 3.5 goals per match during an era where the league average is 4.5 goals per match represents a lower relative dominance than scoring 3.0 goals per match when the league average is 2.2.

3. Club recurrence and the competitive meaning of non-champion seasons

The recurrence results show that a relatively small group of clubs appears repeatedly among the selected elite performances, but the concentration varies by league and indicator (Figs. 33–36). Bayern Munich, Barcelona, Real Madrid, Juventus, Benfica, Porto, Ajax and several Premier League clubs recur in different combinations. Crucially, the club most associated with points performance is not always the club most associated with attacking or defensive records. Recurrence therefore reveals distinct club profiles rather than one fixed hierarchy of historical greatness. The non-champion points ranking adds a particularly useful competitive perspective (Fig. 34). It identifies seasons in which a club accumulated an exceptional points total but did not win the league. Liverpool's 97 points in 2018–19, one point behind Manchester City, illustrates how a historically strong campaign can be overshadowed by an even stronger rival. We observed the same thing in Spain between Real Madrid and Barcelona or in Portugal with Benfica and Porto. These seasons are not failed versions of championship seasons; they show the strength of the competition at the top of the table. Comparing overall points with non-champion points can therefore help distinguish a club's own performance from its position relative to its strongest opponent. Raw recurrence counts nevertheless require care. The report uses different cut-offs for standard Top 10 rankings, individual all-time Top 15 rankings and the combined Big Five all-time Top 20, and some historical periods overlap. The same club-season may consequently appear in more than one ranking. The counts measure appearances under this framework, not independent seasons or a complete measure of long-term dominance. Nor can recurrence alone demonstrate that financial resources caused a club's presence at the top; budgets, wages, transfers and squad depth are not directly analysed here.

4. Interpreting the relationships between points, attack and defence

The scatter plots and correlation heatmap show that the association between points per match and goals scored per match differs across league-period combinations (Figs. 37–38). Some selected groups show positive associations, while others are weak or negative. The correct interpretation is not that scoring more goals reduces points. Rather, within the restricted set of team-seasons already selected for elite points and attacking rankings, the most extreme attacking output does not always coincide with the highest points rate. This decoupling is further explained by macro-historical, regulatory, and strategic factors.

On a regulatory level, successive offside rule revisions directly altered tactical space and offensive productivity. Beyond the landmark 1925 change, IFAB's 1990 amendment which established that an attacker level with the second-to-last defender is not offside was explicitly

designed to counter defensive trapping like in Arrigo Sacchi's AC Milan and low-scoring trends following the 1990 World Cup. Later, the 2005 refinement clarifying "active play" and "seeking to gain an advantage" further favored attacking movement by allowing passive attackers to remain on the pitch without stopping play. However, these offensive rule adjustments did not lead to a return to pre-1950 goal volumes (such as 5 or 6 goals per match). Instead, they coincided with the post-1995 Bosman ruling and the rise of modern superteams, which converted tactical fluidity into territorial dominance, high possession, and systematic match control rather than reckless goal-chasing. However, we see a rise of the goal scored and especially by the top teams, notably during the 2010's and since then.

Additionally, the strategic shift from two to three points for a win altered incentives: under two points, a draw retained 50% of a win's value, encouraging conservative away setups, whereas three points reduced a draw to 33%, forcing dominant sides to play aggressively for all three points. Once in control, modern elite teams actively manage game states. After securing a two or three-goal lead, dominant teams often prioritize rhythm control, ball possession, and physical preservation for European fixtures rather than chasing historical goal totals. In contrast, pre-1950 teams operated without modern workload management, translating technical superiority into unfiltered high-scoring scorelines. Modern superteams in the 2010s thus achieve unprecedented points totals (2.6 to 2.8 pts/match) through financial polarization, squad depth, and game-state management rather than extreme goal totals alone.

From a statistical standpoint, these findings are subject to inherent methodological boundaries such as selection bias, small sample sizes, and linearity assumptions which require contextual historical knowledge to fully interpret. The correlation sample consists strictly of team-seasons appearing in both corresponding Top 10 or Top 15 rankings, rather than all teams across a full league season. Restricting the range to extreme outliers artificially flattens or reverses linear relationships. At a league-wide level, scoring more goals strongly predicts accumulating more points, whereas within an elite-only subsample, this relationship plateaus or becomes noisy. Pearson correlation also captures linear association, is sensitive to extreme observations, and does not establish causation. The defensive scatter plot offers a visual comparison but, without a corresponding set of coefficients and sample sizes, should be interpreted descriptively (Fig. 39). These limitations qualify the statistical claims without removing the value of comparing the selected elite seasons.

5. Scope, robustness and the next stage of the project

The scope of the project is deliberately focused: eight domestic leagues, selected historical windows and four performance indicators. This makes a broad historical comparison possible, but the leagues do not share identical coverage. The Bundesliga and Eredivisie began after 1955, so their 1955+ and All-Time periods represent the same underlying league histories; the Premier League window begins in 1992 even though three points for a win had already been used in the English top flight since 1981. These differences affect which seasons can enter the analysis and mean that period comparisons are not always comparisons of equivalent samples.

The historical points normalization, described in Section 2.2, is a practical compromise rather than a complete reconstruction of modern football. It expresses original points relative to the maximum possible under the system of the time, but does not recalculate every season under a uniform three-point system. Likewise, per-match rates reduce the effect of different season lengths but cannot remove differences caused by unusual competition formats or historical playing conditions. These choices make the dataset more comparable than raw seasonal totals would be, while leaving limits that should remain visible when interpreting the rankings.

To build upon this work, the most valuable next step would be to move beyond the selected extremes and build a comprehensive team-season dataset covering all available observations across history. Moving beyond Top 10 or Top 15 cut-offs would eliminate selection bias, enable full-sample correlation models, and test whether the 2010s concentration of points performances reflects a league-wide shift or an isolated polarization among a few dominant clubs. It would also clarify whether attacking and defensive relationships among elite seasons resemble those across the full distribution, and whether diversity at the top corresponds to greater competitive balance overall. Testing alternative ranking cut-offs and alternative points-normalization methods could further confirm how robust the main patterns are. For the correlation analysis, reporting exact sample sizes for all overlaps, examining influential leverage points, and comparing Pearson correlations with rank-based methods such as Spearman's rank correlation would clarify whether the findings depend on linearity or extreme outliers.

A further extension would test historical explanations quantitatively rather than simply placing them alongside the results. Structural shifts most notably the 1995 Bosman ruling, the expansion of the UEFA Champions League, and the concentration of global broadcasting revenues in the 2010s created an unprecedented financial polarization. This allowed modern 'superteams' such as Bayern Munich, Real Madrid, Barcelona, Juventus, and PSG to hoard elite talent, deepen squad depth, and achieve historically high points-per-match ratios. Future work could connect the performance dataset directly to financial metrics such as television revenue distribution, wage bills, and squad market values, as well as match-level records and documented rule changes like the 1925 offside rule or the introduction of three points for a win. Furthermore, expressing goals scored and conceded as standard deviations or Z-scores relative to each league-season average would isolate true performance dominance from era-specific scoring environments, making it possible to distinguish an absolute historical record from achievement relative to contemporary competitive balance.

Overall, the project's main contribution is to show that elite football history cannot be reduced to one record, one league hierarchy or one measure of dominance. Points performance is comparatively stable across periods, extreme attacking records are disproportionately historical, defensive records follow a different timeline, and recurring clubs do not dominate every metric in the same way. These are not contradictions to be resolved by selecting a single "best" team; they are complementary findings about different dimensions of football performance. The analysis establishes where the exceptional seasons are and how they are distributed. Explaining why they occurred requires a broader sample and additional variables, which provides a clear and realistic direction for further research.

Supplementary Information

S1. Boxplot for points per match……S1

S2. Boxplot for points per match among non-champions……..S2

S3. Boxplot for goals scored per match……..S3

S4. Boxplot for goals conceded per match ……S4

S5. Average ratio of points per match…….S5

S6. Average ratio of points per match among non-champions……...S6

S7. Average ratio of goals scored per match……...S7

S8. Average ratio of goals conceded per match…...……...S8

S9. Persistence of clubs for points per match…….S9

S10. Persistence of clubs for points per match among non-champions…….S10

S11. Persistence of clubs for goals scored per match……...S11

S12. Persistence of clubs for goals conceded per match……..S12

S13. Recurrence of clubs for points per match…….S13

S14. Recurrence of clubs for points per match among non-champions…….S14

S15. Recurrence of clubs for goals scored per match……...S15

S16. Recurrence of clubs for goals conceded per match…….....S16

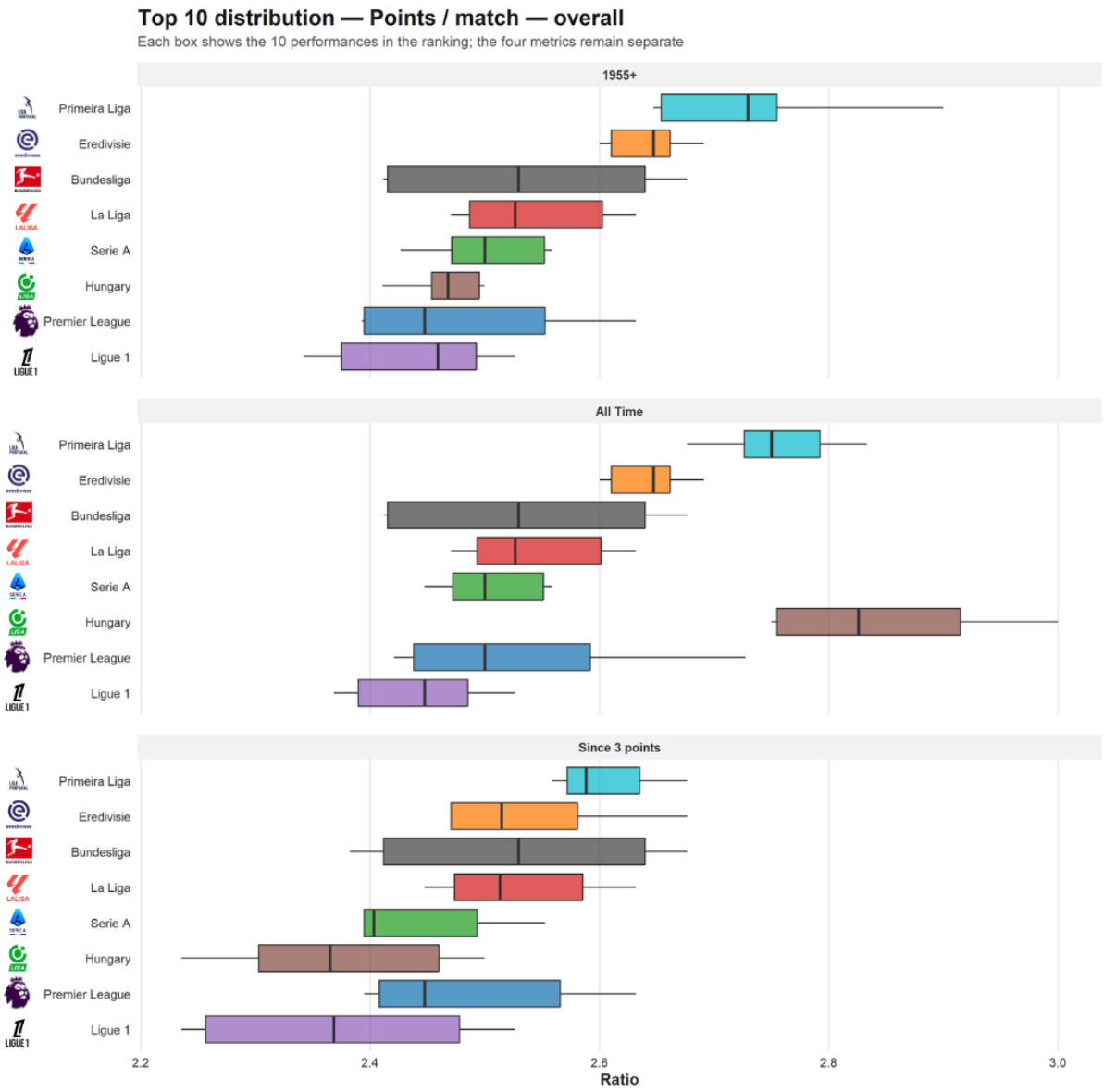


Fig. S1: Boxplot of the Top rankings for points per match for each domestic league and period.

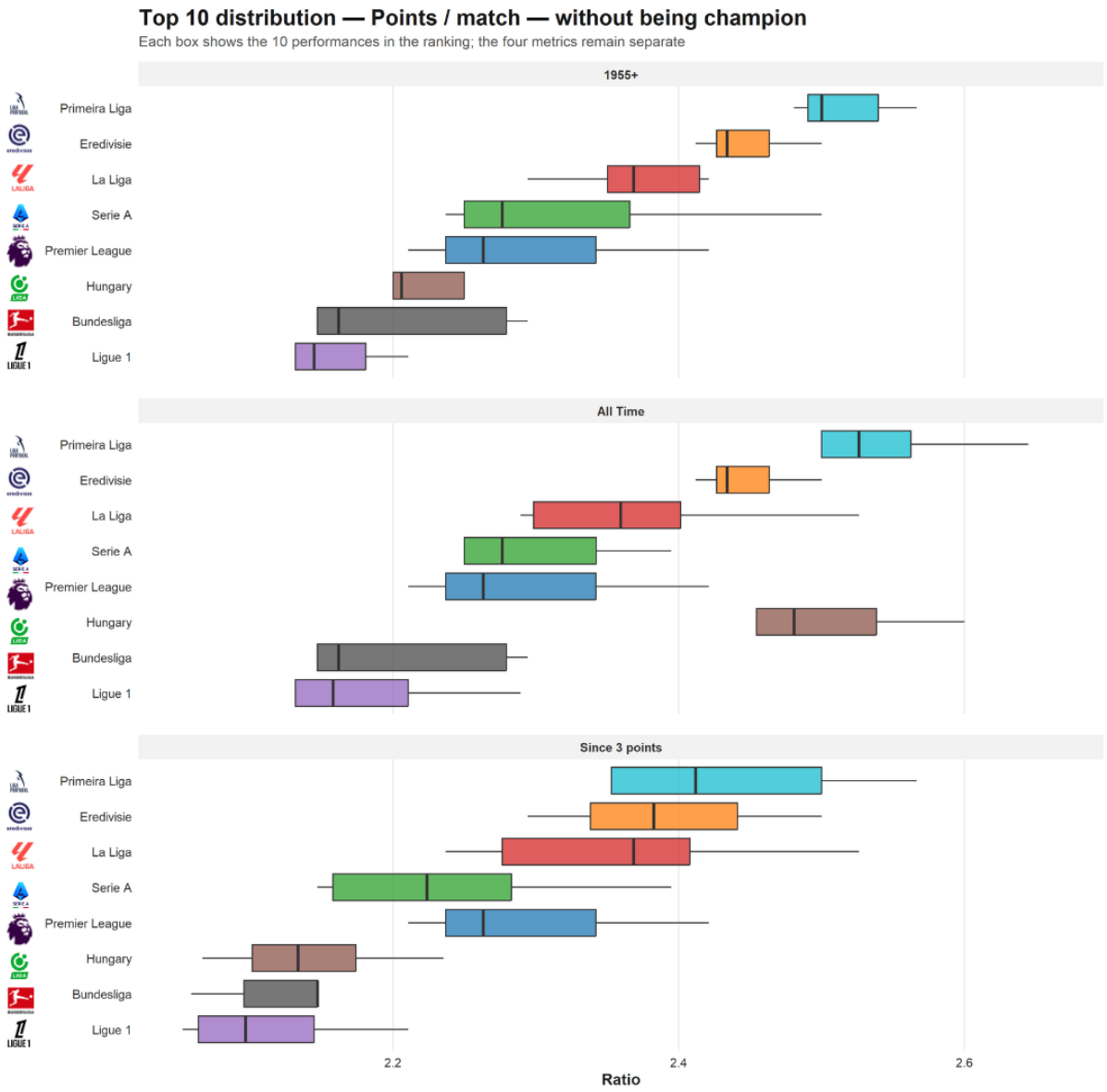


Fig. S2: Boxplot of the Top rankings for points per match among non-champions for each domestic league and period.

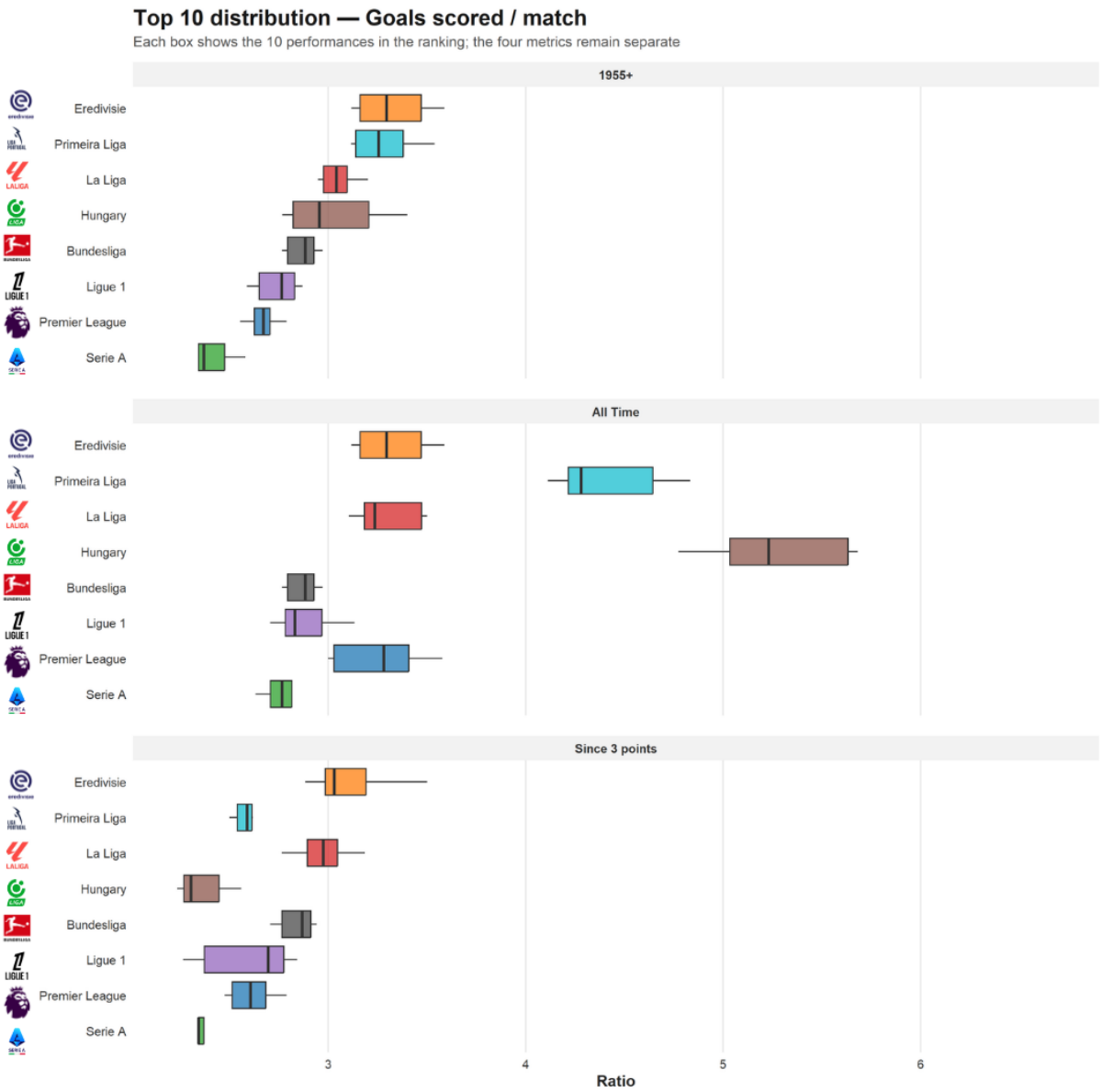


Fig. S3: Boxplot of the Top rankings goals scored per match for each domestic league and period.

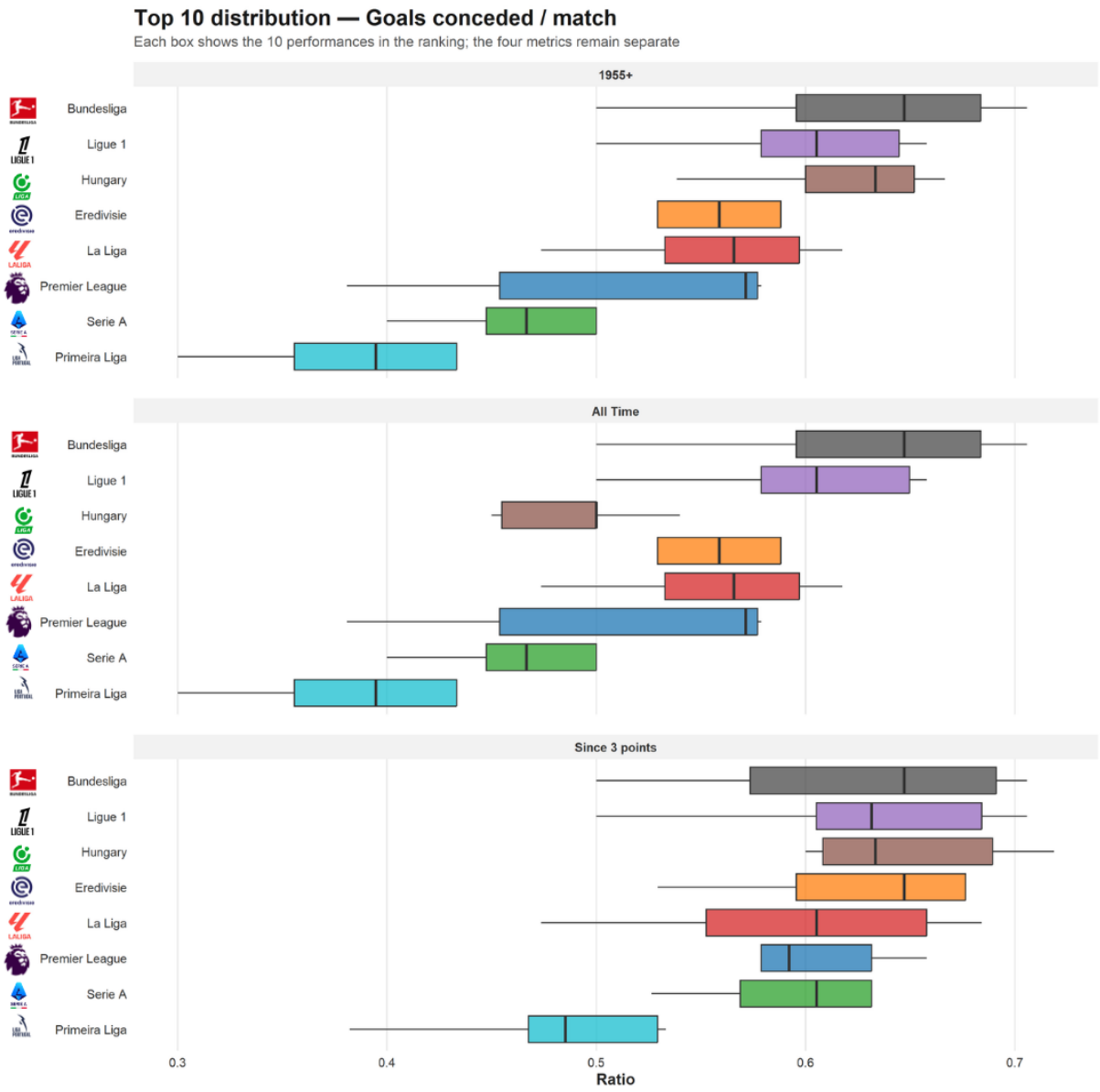


Fig. S4: Boxplot of the Top rankings goals conceded per match for each domestic league and period.

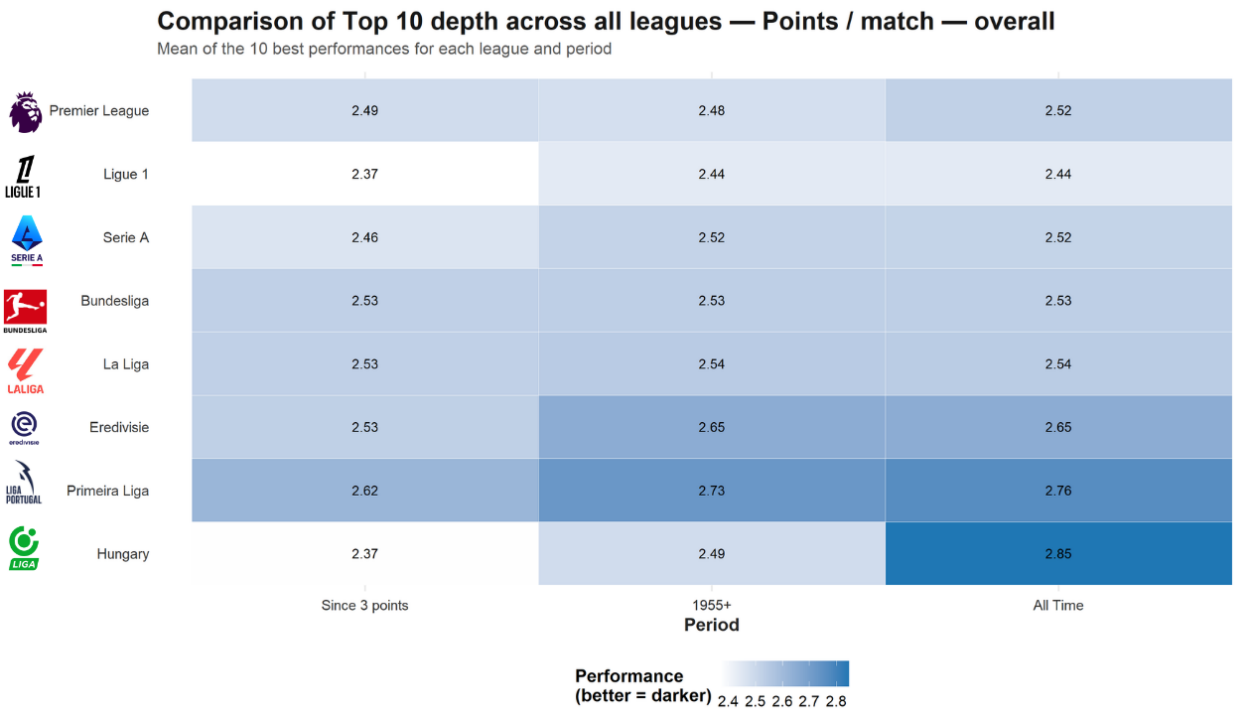


Fig. S5: Heatmap for the average ratio of the rankings of points per match for each league and period.

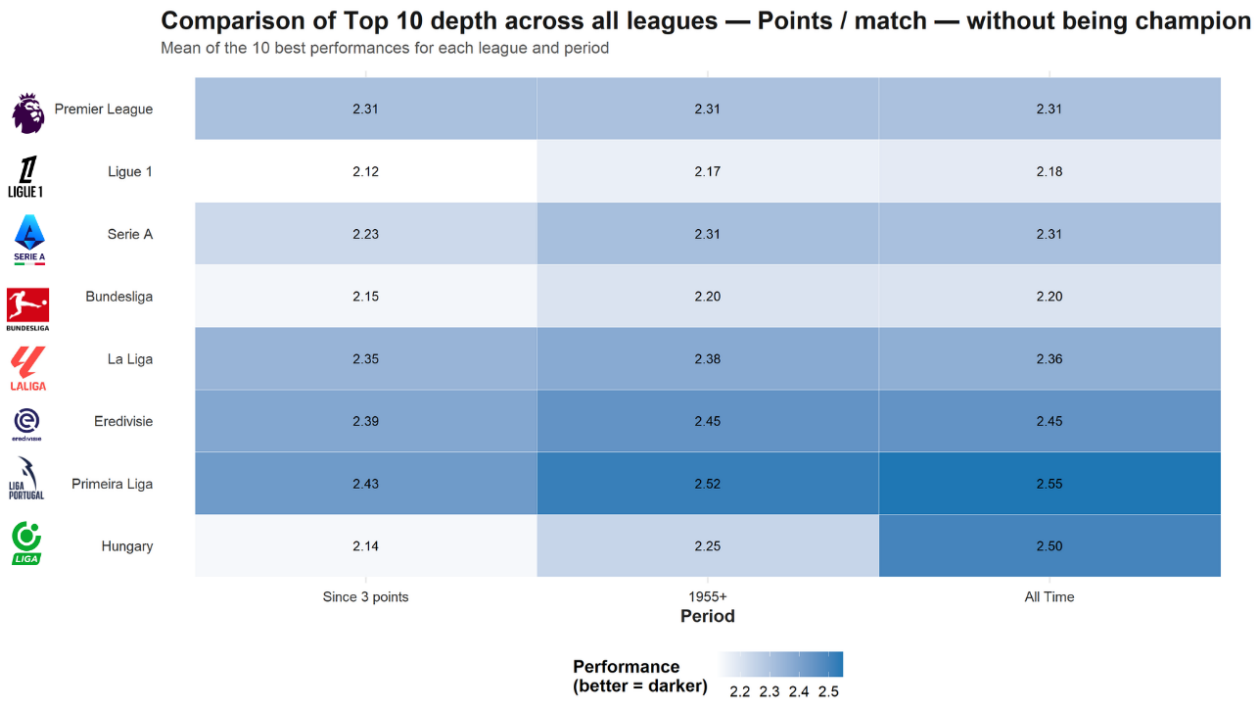


Fig. S6: Heatmap for the average ratio of the rankings of points per match among non-champions for each league and period.

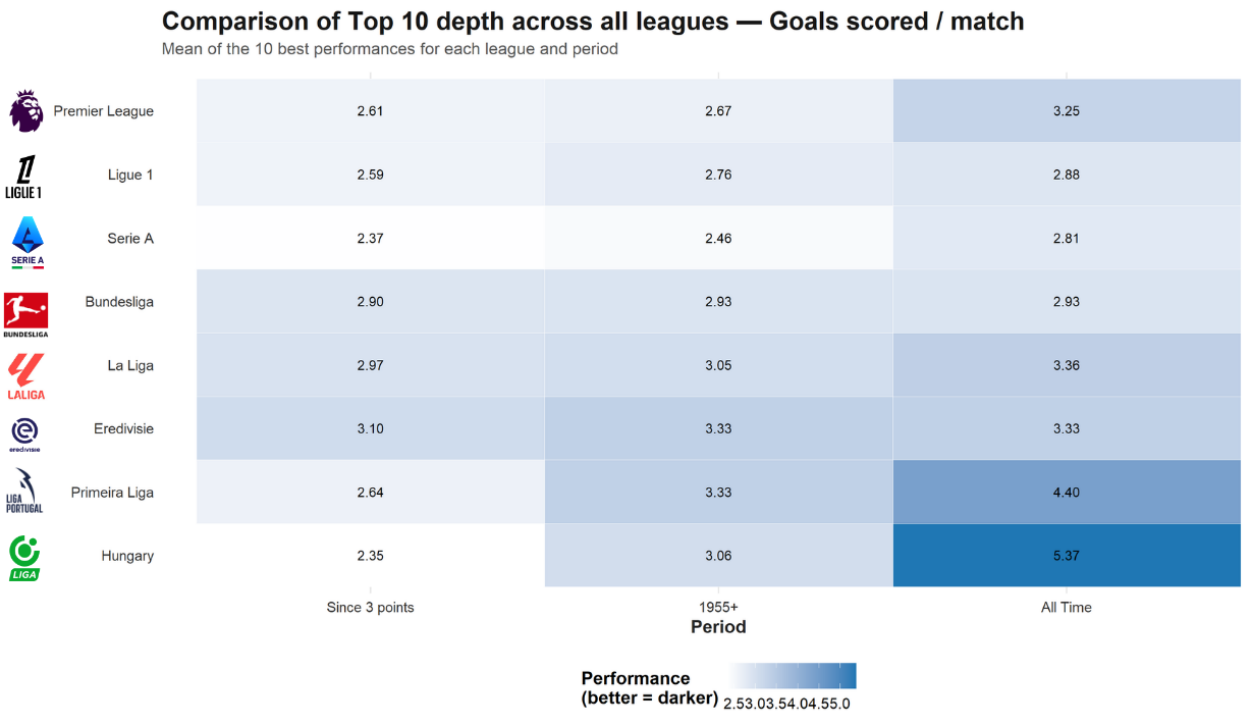


Fig. S7: Heatmap for the average ratio of the rankings of goals scored per match for each league and period.

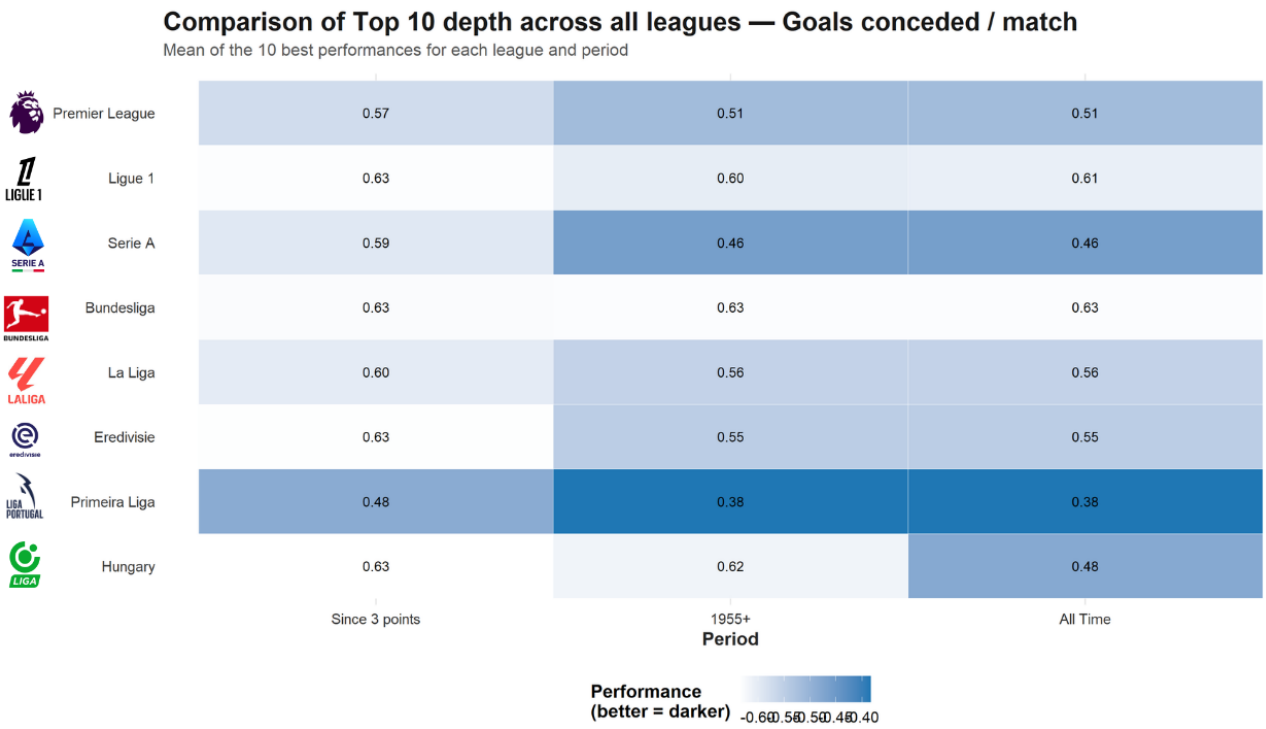


Fig. S8: Heatmap for the average ratio of the rankings of goals conceded per match for each league and period.

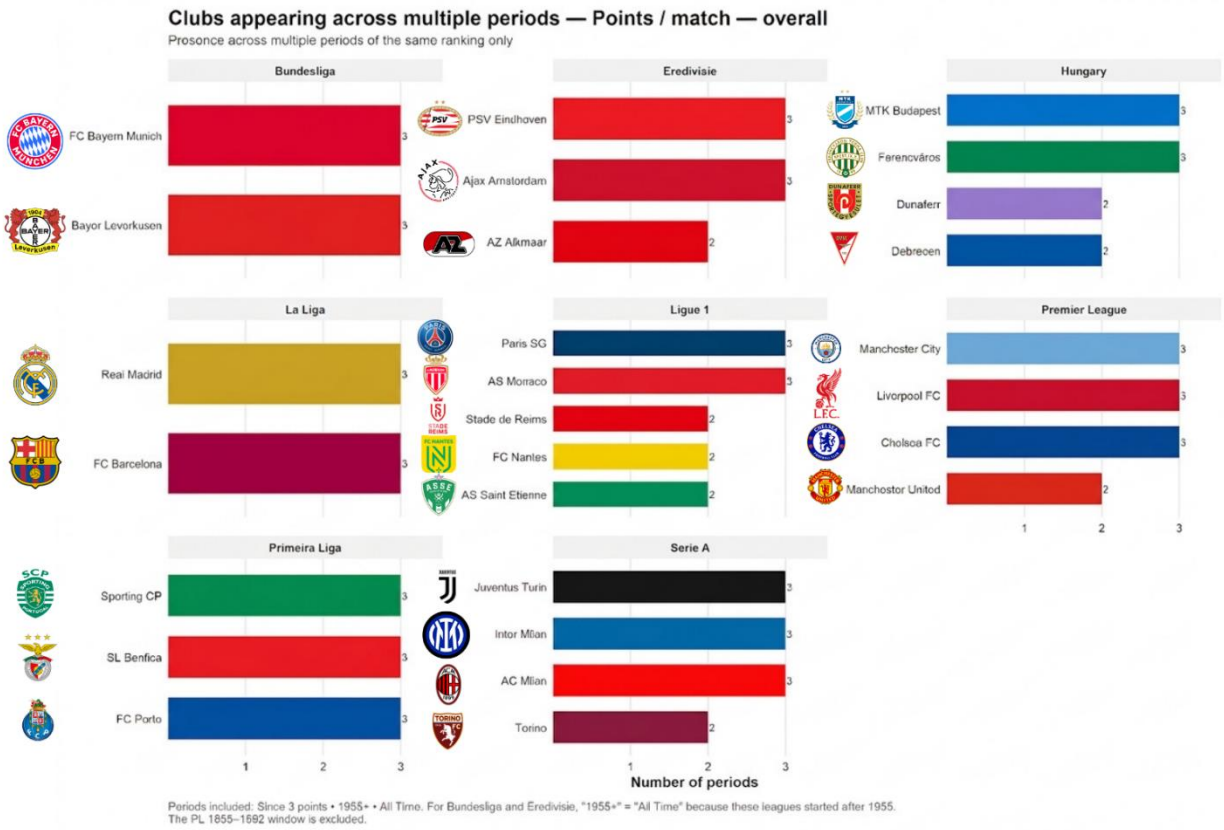


Fig. S9: Persistence of clubs in the rankings of points per match for each period in each league

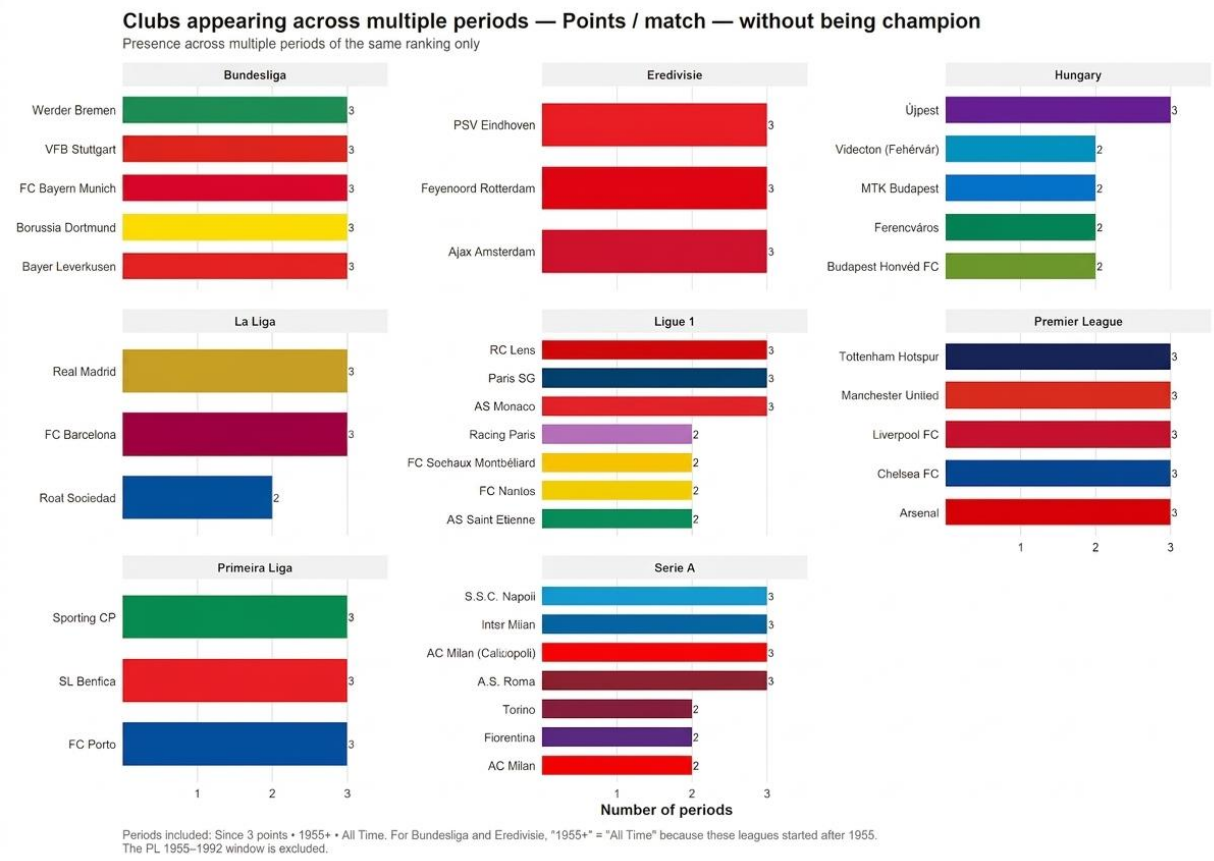


Fig. S10: Persistence of clubs in the rankings of points per match among non-champions for each period in each league

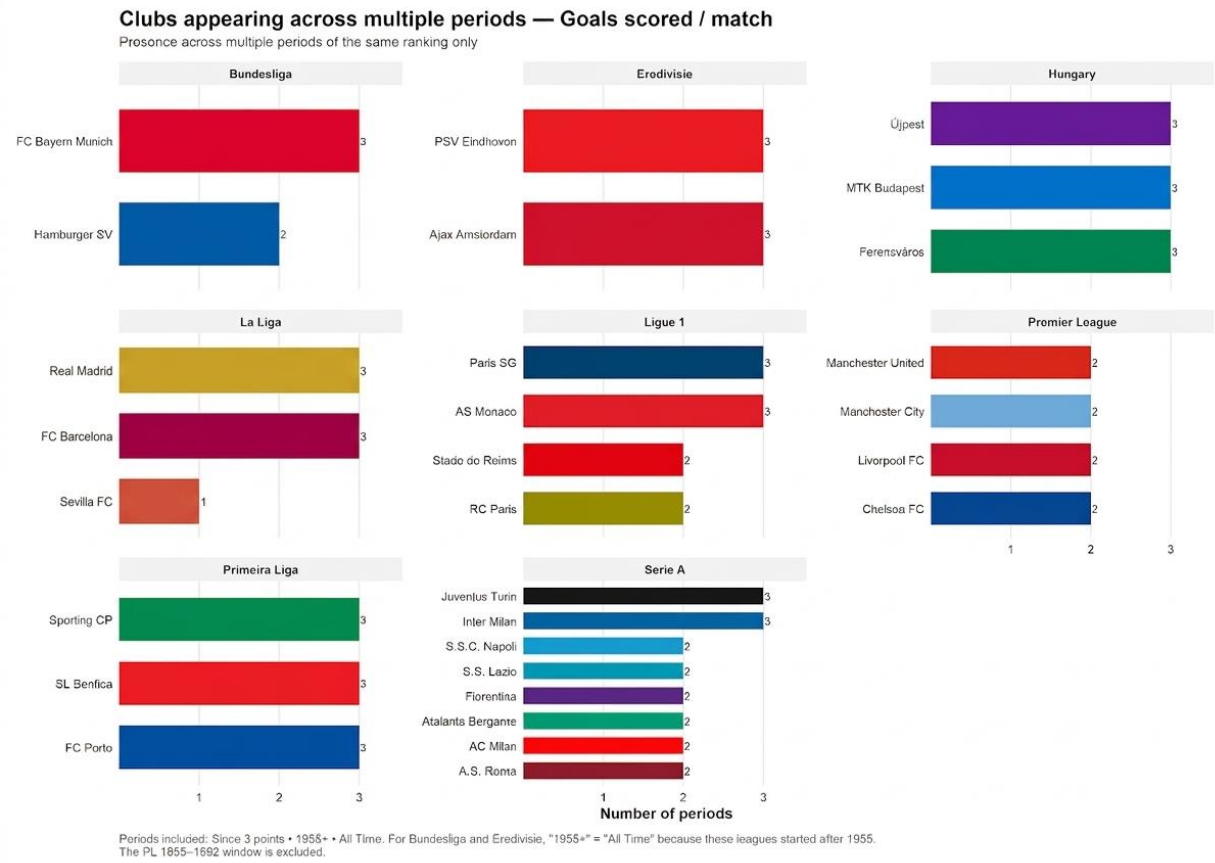


Fig. S11: Persistence of clubs in the rankings of goals scored per match for each period in each league

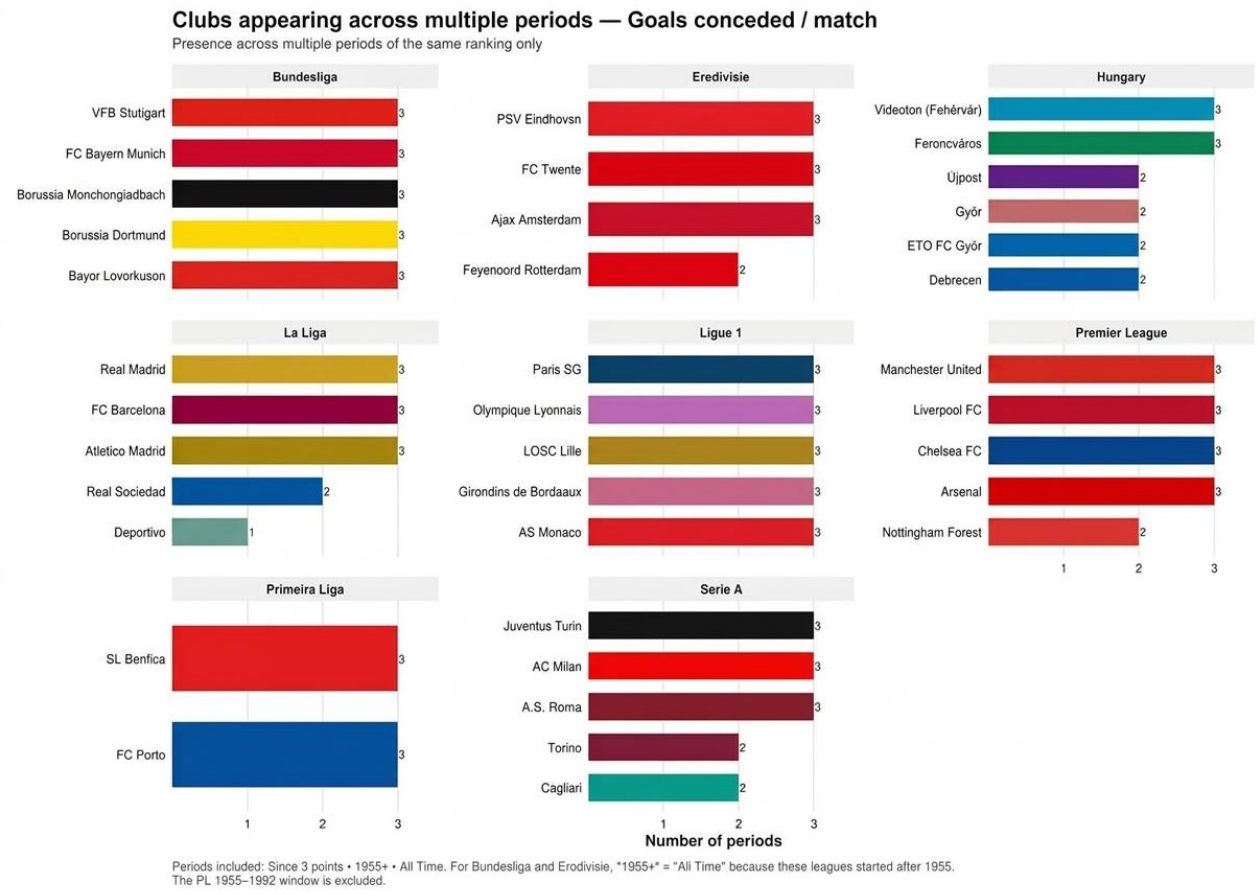


Fig. S12: Persistence of clubs in the rankings of goals conceded per match for each period in each league

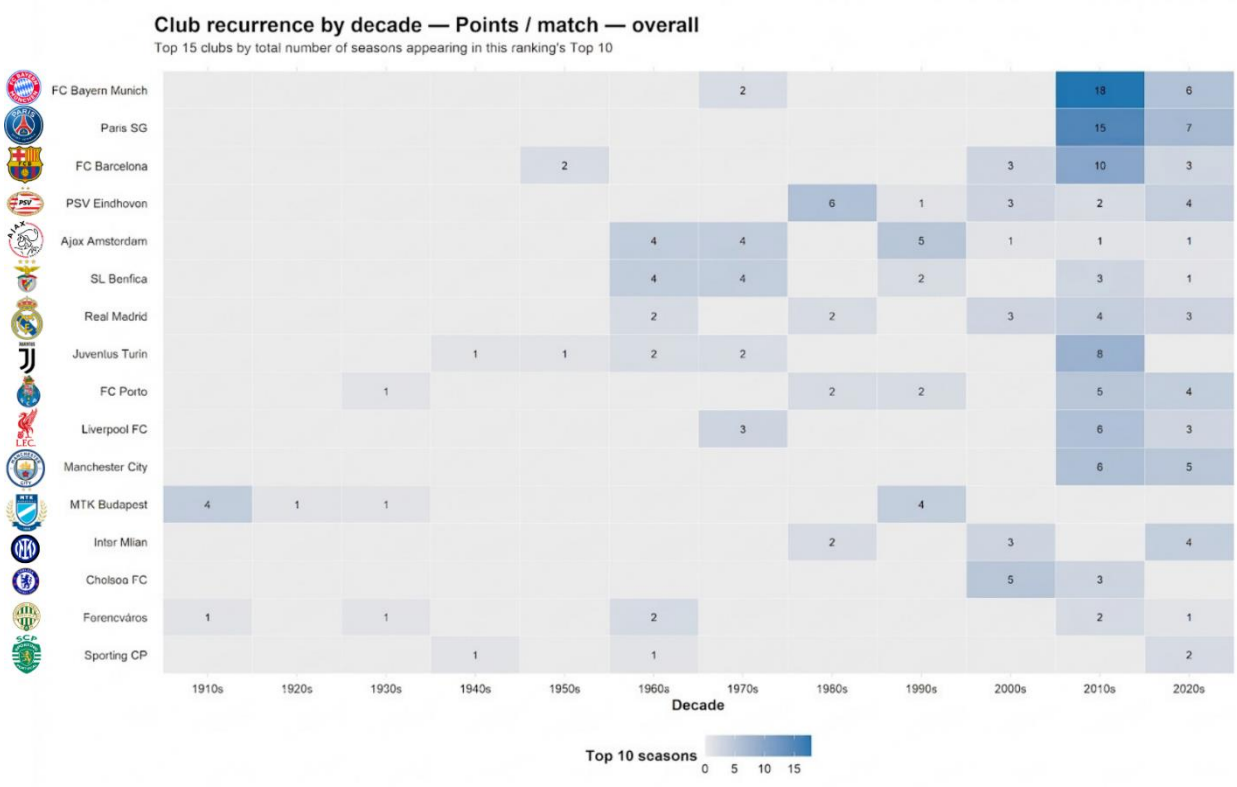


Fig. S13: Recurrence of clubs across historical rankings of points per match, aggregated by decade.

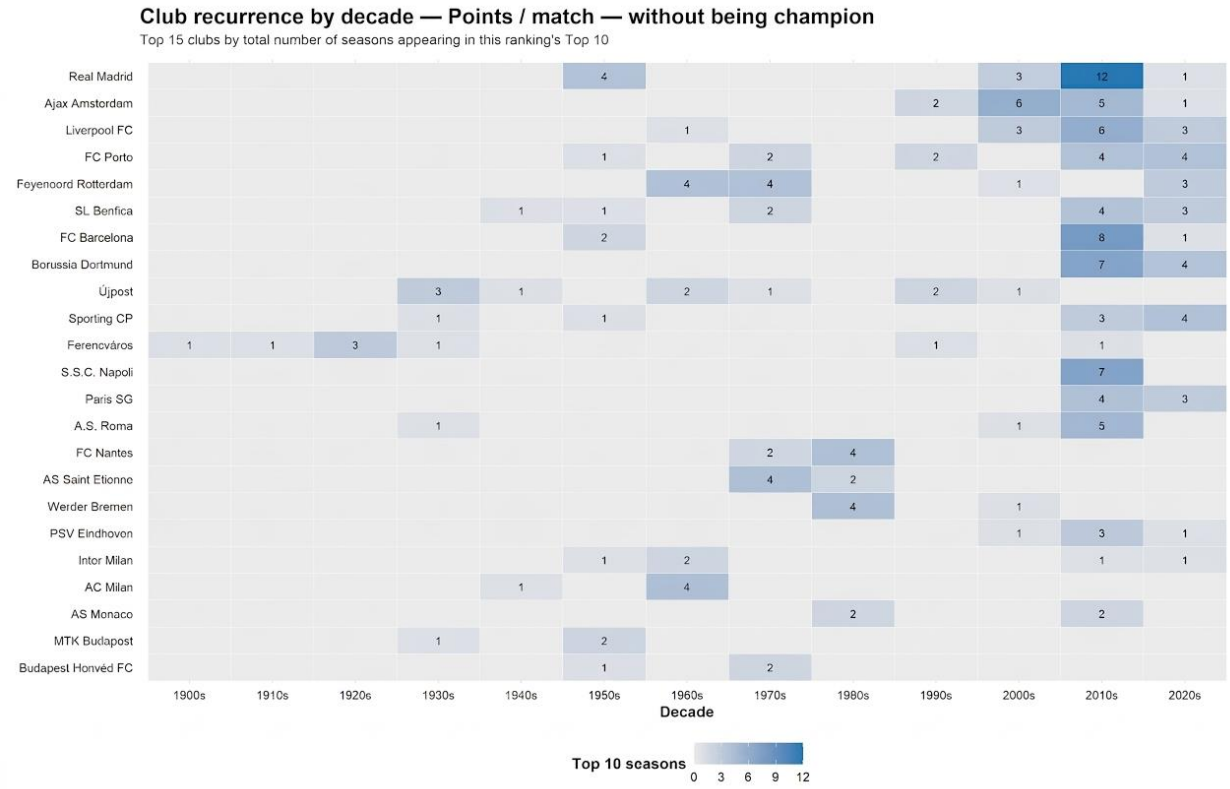


Fig. S14: Recurrence of clubs across historical rankings of points per match among non-champions, aggregated by decade.

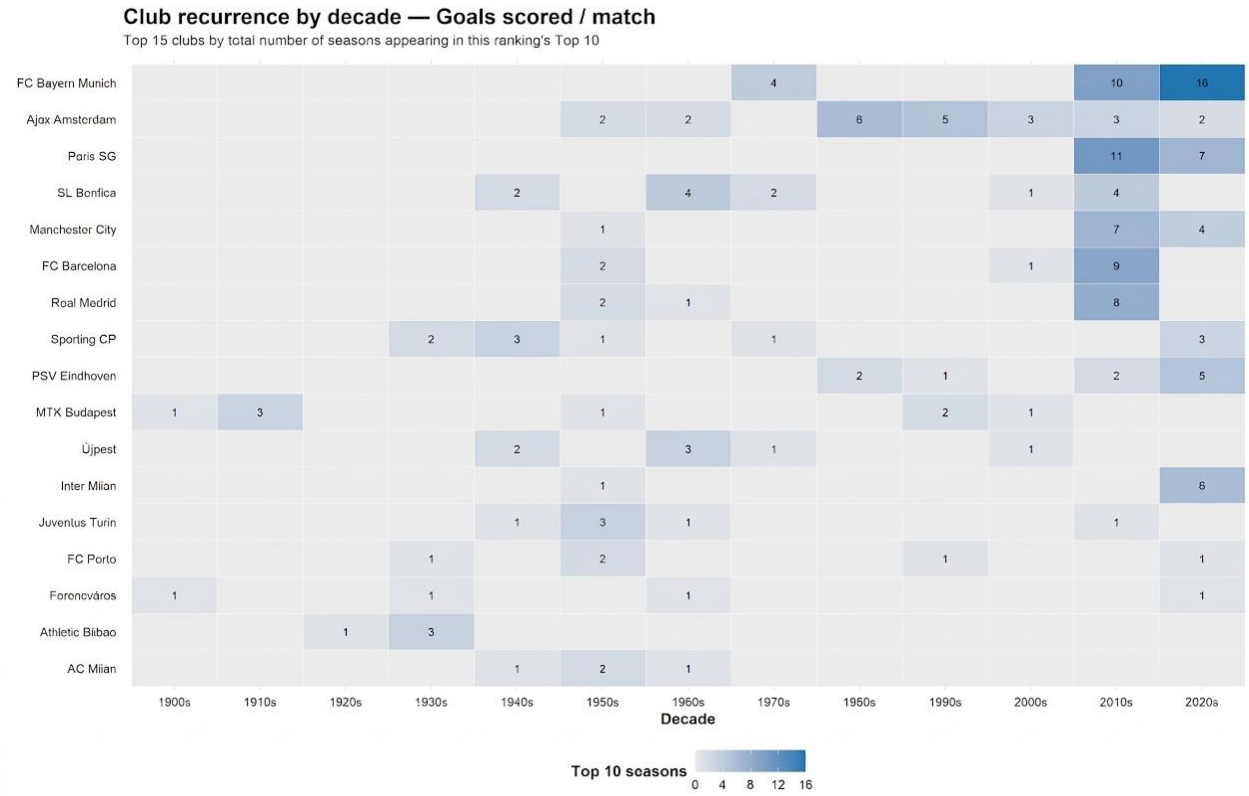


Fig. S15: Recurrence of clubs across historical rankings of goals scored per match aggregated by decade.

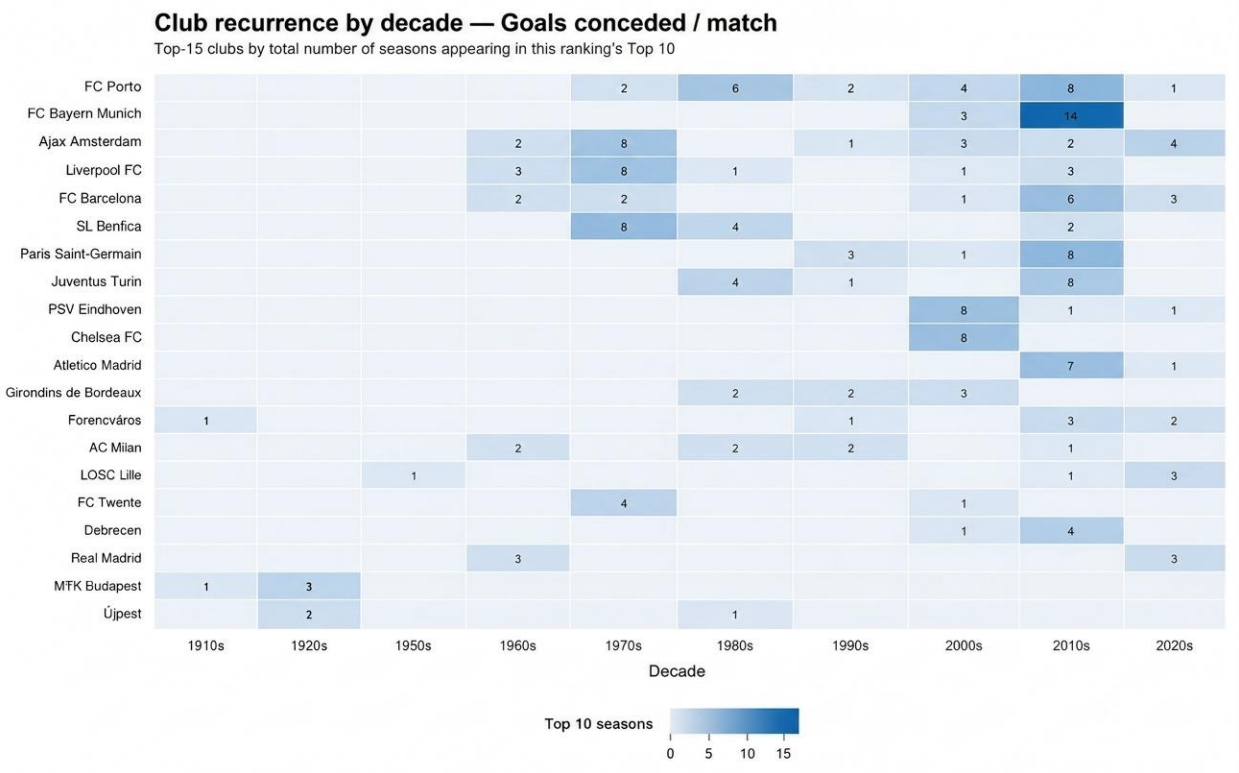


Fig. S16: Recurrence of clubs across historical rankings of goals conceded per match aggregated by decade.